



US 20250277230A1

(19) **United States**

(12) **Patent Application Publication**
ARLING et al.

(10) **Pub. No.: US 2025/0277230 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **PARTHENOGENESIS METHODS AND COMPOSITIONS**

Related U.S. Application Data

(60) Provisional application No. 63/363,180, filed on Apr. 19, 2022.

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Publication Classification

(51) **Int. Cl.**
C12N 15/82 (2006.01)
C12N 9/22 (2006.01)
C12N 15/11 (2006.01)

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(52) **U.S. Cl.**
CPC **C12N 15/8287** (2013.01); **C12N 9/222** (2025.05); **C12N 15/11** (2013.01); **C12N 15/8205** (2013.01); **C12N 2310/20** (2017.05)

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(57) **ABSTRACT**

Provided herein are methods and compositions for inducing parthenogenesis by expressing, in a female gametophyte, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell. Also included herein are promoters that express in a plant egg cell and methods of expressing a gene product in an egg cell of a plant using these promoters.

(21) Appl. No.: **18/857,700**

(22) PCT Filed: **Apr. 18, 2023**

(86) PCT No.: **PCT/US23/65923**

§ 371 (c)(1),

(2) Date: **Oct. 17, 2024**

Specification includes a Sequence Listing.

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Consensus  MAKXNMWLGFSLSXQXNPXYQDSPL-AAIDY-XX---KXXXXX-X---KDX-XXX-X-  48
TA-BBM6A  MASNNDWLGFSLSQDENPQPHQDSBP-ADIDACASGFYGLPTQPA--PDAQLVPCW  56
OS-BBM2   MASANNWLGFSLSQDENPQPHQDSBP-ADIDVSGAGDFYGLPTQPTAADAHLGVACH  58
SV-BBM2   MASAKNWLGFSLSQDENPQPHQDSBPAAAGIDISCATDFYGLPTQPG--SDGHLGVPCW  57
ZM-BBM2   MASANNWLGFSLSQDENPQPHQDSBPAAAGIDISGASDFYGLPTQPG--SDGHLGVPCW  56
TA-BBM3A  MATVNNWLGFSLSQDELPSAAS-----GTV-----  27
OS-ODP2   MATMNNWLGFSLSQDELPPSQTKSTLISAAAT-----TTTAGDS-----  40
SV-BBM    MATVNNWLGFSLSQDELPPSQTKSTLISAAAT-----TDE-----  34
ZM-ODP2   MATVNNWLGFSLSQDELPPSQTKSTLISAAAT-----ADH-----  35

Consensus  -----XXXY-EAKXKCFNXPQDWNMRG-XXXXXX-SELSKLYXXX-----VA  87
TA-BBM6A  H--QPSYGTAEAFNRGAHEIHDWNNMRGDLDYNGGASELSMLVGS SAVGG-KRAAMEAET  112
OS-BBM2   H-HNASYGIMEAFNRGAQAQDWNMRG-LDYNGGASELSMLVGS SGGK-----RAAAVEET  112
SV-BBM2   RDDHASYGIMEAFNRSCQETQDWNMRG-LDYNGGASELSMLVGS S-----GKRAVEDG  110
ZM-BBM2   RDDHASYGIMEAYNRVQETQDWNMRG-LDYNGGASELSMLVGS SGGGGGNGKRAVEDS  114
TA-BBM3A  -----SGADVVCYNIPQDWNMRG-----SELSAL-----VA  52
OS-ODP2   -----STGDVVCFNIPQDWSMRG-----SELSAL-----VA  65
SV-BBM    -----VSGDVVCFNIPQDWSMRG-----SELSAL-----VA  59
ZM-ODP2   -----VSGDVVCFNIPQDWSMRG-----SELSAL-----VA  60

Consensus  EPKLEDFLGGSFVSZQDX-XXGGXLFSGVPM-----XX--XX--X-XX-----  126
TA-BBM6A  EPKLEDFLVGNSFVSEQDQ--AGGFLFSGVPM-----  143
OS-BBM2   EPKLEDFLGNSFVSEQDQ--AGGFLFSGVPM-----  145
SV-BBM2   EPKLEDFLGNSFVSEQDQ--SGGYLFSGVPM-----  141
ZM-BBM2   EPKLEDFLGNSFVSEQDQ--SGGYLFSGVPM-----  145
TA-BBM3A  EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
OS-ODP2   EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
SV-BBM    EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
ZM-ODP2   EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145

Consensus  EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
TA-BBM6A  EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
OS-BBM2   EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
SV-BBM2   EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
ZM-BBM2   EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
TA-BBM3A  EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
OS-ODP2   EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
SV-BBM    EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
ZM-ODP2   EPKLEDFLGGSFVSZQDQ--SGGYLFSGVPM-----  145
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```
Consensus  --X--X-XXXXXXX-X-----SSTNSNGSXTXXLSMIKXWLR  160
TA-BBM6A  -----TSTNSNGSXTXXLSMIKXWLR  166
OS-BBM2   -----SSTNSNGSXTXXLSMIKXWLR  168
SV-BBM2   -----SSTNSNGSXTXXLSMIKXWLR  164
ZM-BBM2   -----SSTNSNGSXTXXLSMIKXWLR  168
TA-BBM3A  PSSASLQFADSVMVASSGGGVHHDGAGIMANTANG-DLNNAGSGGGGIGLSMIKXWLR  165
OS-ODP2   PSSASLQFADSVMVASSGGGVHHDGAGIMANTANG-DLNNAGSGGGGIGLSMIKXWLR  170
SV-BBM    PSSASLQFADSVMVASSGGGVHHDGAGIMANTANG-DLNNAGSGGGGIGLSMIKXWLR  165
ZM-ODP2   PSSASLQFADSVMVASSGGGVHHDGAGIMANTANG-DLNNAGSGGGGIGLSMIKXWLR  167
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Consensus  XPXPQCPX---X-----AXAQ---GLXKSXBYX---MXGCS---XX-----NMX  191
TA-BBM6A  NQVPQAPHPQSC-----TGAPAPQPPHEEMGIDASS--F---DPLGR-----NGA  207
OS-BBM2   NQVPQAPHPQSC-----TGAPAPQPPHEEMGIDASS--F---DPLGR-----NGA  207
SV-BBM2   NQVPQAPHPQSC-----TGAPAPQPPHEEMGIDASS--F---DPLGR-----NGA  205
ZM-BBM2   NQVPQAPHPQSC-----TGAPAPQPPHEEMGIDASS--F---DPLGR-----NGA  207
TA-BBM3A  QPSPAPPPPPQ---AD-AAAQ---GLSLSMN---MACMP-----P  197
OS-ODP2   QPSPAPPPPPQ---AD-AAAQ---GLSLSMN---MACMP-----P  199
SV-BBM    QPSPAPPPPPQ---AD-AAAQ---GLSLSMN---MACMP-----P  204
ZM-ODP2   QPSPAPPPPPQ---AD-AAAQ---GLSLSMN---MACMP-----P  202
```

```
Consensus  LXAGESS---QSLALSMSTGXQXHLPMVVGGA--XXSGGXSESXSNXNKRASGAMDS  244
TA-BBM6A  LVVAGSS---QSLALSMSTGXQXHLPMVVGGA--XXSGGXSESXSNXNKRASGAMDS  259
OS-BBM2   AAGGTS---QSLALSMSTGXQXHLPMVVGGA--XXSGGXSESXSNXNKRASGAMDS  280
SV-BBM2   VAGGTS---QSLALSMSTGXQXHLPMVVGGA--XXSGGXSESXSNXNKRASGAMDS  258
ZM-BBM2   VAGGTS---QSLALSMSTGXQXHLPMVVGGA--XXSGGXSESXSNXNKRASGAMDS  257
TA-BBM3A  LVGCRCVPE---LAIV-----RKDDTAGCSAG--SGAVVAGGADST  236
OS-ODP2   LVGCRCVPE---LAIV-----RKDDTAGCSAG--SGAVVAGGADST  254
SV-BBM    LVGCRCVPE---LAIV-----RKDDTAGCSAG--SGAVVAGGADST  249
ZM-ODP2   LVGCRCVPE---LAIV-----RKDDTAGCSAG--SGAVVAGGADST  249
```

Consensus	MAXNNWLGFSLSXQXNPXPQDSSPL-AAIDX-XX--XXXXXX-X--XDX-XXX-X-	48
TA-BBM6A	MASGNDWLGFSLSGQENPQPHQDSSPP-ADIDIAGSGFYGLPTQPA--PDAQLGVPGH	56
OS-BBM2	MASANNNLGFSLSGQENPQPHQDSSPP-AAIDVSGAGDFYGLPTSQPTAADAHLGVAGH	58
SV-BBM2	MASANNNLGFSLSGQDNPOANQDSSPAAAGIDISGATDFYGLPTQQG--SDGHLGVPGL	57
ZM-BBM2	MASANNNLGFSLSGQDNPOPNQDSSPA-AGIDISGASDFYGLPTQQG--SDGHLGVPGL	56
TA-BBM3A	MATVNNWLGFSLSPQELPPSAAS-----GDV-----	27
OS-ODP2	MATMNNWLAFSLSPQDQLPPSQTNSTLISAAAT-----TTTAGDS-----	40
SV-BBM	MATVNNWLAFSLSPQDLPPSQT-DSTLISATA-----TDE-----	34
ZM-ODP2	MATVNNWLAFSLSPQELPPSQTTDSTLISAAAT-----ADH-----	35
Consensus	-----XXXX-EAXBXCFNXPQDWNMRG-XXXXXX-SELSXLXXXX-----VA	87
TA-BBM6A	H--QPSYGITEAFNRGAHEIHDWNMRGDLDYNGGASELSMLVGSSAVGG-KRAAMEAET	112
OS-BBM2	H-HNASYGIMEAFNRGAQEAQDWNMRG-LDYNGGASELSMLVGSSGK--RAAAVEET	112
SV-BBM2	RDDHASYGIMEAFNRSQQETQDWNMRG-LDYNGGASELSMLVGSSG--GKRAVEDG	110
ZM-BBM2	RDDHASYGIMEAYNRVPOETQDWNMRG-LDYNGGSELSMLVGSSGGGNGKRAVEDS	114
TA-BBM3A	-----SGADVVCYNIPQDWNMRG-----SELSAL-----VA	52
OS-ODP2	-----STGDVVCFNIPQDWSMRG-----SELSAL-----VA	65
SV-BBM	-----VSGDVVCFNIPQDWSMRG-----SELSAL-----VA	59
ZM-ODP2	-----VSGDVVCFNIPQDWSMRG-----SELSAL-----VA	60
Consensus	EPKLEDFLGXSFXSZQDX-XXGGXLFSGVPMA-----XX-XX-X-XX-----	126
TA-BBM6A	EPKLEDFLVGNSFVSEQDQ--AGGFLFSGVPMA-----	143
OS-BBM2	EPKLEDFLGNSFVSEQDHHAGGFLFSGVPMA-----	145
SV-BBM2	EPKLEDFLGNSFVSEQDQ--SGYLFSGVPMA-----	141
ZM-BBM2	EPKLEDFLGNSFVSDQDQ--SGYLFSGVPIA-----	145
TA-BBM3A	EPKLEDFLGGISYSDHHHHKQAGGNMVMVPAGSGGGAACYTTSGGSSVGY----LYH	107
OS-ODP2	EPKLEDFLGGISFSEQQ--HHHGKGGVIP-S--SAAACYASSGS-SVGY--LYPP	113
SV-BBM	EPKLEDFLGGINFSEQH--HKA--NLNVIPSS--SNACYASSGA-STGYHHQLYHH-	107
ZM-ODP2	EPKLEDFLGGISFSEQH--HKA--NCNMIPST--SSTVCYASSGA-STGYHHQLYHQ	110

FIG. 1

Consensus	--X--X-XXXXXXX-X-	SSTNSNSGSXTXXLSMIKXWLRS	160
TA-BBM6A	-----	TSTNSNSRNTMELSMIKSWLRD	166
OS-BBM2	-----	SSTNSNSGSNTMELSMIKTWLRN	168
SV-BBM2	-----	SSTNSNSGSNTMELSMIKTWLRN	164
ZM-BBM2	-----	SSANSNSGSNTMELSMIKTWLRN	168
TA-BBM3A	PSSASLQFADSVMVASSGGGVHHDGAGIMANTTANG-DLNNSGGGGIGLSMIKSWLRS	165	
OS-ODP2	PSSSSLQFADSVMTSSPVAHDGVS--GGMVSAAAAAASGNGGIGLSMIKSWLRS	170	
SV-BBM	-QSSALHFADSVMVASSAGVHDGGASMLSAATVNGGAGAAASANGGIGLSMIKSWLRS	165	
ZM-ODP2	PTSSALHFADSVMVASSAGVHDGGAMLSA--AAANGVAGAAASANGGIGLSMIKSWLRS	167	
Consensus	XPXPQXP----	AXAQ-----GLXXSXBXX--MXGCS-----XX-----NMX	191
TA-BBM6A	NQVPQAPHPQSQ-----	TCAPAQPPQPHHEMGTDASS--F-----DPLGR-----NGA	207
OS-BBM2	NGQVPAGHPQPPQPPAAAAAAQQAHEAEMSTDASASFCCSSDAMGRSNNGGAVSA	227	
SV-BBM2	NQVPQ-----PQPP-----	APHQAAPQPEEMSTDASASFCCSDS-LGR-----NGT	205
ZM-BBM2	NQVAQ-----PQPP-----	APHQ--PQPEEMSTDASGSFCCSDS-MGR-----NSM	207
TA-BBM3A	QPSAPQPEQQR-----AD-AAAQ-----GLSLSMN-----MAACMP-----P	197	
OS-ODP2	QPAPQP-----	-AQ-----ALSLSMN-----MAGTTTA-----QGGGAMA	199
SV-BBM	QPAPPLPQPRVAAEGA-QAAQ-----GLSLSMN-----MAGAO-----GAGMP	204	
ZM-ODP2	QPAPMQPRVA-----AA-EGAQ-----GLSLSMN-----MAGTT-----QGAAGMP	202	
Consensus	LXAGESS----	QSLALSMSTGXQXHLPMVVXGGA--XXSGGXSESXXXNXRASGAMDS	244
TA-BBM6A	LVVAGSS----	QSLALSMSTGSGSHLPMVAVLGGs--PSGGVSESTSENKRASGAMDS	259
OS-BBM2	AAGGTSS----	QSLALSMSTGSHSLPIVVAGCG--NASGGAESTSENKRASGAMDS	280
SV-BBM2	VAAAG-SS----	QSLALSMSTGPQ-HLPMVVAGGGGASGAAASESTSENKRASGAMDS	258
ZM-BBM2	VAAAGSS----	QSLALSMSTGS--HLPMVVPSPA--ASGAASESTSENKRASGAMDS	257
TA-BBM3A	LVGGERGVPE-----LAIV-----	RKDDTAGGSSAG-SGAVVSAGGADST	236
OS-ODP2	LLAGAGERGRTTPASELSLTSAHGATTATMAGGRKEINE-EGSGS-AGAVVAVG--SES	254	
SV-BBM	LLAGERG--R--APESVSTSAQGGAVA-----ARKEDS-GGSGG-AGALVAVS--TDT	249	
ZM-ODP2	LLAGERA--R--APESVSTSAQGGAVVVT--APKEDS-GGSGV-AGALVAVS--TDT	249	

FIG. 1 (Cont'd)

Consensus	XXX---PX--AAEAVARKSIDTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQXRK	298
TA-BBM6A	-----PGG-AVEAVARKSIDTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQSRK	311
OS-BBM2	-----PGGAI EAVPRKSIDTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQSRK	333
SV-BBM2	-----PSSGAIEAVPRKSIDTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQSRK	311
ZM-BBM2	-----PGS-AVEAVPRKSIDTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQSRK	309
TA-BBM3A	GGSSGVV---VETPASRKTADTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQTRK	292
OS-ODP2	GGSGAVVEAGAAAARKSVDTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQTRK	313
SV-BBM	GGSGT-V----AETAARKTVDTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQTRK	303
ZM-ODP2	GGSGGAS----ADNTARKTVDTFFGQRTSIYRGVTRHRWTGRYE AHLWDNSCRREGQTRK	304
Consensus	GRQVYLG GYDKEXKAARAYDLAALKYWGT TTTTNFPXSNYEKELEEMKHMTRQEXXAXL	357
TA-BBM6A	GRQVYLG GYDKEDKAARAYDLAALKYWGT TTTTNFPINTYEKEVDEM KHMTRQEYIAYL	370
OS-BBM2	GRQVYLG GYDKEDKAARAYDLAALKYWGT TTTTNFPISNYEKELDEM KHMTRQEYIAYL	392
SV-BBM2	GRQVYLG GYDKEDKAARAYDLAALKYWGT TTTTNFPISNYEKELEEMKHMTRQEYIAYL	370
ZM-BBM2	GRQVYLG GYDKEDKAARAYDLAALKYWGT TTTTNFPISNYEKELEEMKHMTRQEYIAYL	368
TA-BBM3A	GRQVYLG GYDK EEKAARAYDLAALKYWGT TTTTNFPVNNYEKELEEMKHMTRQEFFVASL	351
OS-ODP2	GRQVYLG GYDK EEKAARAYDLAALKYWGT TTTTNFPVNNYEKELEEMKHMTRQEFFVASL	372
SV-BBM	GRQVYLG GYDK EEKAARAYDLAALKYWGT TTTTNFPVSNYEKELEEMKHMTRQEFFVASL	362
ZM-ODP2	GRQVYLG GYDK EEKAARAYDLAALKYWGT TTTTNFPVSNYEKELEDMKHMTRQEFFVASL	363
Consensus	RRXSSGFSRGASXYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTZEEAAEAYDIAAI	416
TA-BBM6A	RRNSSGFSRGASKYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTEEAAEAYDIAAI	429
OS-BBM2	RRNSSGFSRGASKYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTEEAAEAYDIAAI	451
SV-BBM2	RRNSSGFSRGASKYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTEEAAEAYDIAAI	429
ZM-BBM2	RRNSSGFSRGASKYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTEEAAEAYDIAAI	427
TA-BBM3A	RRKSSGFSRGAS IYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTQEEAAEAYDIAAI	410
OS-ODP2	RRKSSGFSRGAS IYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTQEEAAEAYDIAAI	431
SV-BBM	RRKSSGFSRGAS IYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTQEEAAEAYDIAAI	421
ZM-ODP2	RRKSSGFSRGAS IYRGVTRHHQHGRWQARI GRVAGNKDLYLGTFTSTQEEAAEAYDIAAI	422

FIG. 1 (Cont'd)

Consensus	KFRGLNAVTFNFDMSRYDVKSILXSSXLPVGGAAAXRLKXAEXXAE-AGATIW--R---AX	469
TA-BBM6A	KFRGLNAVTFNFDMSRYDVKTILESSTLPVGGAAARRPKEVADHPE-AGATIW--R---AG	482
OS-BBM2	KFRGLNAVTFNFDMSRYDVKSILESSTLPVGGAAARRLKEADHAEAGATIW--R---AA	505
SV-BBM2	KFRGLNAVTFNFDMSRYDVKSILESSTLPVGGAAARRLKDAVDHVE-AGATIW--R---AD	482
ZM-BBM2	KFRGLNAVTFNFDMSRYDVKSILESSTLPVGGAAARRLKDAVDHVE-AGATIW--R---AD	480
TA-BBM3A	KFRGLNAVTFNFDMSRYDVKSILDSTALPIGSAAKRLKDAEAAATASSLALQQQHHAGVVS	469
OS-ODP2	KFRGLNAVTFNFDMSRYDVKSILDSAALPVGTAAKRLKDAEAA-----A	474
SV-BBM	KFRGLNAVTFNFDMSRYDVKSILDSALPIGSAAKRLKEAEAAASA-----Q---HAG-VV	472
ZM-ODP2	KFRGLNAVTFNFDMSRYDVKSILDSALPIGSAAKRLKEAEAAASA-----QHHAG-VV	475
Consensus	XYDXGXVXSQX--GMGAYAAAXYXXHXH-----HXWPTIAFQQPSPL-----GVXXP	514
TA-BBM6A	M--DGGVISQLTDVGMGTYAS-----YH-----HGWPTIAFQQPSPL-----SVHYP	522
OS-BBM2	DMDGAGVISGLADVGMGAYAAASYHHHH-----HGWPTIAFQQPPPL-----AVHYP	552
SV-BBM2	-M-DAGVISQLADAGMGAYAS-----YGH-----HGWPTIAFQQPSPL-----TVHYP	523
ZM-BBM2	-M-DGAVISQLAEAGMGYAS-----YGH-----HGWPTIAFQQPSPL-----SVHYP	521
TA-BBM3A	GYDMGALA-----AYGAAYHHHPSVAGAVAWPTIAFQAPPHQVSSGGGHMYHP	519
OS-ODP2	AYDVGRIASHLG--GDGAYAAHYCHHH--SAAAAWPTIAFQAAAAPP--PHAAGLYHP	527
SV-BBM	SYDVGRIASHLG--DGGALA-AYGTHYH--AAAAAWPTIAFQPS-----AAAAGLYHP	520
ZM-ODP2	SYDVGRIASQLG--DGGALAAAYGAHYH--G--AAWPTIAFQPG-----AASTGLYHP	522
Consensus	YAQ-X--PSRGWCKXEQDXAVIAAAHSLQDLQXLXLGAA-XAHBFFXAS-----X	560
TA-BBM6A	YVA---QPSRGWCKPEQDASVA--TQSLQDLQQLHLGTA--AHNYFQAS-----	564
OS-BBM2	YQAPAAPSRGWCKPEQDAVAATAAAHSLQDLQQLHLGSA-AAHNFFQAS-----	600
SV-BBM2	YGQ----PSRGWCKPEQDAVAATAAAHSLQDLQQLHLGSA--AHNFFQAS-----	566
ZM-BBM2	YGQ----PSRGWCKPEQDAVAATAAAHSLQDLQQLHLGSA--AHNFFQAS-----	563
TA-BBM3A	YAH-A-QPLRGWCKQEPDHAVIAAAHSLQELHHLNLGAGGGAHDFFSQHAQAMQQQHGG	576
OS-ODP2	YAQ-PL---RGWCKQEQDHAVIAAAHSLQDLHHLNLGAAAAAHDFFSQAM---QQQHGG	578
SV-BBM	YAQ-P-M--RGWCKQEQDHAVIAAAHSLHHLNLGAAACAHDFFSAGQ---AAMHG	571
ZM-ODP2	YAQ-QPMRGGWCKQEQDHAVIAAAHSLQDLHHLNLGA-AGAHDFFSAGQQAAMHG	579

FIG. 1 (Cont'd)

Consensus	XXXX- - - - XXXXXXXXSSXVYNGGGG-X-G- - - -X- -GGYFILPMPSTVVAXXXQGH	606
TA-BBM6A	- - - - -SSSTVYNGGGYQQG- - - - -TSGNAFLMPASTVVD- -EQGH	597
OS-BBM2	- - - - -SSSTVYNGGGG- - -GYQGL- - -GGNAFLMPASTVVA- -DQGH	634
SV-BBM2	- - - - -SSSTVYNGGGASAGYQGLSGGGGFLMPSSTVV- -ADQGH	606
ZM-BBM2	- - - - -SSSTVYNGGAGASGGYQGLG- -GGSSFLMPSSTVVAAADQGH	603
TA-BBM3A	LGSIHNAAGASLEHSTGSNSVYNNGGV- - - - -ESY-MLPMSTTTATASHA	622
OS-ODP2	LGSIDN- - -ASLEHSTGSNSVYNGDNG- - - - -GGGGYIMAPMSAVSATATAVA	625
SV-BBM	LGSIDN- - -ASLEHSTGSNSVYNG-VGDSNG- - - -GAVGGGYM-MPMSAAAAT-TTAM	620
ZM-ODP2	LGSIDS- - -ASLEHSTGSNSVYNGGVGDSNGASAVGGSGGYM-MPMSAAGATTTSAM	634
Consensus	SS- -XN- - - -QGSTCSY- - -GDG-XQGKXIGYDAYLVAAAAGG- - - - -GGRMXSW	647
TA-BBM6A	SSTATN- - - -QGSTCSY- - -GDEE-GKLNIGYDAMAMAS-TGADPYAA- - - - -	636
OS-BBM2	SSTATN- - - -HGNTCSY- - -GNE-EQGKLIgyDAMAMASGA- - - - -AGGYQLSQ	676
SV-BBM2	SS-TAN- - - -QGSTCSY- - -GDD-QDGKLIgyDAM- - -AAAGGDPYAAARSgyQFSQ	651
ZM-BBM2	SS-TAN- - - -QGSTCSY- - -GDDHQEGKLIgyDAAMVATAAGDPYAAARNgyQFSQ	652
TA-BBM3A	REQVAHAHARAVAAQAYSGGNMSRDHDDGKIAYENYLvSTAEAYG- - - - -GGRMAAW	675
OS-ODP2	S- - - - -SH- - - - -DHGGDGKQVQMgyDSYLvGADAYCG- - - - -GGAGRMPsw	663
SV-BBM	V- - - - -SHEQVHARSHQGEHDEATKQAAQMgyESYLvNAEAAyG- - - - -GGRMPsw	666
ZM-ODP2	V- - - - -SHEQVHARAY- - - - -DEAKQAAQMgyESYLvNAEN-NG- - - - -GGRMSAW	674
Consensus	GSASTVSIARA- - - - -XXXXFSXWXXPFNGXMG	675
TA-BBM6A	- -APTvSIARA- - - - -NGYSNNWSSPFNGM- - -	659
OS-BBM2	GSASTVSIARA- - - - -NGYSANWSSPFNGAMG	703
SV-BBM2	GSGSTVSIARA- - - - -NGYSNNWSSPFNG-MG	677
ZM-BBM2	GSGSTVSIARA- - - - -NGYANNWSSPFNNGMG	679
TA-BBM3A	AQASAPPATSS- - - - -SDMTGAQLFSVWNDN- - - - -	702
OS-ODP2	AMTPASAPAAATSSS- - -DMTGVC-HGAQLFSVWNDT- - - - -	695
SV-BBM	TPASAP- - -AAASSNDN-MAGVGHGGAQLFSVWNDT- - - - -	698
ZM-ODP2	GTVVSAAAAAAASSNDNMAADVGHGGAQLFSVWNDT- - - - -	710

FIG. 1 (Cont'd)

Consensus	MAXXNBWLGFSLSXQEXPXXXXXXSXXXXXXXXDXXGAXXXYXJPXXXXXXXXXXXXXXXXXXXX	59
TA-BBM-3A	MATVNNWLGFSLSPQELPPSAAS-----GDVSGADVVCYNIP-----	37
TA-BBM-3A (ALT1)	-----	
TA-BBM3A (ALT2)	-----	
TA-BBM-6A	MASGNDWLGFSLSGQENPQPHQDSSPPADIDIAGASGFYGLPTQPAPDAQLGVPGHHQP	59
TA-BBM-6A (ALT1)	-----	
TA-BBM-6A (ALT2)	-----	
Consensus	XXXXXXXXXXXXXXXXXDWNMRGSEL-----QDWNMRGSEL-----SAL-----	118
TA-BBM-3A	-----VAEPKLED	58
TA-BBM-3A (ALT1)	-----MAEPKLED	8
TA-BBM3A (ALT2)	-----MAEPKLED	8
TA-BBM-6A	SYGITEAFNRGAHEIHDWNMRGDLDYNGGASELSMLVGSSAVGGKRAAMEAETEPKLED	118
TA-BBM-6A (ALT1)	-----MAETEPKLED	10
TA-BBM-6A (ALT2)	-----MAETEPKLED	10
Consensus	FLXGSXX	177
TA-BBM-3A	FLGGSYSDDHHHHKQAGGNMVMVPAGSGSGGAACYTTSGGSSVGILYHPSSASLQFAD	117
TA-BBM-3A (ALT1)	FLGGSYSDDHHHHKQAGGNMVMVPAGSGSGGAACYTTSGGSSVGILYHPSSASLQFAD	67
TA-BBM3A (ALT2)	FLGGSYSDDHHHHKQAGGNMVMVPAGSGSGGAACYTTSGGSSVGILYHPSSASLQFAD	67
TA-BBM-6A	FLVGNSFVS-----	127
TA-BBM-6A (ALT1)	FLVGNSFVS-----	19
TA-BBM-6A (ALT2)	FLVGNSFVS-----	19
Consensus	XXXXXXXXXGGXXXXXXSXXXXXXXXXXXXXXXXXLSMIKSWLRXXXXXXPPXXQX	236
TA-BBM-3A	SVMVASSGGGVHHDGAGIMANTTANGDLNNGSGGGIGLSMIKSWLRSQSPAPQPEQQ	176
TA-BBM-3A (ALT1)	SVMVASSGGGVHHDGAGIMANTTANGDLNNGSGGGIGLSMIKSWLRSQSPAPQPEQQ	126
TA-BBM3A (ALT2)	SVMVASSGGGVHHDGAGIMANTTANGDLNNGSGGGIGLSMIKSWLRSQSPAPQPEQQ	126
TA-BBM-6A	---EQDQAGGFLFSGVPM-----ATSTNSNSRNTMELSMIKSWLRDNQVPQAPHQPS	177
TA-BBM-6A (ALT1)	---EQDQAGGFLFSGVPM-----ATSTNSNSRNTMELSMIKSWLRDNQVPQAPHQPS	69
TA-BBM-6A (ALT2)	---EQDQAGGFLFSGVPM-----ATSTNSNSRNTMELSMIKSWLRDNQVPQAPHQPS	69

FIG. 2

Consensus	XXXXXAXQXXXXXX	295
TA-BBM-3A	RADAAAQGLSLSMNMA--ACMPPL	213
TA-BBM-3A (ALT1)	RADAAAQGLSLSMNMA--ACMPPL	163
TA-BBM-3A (ALT2)	RADAAAQGLSLSMNMA--ACMPPL	163
TA-BBM-6A	QTGAPAQOPQPOHEEMGTDA	236
TA-BBM-6A (ALT1)	QTGAPAQOPQPOHEEMGTDA	128
TA-BBM-6A (ALT2)	QTGAPAQOPQPOHEEMGTDA	128
Consensus	XXXXGGXSXXXXXX	354
TA-BBM-3A	DDTAGGSSA--GSGAVVSAGGADST	270
TA-BBM-3A (ALT1)	DDTAGGSSA--GSGAVVSAGGADST	220
TA-BBM-3A (ALT2)	DDTAGGSSA--GSGAVVSAGGADST	220
TA-BBM-6A	-SPSGGVSESTSENKRASGAMDSPGGA	289
TA-BBM-6A (ALT1)	-SPSGGVSESTSENKRASGAMDSPGGA	181
TA-BBM-6A (ALT2)	-SPSGGVSESTSENKRASGAMDSPGGA	181
Consensus	WTGRYEAHLWDNSCRREGQ	413
TA-BBM-3A	WTGRYEAHLWDNSCRREGQ	329
TA-BBM-3A (ALT1)	WTGRYEAHLWDNSCRREGQ	279
TA-BBM-3A (ALT2)	WTGRYEAHLWDNSCRREGQ	279
TA-BBM-6A	WTGRYEAHLWDNSCRREGQ	348
TA-BBM-6A (ALT1)	WTGRYEAHLWDNSCRREGQ	240
TA-BBM-6A (ALT2)	WTGRYEAHLWDNSCRREGQ	240
Consensus	NYXEKEXXEMKHMTRQ	472
TA-BBM-3A	NNYEKELEEMKHMTRQ	388
TA-BBM-3A (ALT1)	NNYEKELEEMKHMTRQ	338
TA-BBM-3A (ALT2)	NNYEKELEEMKHMTRQ	338
TA-BBM-6A	NTYEKEVDEMKHMTRQ	407
TA-BBM-6A (ALT1)	NTYEKEVDEMKHMTRQ	299
TA-BBM-6A (ALT2)	NTYEKEVDEMKHMTRQ	299

FIG. 2 (Cont'd)

Consensus		DLYLGTFTSTZEEAAEAYDIAAIKFRGLNAVTFDMXRYDVXKILXSXXLPXGXAAXRXK	531
TA-BBM-3A		DLYLGTFTSTQEEAAEAYDIAAIKFRGLNAVTFDMTRYDVKSILDSTALPIGSAAKRLK	447
TA-BBM-3A	(ALT1)	DLYLGTFTSTQEEAAEAYDIAAIKFRGLNAVTFDMTRYDVKSILDSTALPIGSAAKRLK	397
TA-BBM3A	(ALT2)	DLYLGTFTSTQEEAAEAYDIAAIKFRGLNAVTFDMTRYDVKSILDSTALPIGSAAKRLK	397
TA-BBM-6A		DLYLGTFTSTEEAAEAYDIAAIKFRGLNAVTFDMTRYDVKTILESTLPVGGAAARRPK	466
TA-BBM-6A	(ALT1)	DLYLGTFTSTEEAAEAYDIAAIKFRGLNAVTFDMTRYDVKTILESTLPVGGAAARRPK	358
TA-BBM-6A	(ALT2)	DLYLGTFTSTEEAAEAYDIAAIKFRGLNAVTFDMTRYDVKTILESTLPVGGAAARRPK	358
Consensus		XXXXXXXXXXJXXXXXXXXXGVXSXXXXXXXXXXAXYXXXXXXXXXXXXXWPTIAFQXPX	590
TA-BBM-3A		DAEATATASSLALQ-QQQHAGVVSQYDMGALAAAGAAAYHHHHHPSVAGAVAWPTIAFQAPP	505
TA-BBM-3A	(ALT1)	DAEATATASSLALQ-QQQHAGVVSQYDMGALAAAGAAAYHHHHHPSVAGAVAWPTIAFQAPP	455
TA-BBM3A	(ALT2)	DAEATATASSLALQ-QQQHAGVVSQYDMGALAAAGAAAYHHHHHPSVAGAVAWPTIAFQAPP	455
TA-BBM-6A		EVADHPEAGATIWRAQMDGGVISQLTVDVGMGTY-ASY-----HHGWPTIAFQQPS	515
TA-BBM-6A	(ALT1)	EVADHPEAGATIWRAQMDGGVISQLTVDVGMGTY-ASY-----HHGWPTIAFQQPS	407
TA-BBM-6A	(ALT2)	EVADHPEAGATIWRAQMDGGVISQLTVDVGMGTY-ASY-----HHGWPTIAFQQPS	407
Consensus		XXXXXXXXXXHXXAQPXRGWCKXEXDXXVXXXXXSLQXLXLLGXAXHXBFX	649
TA-BBM-3A		HQVSSGGGHHMYPYAHQAQPLRGWCKQEPDHAVIAAAHSLQELHHLNLGAGGAHDFFS	564
TA-BBM-3A	(ALT1)	HQVSSGGGHHMYPYAHQAQPLRGWCKQEPDHAVIAAAHSLQELHHLNLGAGGAHDFFS	514
TA-BBM3A	(ALT2)	HQVSSGGGHHMYPYAHQAQPLRGWCKQEPDHAVIAAAHSLQELHHLNLGAGGAHDFFS	514
TA-BBM-6A		PLS-----VHYPYVAQPSRGWCKPEQDASVA--TQSLQDLQQLHLGT--AAHNYFQ	562
TA-BBM-6A	(ALT1)	PLS-----VHYPYVAQPSRGWCKPEQDASVA--TQSLQDLQQLHLGT--AAHNYFQ	454
TA-BBM-6A	(ALT2)	PLS-----VHYPYVAQPSRGWCKPEQDASVA--TQSLQDLQQLHLGT--AAHNYFQ	454
Consensus		XXXXXXXXXXGXXXXXGXXXXXGXXXXXGXXXXXGXXXXXGXXXXXGXXXXXGXXXXX	708
TA-BBM-3A		QHAQAMQOQHGGGLGSIHNAAGASLEHSTGNSVYNNNGGVESYMLPMSTTTTATASHAR	623
TA-BBM-3A	(ALT1)	QHAQAMQOQHGGGLGSIHNAAGASLEHSTGNSVYNNNGGVESYMLPMSTTTTATASHAR	573
TA-BBM3A	(ALT2)	QHAQAMQOQHGGGLGSIHNAAGASLEHSTGNSVYNNNGGVESYMLPMSTTTTATASHAR	573
TA-BBM-6A		ASSSSTVYNGGGYQQG--TS-----GNAFLMPASTVVDEQGH---	597
TA-BBM-6A	(ALT1)	ASSSSTVYNGGGYQQG--TS-----GNAFLMPASTVVDEQGH---	489
TA-BBM-6A	(ALT2)	ASSSSTVYNGGGYQQG--TS-----GNAFLMPASTVVDEQGH---	489

FIG. 2 (Cont'd)

Consensus	XXXXXXXXXXXXXXXXXXXXXGKJXXXXXX	758
TA-BBM-3A	EQVAHAHARAVAAQAAYSGNMSRDHDDGKIA--YENYLVST--AEAYG--	668
TA-BBM-3A (ALT1)	EQVAHAHARAVAAQAAYSGNMSRDHDDGKIA--YENYLVST--AEAYG--	618
TA-BBM-3A (ALT2)	EQVAHAHARAVAAQAAYSGNMSRDHDDGKIA--YENYLVST--AEAYG--	618
TA-BBM-6A	-----SSTA--TNQGSTCSYGDEEGKLNLCYDAMAMASTGADPYAAAPTYSIARA	645
TA-BBM-6A (ALT1)	-----SSTA--TNQGSTCSYGDEEGKLNLCYDAMAMASTGADPYAAAPTYSIARA	537
TA-BBM-6A (ALT2)	-----SSTA--TNQGSTCSYGDEEGKLNLCYDAMAMASTGADPYAA--	537
Consensus	GGYSNNWSSPF-----NGMQLFVWNDSN	782
TA-BBM-3A	GGRMAAWAQASAPPATSSSDMTGAQLFSVWNDSN	702
TA-BBM-3A (ALT1)	GG-----	620
TA-BBM-3A (ALT2)	GG-----	620
TA-BBM-6A	NGYSNNWSSPF-----NGM--	659
TA-BBM-6A (ALT1)	NGYSNNWSSPF-----NGM--	551
TA-BBM-6A (ALT2)	-----	528

FIG. 2 (Cont'd)

PARTHENOGENESIS METHODS AND COMPOSITIONS

CROSS REFERENCE

[0001] This application claims the benefit of U.S. Provisional Application No. 63/363,180 filed Apr. 19, 2022, which is incorporated herein by reference in its entirety.

FIELD

[0002] The present disclosure relates to the field of plant molecular biology and plant breeding.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

[0003] The official copy of the sequence listing is submitted electronically via EFS-Web as an XML formatted sequence listing with a file named 8893-WO PCT_SEQ_LIST_ST26.XML created on Apr. 11, 2023, and having a size of 287 kilobytes and is filed concurrently with the specification. The sequence listing comprised in this XML formatted document is part of the specification and is herein incorporated by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

[0004] In plants, germ line cells (germline) provide the transgenerational inheritance of genetic information in each subsequent generation by producing spore mother cells during sporogenesis. For example, sporogenesis provides the megaspore mother cell that develops the female gametes, the egg cell and central cell that give rise to the embryo and endosperm, respectively; or the microspore mother cell that develops the male gamete, giving rise to four haploid microspores, wherein each microspore further develops into a mature pollen grain. A key aspect for the unique role of germline cells is providing the genetic information a future offspring receives, wherein half of the genetic contribution is from the female gamete and half of the genetic contribution is from the male gamete. Fertilization of the egg cell with one sperm cell forms a diploid zygote, while a second sperm cell fuses with the two polar nuclei of the central cell to form a triploid endosperm. The endosperm is a terminally nourishing tissue for the embryo yet does not contribute to the germline. After fertilization, the zygote gives rise to an embryo, a process referred to as zygotic embryogenesis that is characteristic of sexual reproduction. A newly formed embryo undergoing such an embryogenesis developmental program comprising an underlying regulatory program affected by genetic determinants and epigenetic reprogramming leading from an embryogenic cell state to the acquisition of a differentiated cell fate, or cell fates, ultimately giving rise to a plant with all differentiated tissues thereof.

[0005] Parthenogenesis is a form of asexual reproduction wherein growth and development of female gametes (embryos) occur without fertilization by sperm. The female gamete produced parthenogenetically may be either haploid or diploid.

SUMMARY

[0006] Described herein are methods of expressing a gene product in an egg cell of a plant, which includes introducing into the plant a polynucleotide comprising (a) a polynucleotide sequence of SEQ ID NO: 37, 38, or 39, (b) a poly-

nucleotide sequence having at least 90% sequence identity to SEQ ID NO: 37, 38, or 39; (c) a fragment of the polynucleotide of (a) or (b), where the fragment retains its ability to drive expression of a heterologous polynucleotide sequence in an egg cell, where the polynucleotide sequence drives expression of an operably linked heterologous polynucleotide sequence. In some aspects, the operably linked heterologous polynucleotide sequence is a Babyboom (BBM) polynucleotide that encodes a BBM polypeptide that, when expressed in a female gametophyte, induces haploid induction. Also included herein are plants that include a promoter operably linked to a heterologous polynucleotide sequence, where the promoter is selected from (a) a polynucleotide sequence of SEQ ID NO: 37, 38, or 39, (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 37, 38, or 39, (c) a fragment of the polynucleotide of (a) or (b), where the fragment retains its ability to drive expression of the heterologous polynucleotide sequence in an egg cell of a plant.

[0007] Also described herein are methods of inducing parthenogenesis, where the method includes expressing, in a female gametophyte, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in a plant egg cell, where the Babyboom (BBM) polynucleotide includes a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof, where the female gametophyte is rendered parthenogenic and forms a haploid embryo without pollination, and where the Babyboom (BBM) polynucleotide is not naturally expressed in a female gametophyte. In some aspects, the Babyboom (BBM) polynucleotide or polypeptide or fragment thereof is a BBM1, BBM2, BMN2, BMN3, ODP2, and BBML polynucleotide or polypeptide or fragment thereof. In some aspects, the promoter that expresses in the plant egg cell is an egg cell-specific promoter or an egg cell-preferred promoter.

[0008] In some aspects, the female gametophyte has not been fertilized prior to or during expression of the parthenogenesis factor, e.g. the BBM polynucleotide. In some aspects, BBM is expressed in the egg cell prior to fertilization to induce parthenogenesis, the development of a zygote/embryo in the absence of fertilization. In some aspects, the methods include modifying a regulatory region of an endogenous Babyboom (BBM) polynucleotide so that the Babyboom (BBM) polynucleotide expresses in a female gametophyte. In some aspects, the methods include expressing the Babyboom (BBM) polynucleotide from a modified endogenous genomic BBM locus, where the modified endogenous genomic BBM locus includes a modified regulatory region of an endogenous polynucleotide encoding a BBM polypeptide, where one or more nucleotides in the regulatory region have been modified so that the BBM polypeptide expresses in a female gametophyte. In some aspects, the methods include expressing the Babyboom (BBM) polynucleotide from a modified endogenous genomic egg cell locus, where the egg cell's coding or genomic sequence in the endogenous genomic egg cell locus has been modified so that it encodes a BBM polypeptide, where the BBM polypeptide expresses in a female gametophyte. In some aspects, the female gametophyte is a monocot or dicot female gametophyte. In some aspects, the female gametophyte is wheat, maize, rice, oats, barley, triticale, sorghum, canola, *Arabidopsis*, cotton, sunflower, safflower, tobacco, *cannabis*, sugarcane, soy, turf grass, rye, millet, or a flax female gametophyte. In some aspects, the methods include (a) contacting

the haploid embryo with a chromosome doubling agent for a period sufficient to generate a doubled haploid embryo, (b) isolating the doubled haploid embryo, and (c) regenerating a doubled haploid plant from the doubled haploid embryo of step (b). In some aspects, the methods include (a) regenerating a parthenogenic plant from a haploid embryo comprising the Babyboom (BBM) polynucleotide operably linked to the promoter that expresses in an egg cell; (b) pollinating the parthenogenic plant of (a) with pollen from a non-haploid inducer; and (c) rescuing a haploid embryo from the parthenogenic plant of (b).

[0009] Also included herein are methods for obtaining a wheat plant producing clonal, non-reduced, non-recombined gametes. In some aspects, the method includes (a) suppressing in a wheat plant cell the activity of: (1) all endogenous Spo11 or Prd1, Prd2, or Prd3 polynucleotides or polypeptides; (2) all endogenous Rec8 polynucleotides or polypeptides; (3) all endogenous Osd1 polynucleotides or polypeptides; (b) expressing, in a female gametophyte derived from the wheat plant cell, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell, where the Babyboom (BBM) polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof, where the female gametophyte is rendered parthenogenic and forms an embryo; and (c) obtaining a plant from the embryo, where the embryo comprises the egg cell expressed Babyboom (BBM) polynucleotide and suppressed Spo11, Rec8, Osd1, Prd1, Prd2, or Prd3 polynucleotides or polypeptides, thereby producing a wheat plant producing clonal, non-reduced, non-recombined gametes.

[0010] Also described herein are methods of obtaining a clonal apomictic plant from one or more gametophytic cells in a plant in the absence of egg cell fertilization. In some aspects, the methods include (a) expressing, in one or more gametophytic cell, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell, where the Babyboom (BBM) polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof that retains haploid induction activity, where the Babyboom (BBM) polynucleotide is not naturally expressed in a female gametophyte, (b) developing an embryo from the gametophytic cell in the absence of egg cell fertilization; and (c) obtaining a progeny plant from one or more gametophytic cells where the progeny plant con-

tains the chromosomes from the gametophytic cell of (a), thereby achieving propagation of a flowering plant in the absence of egg cell fertilization. In some aspects, in the methods and compositions described herein, the plant is a monocot or dicot plant. In some aspects,, in the methods and compositions described herein, the plant is a wheat, cotton, sunflower, safflower, tobacco, *Arabidopsis*, soy, barley, oats, rice, maize, triticale, sorghum, canola, *cannabis*, sugarcane, rye, millet, turf grass, or a flax plant.

DESCRIPTION OF THE FIGURES

[0011] FIG. 1 shows an alignment of BBM peptide sequences. (Consensus=SEQ ID NO: 112, TA-BBM6A=SEQ ID NO: 8, OS-BBM2=SEQ ID NO: 25, SV-BBM2=SEQ ID NO: 29; ZM-BBM2=SEQ ID NO: 21; TA-BBM3A=SEQ ID NO: 2; OS-ODP2=SEQ ID NO: 23; SV-BBM=SEQ ID NO: 29; ZM-ODP2=SEQ ID NO: 14). Some of the A motifs in the sequences in FIG. 1 are set forth in SEQ ID NOs: 107, 109, 116, 117, and 118. Some of the B motifs in the sequences in FIG. 1 are set forth in SEQ ID NOs: 108, 110, 119, 120, and 121. Some of the AP2 DNA binding domains in the sequences in FIG. 1 are set forth in SEQ ID NOs: 111 and 122.

[0012] FIG. 2 shows an alignment of wheat TA-BBM native and modified peptide sequences. (Consensus=SEQ ID NO: 113; TA-BBM3A=SEQ ID NO: 2; TA-BBM3A (ALT1)=SEQ ID NO: 42; TA-BBM3A (ALT2)=SEQ ID NO: 44; TA-BBM6A=SEQ ID NO: 8; TA-BBM6A (ALT1)=SEQ ID NO: 46; BBM6A (ALT2)=SEQ ID NO: 48). The A motifs in the sequences in FIG. 2 are set forth in SEQ ID NOs: 107 and 109. The B motifs in the sequences in FIG. 2 are set forth in SEQ ID NOs: 108 and 110. The conserved AP2 DNA binding domain in the sequences in FIG. 2 is set forth in SEQ ID NOs: 111.

BRIEF DESCRIPTION OF THE SEQUENCE LISTING

[0013] The disclosure can be more fully understood from the following detailed description and the accompanying Sequence Listing which form a part of this application.

[0014] Table A presents SEQ ID NOs for various polynucleotide and polypeptide sequences. It is understood, as those skilled in the art will appreciate, that the disclosure encompasses more than these specific exemplary sequences.

TABLE A

SEQ ID NO:	Polynucleotide (DNA) or Polypeptide (PRT)	Name	Description
1	DNA	TA-BBM3	<i>Triticum aestivum</i> BBM3A coding sequence
2	PRT	TA-BBM3A	<i>Triticum aestivum</i> BBM3A amino acid sequence
3	DNA	TA-BBM3B	<i>Triticum aestivum</i> BBM3B coding sequence
4	PRT	TA-BBM3B	<i>Triticum aestivum</i> BBM3B amino acid sequence
5	DNA	TA-BBM3D	<i>Triticum aestivum</i> BBM3D coding sequence
6	PRT	TA-BBM3D	<i>Triticum aestivum</i> BBM3D amino acid sequence
7	DNA	TA-BBM6A	<i>Triticum aestivum</i> BBM6A coding sequence
8	PRT	TA-BBM6A	<i>Triticum aestivum</i> BBM6A amino acid sequence
9	DNA	TA-BBM6B	<i>Triticum aestivum</i> BBM6B coding sequence
10	PRT	TA-BBM6B	<i>Triticum aestivum</i> BBM6B amino acid sequence
11	DNA	TA-BBM6D	<i>Triticum aestivum</i> BBM6D coding sequence
12	PRT	TA-BBM6D	<i>Triticum aestivum</i> BBM6D amino acid sequence
13	DNA	ZM-ODP2	<i>Zea mays</i> ODP2 coding sequence
14	PRT	ZM-ODP2	<i>Zea mays</i> ODP2 amino acid sequence
15	DNA	ZM-DD45 PRO	<i>Zea mays</i> DD45 promoter sequence

TABLE A-continued

SEQ ID NO:	Polynucleotide (DNA) or Polypeptide (PRT)	Name	Description
16	DNA	PV-EGG CELL PRO (TR1)	<i>Panicum virgatum</i> egg cell promoter sequence
17	DNA	EGG MIN PRO	<i>Panicum virgatum</i> egg cell promoter sequence
18	DNA	PV-PRO31696.1 5UTR	<i>Panicum virgatum</i> egg cell promoter sequence
19	DNA	PvEC1 PROMOTER	<i>Panicum virgatum</i> egg cell sequence promoter sequence
20	DNA	ZM-BBM2	<i>Zea mays</i> BBM2 coding sequence
21	PRT	ZM-BBM2	<i>Zea mays</i> BBM2 amino acid sequence
22	DNA	OS-ODP2	<i>Oryza sativa</i> ODP2 coding sequence
23	PRT	OS-ODP2	<i>Oryza sativa</i> ODP2 amino acid sequence
24	DNA	OS-BBM2	<i>Oryza sativa</i> BBM2 coding sequence
25	PRT	OS-BBM2	<i>Oryza sativa</i> BBM2 amino acid sequence
26	DNA	SV-BBM1	<i>Setaria viridis</i> BBM coding sequence
27	PRT	SV-BBM1	<i>Setaria viridis</i> BBM amino acid sequence
28	DNA	SV-BBM2	<i>Setaria viridis</i> BBM2 coding sequence
29	PRT	SV-BBM2	<i>Setaria viridis</i> BBM2 amino acid sequence
30	DNA	ZM-AS2-EME	<i>Zea mays</i> AS2 EME nucleotide sequence
31	DNA	ZM-DD45 PRO:(1X ZM-AS2-20)	DD45 promoter sequence with 1X ZM-AS2-20
32	DNA	ZM-DD45 PRO:(2X ZM-AS2-20)	DD45 promoter sequence with 2X ZM-AS2-20
33	DNA	ZM-DD45 PRO:(3X ZM-AS2-20)	DD45 promoter sequence with 3X ZM-AS2-20
34	DNA	ZM-DD45 PRO:(2X ZM-AS2-20)	DD45 promoter sequence with 2X ZM-AS2-20
35	DNA	ZM-DD45 PRO:(1X ZM-AS2-20)	DD45 promoter sequence with 1X ZM-AS2-20
36	DNA	ZM-DD45 PRO:(2X ZM-AS2-100)	DD45 promoter sequence with 2X ZM-AS2-100
37	DNA	TA-EGG PRO	<i>Triticum aestivum</i> egg cell promoter sequence
38	DNA	TA-EGG4A PRO	<i>Triticum aestivum</i> egg cell promoter sequence
39	DNA	TA-EGG4B PRO	<i>Triticum aestivum</i> egg cell promoter sequence
40	DNA	TA-EC PRO	<i>Triticum aestivum</i> egg cell promoter sequence
41	DNA	TA-BBM3A (ALT1)	<i>Triticum aestivum</i> BBM3A coding sequence
42	PRT	TA-BBM3A (ALT1)	<i>Triticum aestivum</i> BBM3A amino acid sequence
43	DNA	TA-BBM3A (ALT2)	<i>Triticum aestivum</i> BBM3A coding sequence
44	PRT	TA-BBM3A (ALT2)	<i>Triticum aestivum</i> BBM3A amino acid sequence
45	DNA	TA-BBM6A (ALT1)	<i>Triticum aestivum</i> BBM6A coding sequence
46	PRT	TA-BBM6A (ALT1)	<i>Triticum aestivum</i> BBM6A amino acid sequence
47	DNA	TA-BBM6A (ALT2)	<i>Triticum aestivum</i> BBM6A coding sequence
48	PRT	TA-BBM6A (ALT2)	<i>Triticum aestivum</i> BBM6A amino acid sequence
49	DNA	TA-BBM3A gRNA1	Nucleotide sequence of gRNA that targets wheat genomic BBM3A
50	DNA	TA-BBM3A gRNA2	Nucleotide sequence of gRNA that targets wheat genomic BBM3A
51	DNA	TA-BBM3A gRNA3	Nucleotide sequence of gRNA that targets wheat genomic BBM3A
52	DNA	TA-BBM3A gRNA4	Nucleotide sequence of gRNA that targets wheat genomic BBM3A
53	DNA	TA-BBM3A gRNA5	Nucleotide sequence of gRNA that targets wheat genomic BBM3A
54	DNA	TA-BBM3A gRNA6	Nucleotide sequence of gRNA that targets wheat genomic BBM3A
55	DNA	TA-BBM3A gRNA7	Nucleotide sequence of gRNA that targets wheat genomic BBM3A
56	DNA	TA-BBM3A gRNA8	Nucleotide sequence of gRNA that targets wheat genomic BBM3A
57	DNA	TA-SPO11-1 A genomic	<i>Triticum aestivum</i> Spo11-1 A genomic sequence
58	DNA	TA-SPO11-1 B genomic	<i>Triticum aestivum</i> Spo11-1 B genomic sequence
59	DNA	TA-SPO11-1 D genomic	<i>Triticum aestivum</i> Spo11-1 D genomic sequence
60	DNA	TA-REC8 A genomic	<i>Triticum aestivum</i> REC8 A genomic sequence
61	DNA	TA-REC8 B genomic	<i>Triticum aestivum</i> REC8 B genomic sequence
62	DNA	TA-REC8 D genomic	<i>Triticum aestivum</i> REC8 D genomic sequence
63	DNA	TA-OSD A genomic	<i>Triticum aestivum</i> OSD A genomic sequence

TABLE A-continued

SEQ ID NO:	Polynucleotide (DNA) or Polypeptide (PRT)	Name	Description
64	DNA	TA-OSD B genomic	<i>Triticum aestivum</i> OSD B genomic sequence
65	DNA	TA-OSD D genomic	<i>Triticum aestivum</i> OSD D genomic sequence
66	DNA	TA-SPO11-1 gRNA1	Nucleotide sequence of gRNA that targets wheat genomic SPO11-1
67	DNA	TA-SPO11-1 gRNA2	Nucleotide sequence of gRNA that targets wheat genomic SPO11-1
68	DNA	TA-REC8 gRNA1	Nucleotide sequence of gRNA that targets wheat genomic REC8
69	DNA	TA-REC8 gRNA2	Nucleotide sequence of gRNA that targets wheat genomic REC8
70	DNA	TA-REC8 gRNA3	Nucleotide sequence of gRNA that targets wheat genomic REC8
71	DNA	TA-REC8 gRNA4	Nucleotide sequence of gRNA that targets wheat genomic REC8
72	DNA	TA-OSD gRNA1	Nucleotide sequence of gRNA that targets wheat genomic OSD
73	DNA	TA-OSD gRNA2	Nucleotide sequence of gRNA that targets wheat genomic OSD
74	DNA	TA-BBM3A genomic	<i>Triticum aestivum</i> TA-BBM3A genomic sequence
75	DNA	TA-BBM3A genomic coding	<i>Triticum aestivum</i> BBM3A genomic coding sequence
76	DNA	TA-BBM3B genomic	<i>Triticum aestivum</i> BBM3B genomic sequence
77	DNA	TA-BBM3B genomic coding	<i>Triticum aestivum</i> TA-BBM3B genomic sequence
78	DNA	TA-BBM3D genomic	<i>Triticum aestivum</i> TA-BBM3D genomic sequence
79	DNA	TA-BBM3D genomic coding	<i>Triticum aestivum</i> TA-BBM3D genomic coding sequence
80	DNA	TA-BBM6A genomic	<i>Triticum aestivum</i> BBM6A genomic sequence
81	DNA	TA-BBM6A genomic coding	<i>Triticum aestivum</i> BBM6A genomic coding sequence
82	DNA	TA-BBM6B genomic	<i>Triticum aestivum</i> BBM6B genomic sequence
83	DNA	TA-BBM6B genomic coding	<i>Triticum aestivum</i> BBM6B genomic coding sequence
84	DNA	TA-BBM6D genomic	<i>Triticum aestivum</i> BBM6D genomic sequence
85	DNA	TA-BBM6D genomic coding	<i>Triticum aestivum</i> BBM6D genomic coding sequence
86	DNA	ZM-EGG PRO	<i>Zea mays</i> EGG PRO sequence
87	DNA	ZM-ODP2 genomic coding	<i>Zea mays</i> ODP2 genomic coding sequence
88	DNA	ZM-BBM2 genomic coding	<i>Zea mays</i> BBM2 genomic coding sequence
89	DNA	ZM-ODP2 promoter	<i>Zea mays</i> ODP2 promoter sequence
90	DNA	ZM-BBM2 promoter	<i>Zea mays</i> BBM2 promoter sequence
91	DNA	ZM-ODP2 PRO gRNA1	Nucleotide sequence of gRNA that targets <i>Zea mays</i> ODP2 promoter
92	DNA	ZM-ODP2 PRO gRNA2	Nucleotide sequence of gRNA that targets <i>Zea mays</i> ODP2 promoter
93	DNA	ZM-ODP2 PRO gRNA3	Nucleotide sequence of gRNA that targets <i>Zea mays</i> ODP2 promoter
94	DNA	ZM-ODP2 PRO gRNA4	Nucleotide sequence of gRNA that targets <i>Zea mays</i> ODP2 promoter
95	DNA	ZM-BBM2 PRO gRNA1	Nucleotide sequence of gRNA that targets <i>Zea mays</i> BBM2 promoter
96	DNA	ZM-BBM2 PRO gRNA2	Nucleotide sequence of gRNA that targets <i>Zea mays</i> BBM2 promoter
97	DNA	ZM-BBM2 PRO gRNA3	Nucleotide sequence of gRNA that targets <i>Zea mays</i> BBM2 promoter
98	DNA	ZM-BBM2 PRO gRNA4	Nucleotide sequence of gRNA that targets <i>Zea mays</i> BBM2 promoter
99	DNA	ZM-EGG gRNA1	Nucleotide sequence of gRNA that targets <i>Zea mays</i> Egg promoter
100	DNA	ZM-EGG gRNA2	Nucleotide sequence of gRNA that targets <i>Zea mays</i> Egg promoter

TABLE A-continued

SEQ ID NO:	Polynucleotide (DNA) or Polypeptide (PRT)	Name	Description
101	DNA	ZM-EGG gRNA3	Nucleotide sequence of gRNA that targets <i>Zea mays</i> Egg promoter
102	DNA	ZM-EGG gRNA4	Nucleotide sequence of gRNA that targets <i>Zea mays</i> Egg promoter
103	DNA	ZM-DD45 gRNA1	Nucleotide sequence of gRNA that targets <i>Zea mays</i> DD45 promoter
104	DNA	ZM-DD45 gRNA2	Nucleotide sequence of gRNA that targets <i>Zea mays</i> DD45 promoter
105	DNA	ZM-DD45 gRNA3	Nucleotide sequence of gRNA that targets <i>Zea mays</i> DD45 promoter
106	DNA	ZM-DD45 gRNA4	Nucleotide sequence of gRNA that targets <i>Zea mays</i> DD45 promoter
107	PRT	TA-BBM3A MOTIF A	<i>Triticum aestivum</i> BBM3A MOTIF A sequence
108	PRT	TA-BBM3A MOTIF B	<i>Triticum aestivum</i> BBM3A MOTIF B sequence
109	PRT	TA-BBM6A MOTIF A	<i>Triticum aestivum</i> BBM6A MOTIF A sequence
110	PRT	TA-BBM6A MOTIF B	<i>Triticum aestivum</i> BBM6A MOTIF B sequence
111	PRT	AP2 conserved domain	<i>Triticum aestivum</i> AP2 DNA binding domain sequence
112	PRT	Consensus sequence from FIG. 1	Consensus BBM peptide sequences
113	PRT	Consensus sequence from FIG. 2	Consensus <i>Triticum aestivum</i> BBM native and modified peptide sequences
114	DNA	ZM-DD45	<i>Zea mays</i> DD45 genomic coding sequence
115	DNA	ZM-EGG	<i>Zea mays</i> EGG genomic coding sequence
116	PRT	Motif A	Motif A consensus sequence
117	PRT	Motif A	Motif A consensus sequence BBM2 family
118	PRT	Motif A	Motif A consensus sequence ODP2 family
119	PRT	Motif B	Motif B consensus sequence
120	PRT	Motif B	Motif B consensus sequence BBM2 family
121	PRT	Motif B	Motif B consensus sequence ODP2 family
122	PRT	AP2 binding domain	AP2 DNA binding domain consensus sequence

DETAILED DESCRIPTION

[0015] The disclosures herein will be described more fully hereinafter with reference to the accompanying figures, in which some, but not all possible aspects are shown. Indeed, disclosures may be embodied in many different forms and should not be construed as limited to the aspects set forth herein; rather, these aspects are provided so that this disclosure will satisfy applicable legal requirements.

[0016] It is to be understood that the disclosures are not to be limited to the specific examples disclosed and that modifications and other aspects are intended to be included within the scope of the appended claims.

[0017] It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting. As used in the specification and in the claims, the term “comprising” can include the aspect of “consisting of.” Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosed methods and compositions belong. In this specification and in the claims which follow, reference will be made to a number of terms which shall be defined herein.

[0018] As used herein the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a cell” includes a plurality of such cells and reference to “the

protein” includes reference to one or more proteins and equivalents thereof known to those skilled in the art, and so forth. All technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure belongs unless clearly indicated otherwise.

[0019] All patents, publications and patent applications mentioned in the specification are indicative of the level of those skilled in the art to which this disclosure pertains. All patents, publications and patent applications are herein incorporated by reference in the entirety to the same extent as if each individual patent, publication or patent application was specifically and individually indicated to be incorporated by reference in its entirety.

[0020] The present disclosure provides methods and compositions for inducing parthenogenesis that includes expressing, in a female gametophyte, a BBM polypeptide or a fragment thereof encoded by a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell. As shown herein in Examples 1, expression of wheat, maize, rice and *Setaria viridis* BBM polypeptides in a female gametophyte of a wheat plant were useful for inducing maternal haploid induction, resulting in the production of haploid embryos without the use of pollination.

[0021] Any suitable polynucleotide that encodes BBM polypeptide or fragments thereof that are able to induce maternal haploid induction in female gametophytes when

expressed by a promoter that expresses in an egg cell may be utilized in the methods and compositions of the present disclosure. In some examples, the BBM polynucleotide comprises a nucleotide sequence encoding a BBM1, BBM2, BBM3, BBM6, BMN2, BMN3, ODP2, or BBML polypeptide. In some embodiments, BBM polynucleotides and BBM polypeptides include any of those disclosed in Publication No. US20210180077, incorporated herein by reference in its entirety. In some aspects, the BBM polynucleotide encoding a BBM polypeptide comprises a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88. In other aspects, the BBM polynucleotide comprises a nucleotide sequence having at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleotide sequence set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88 when aligned across its full length.

[0022] Fragments of polynucleotide sequences encoding BBM polypeptides are also encompassed by the embodiments. “Fragment” as used herein in reference to BBM refers to a portion of the nucleotide sequence encoding a BBM polypeptide. The fragment may encode a biologically active portion of a BBM polypeptide. Nucleic acid molecules that are fragments of a nucleotide sequence encoding a BBM polypeptide comprise at least about 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1050, 1100, 1150, 1200, 1250, 1300, 1350, 1400, 1450, 1500, 1550, 1600, 1650, 1700, 1750, 1800, 1850, 1900, 1950, 2000, 2050, 2100, 2150, 2200, 2250, 2300, 2350, 2400, 2450, 2550, 2600, 2650, 2700, 2750, 2800, 2850, 2900, 2950, 3000, 3100, 3200, 3300, 3400, 3500, 3600, 3700, 3800, 3900, 4000, 4100, 4200, 4300, 4400, 4500, 4600, 4700, 4800, 4900, 5000, 5100, 5200, 5300, 5400, 5500, 5600, 5700, 5800, 5900, 6000, 6100, 6200, 6300, 6400, 6500, 6600, 6700, 6800, 6900, 7000, 7100, 7200, 7300, 7400, 7500, 7600, 7700, 7800, 7900, 8000, 8100, or 8200 contiguous nucleotides or up to the number of nucleotides present in a full-length nucleic acid sequence encoding a BBM polypeptide disclosed herein, for example, those in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88. “Contiguous nucleotides” is used herein to refer to nucleotide residues that are immediately adjacent to one another. Fragments of the nucleic acid sequences of the embodiments that encode polypeptide fragments that retain the haploid induction activity of the BBM polypeptide and, hence, retain its ability to induce maternal haploid induction in a female gametophyte when expressed by a promoter that expresses in an egg cell. In some examples, the BBM polypeptides or polypeptide fragments include one or more of following: the A motif, the B motif, and/or the AP2 DNA binding domain. Non-limiting examples of A motifs are set forth in FIG. 1 and FIG. 2, including in SEQ ID NOs: 107, 109, 116, 117, and 118. Non-limiting examples of B motifs are set forth in FIG. 1 and FIG. 2, including in SEQ ID NO: 108, 110, 119, 120, and 121. Non-limiting examples of the AP2 DNA binding domain is set forth in FIG. 1 and FIG. 2, including in SEQ ID NOs: 111 and 122.

[0023] In some embodiments, the BBM polynucleotide is altered to create a nucleotide sequence that encodes a variant

BBM polypeptide. In some aspects, the BBM polynucleotide encoding a variant BBM polypeptide is derived from SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88 by alteration of one or more nucleotides by deletion, substitution, addition insertion, or combinations thereof. In some examples, nucleotide or amino acid sequences are maintained in one or more conserved regions of BBM polypeptides, such as the A motif, the B motif, or the AP2 DNA binding domain, and altered in non-conserved regions. See, for example, the alignment in FIGS. 1 and 2 and SEQ ID NOs: 107, 108, 109, 110, 111, 116, 117, 118, 119, 120, 121, and 122. In some examples, nucleotide or amino acid sequences are maintained in one or more consensus regions of BBM polypeptides, including in the A motif, the B motif, or the AP2 DNA binding domain, and altered in non-conserved regions. Variants of a BBM polypeptide useful in the methods and compositions described herein retain their ability to induce haploid induction in a female gametophyte when expressed by a promoter that expresses in an egg cell. Fragments and variants may be obtained via methods including but not limited to site-directed mutagenesis and synthetic construction.

[0024] In some aspects, the BBM polypeptide comprises an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48. In other aspects, the BBM polypeptide comprises an amino acid sequence having at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the amino acid set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48 when aligned across the full length of the sequences.

[0025] Fragments of BBM polypeptides are also encompassed by the embodiments. “Fragment” as used herein in reference to BBM refers to a portion of the amino acid sequence for a BBM polypeptide. The fragment may encode a biologically active portion of a BBM polypeptide. Polypeptides that are fragments of the BBM polypeptides comprise at least about 150, 180, 200, 210, 225, 240, 250, 270, 290, 300, 315, 325, 330, 350, 360, 375, 400, 425, 450, 475, 500, 525, 550, 575, 600, 625, 650, 675, 700, 725, or 750 contiguous amino acids or up to the number of amino acids present in a full-length amino acid sequence of a BBM polypeptide disclosed herein, for example, those in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48. “Contiguous amino acids” is used herein to refer to amino acid residues that are immediately adjacent to one another. Polypeptide fragments that retain the haploid induction activity of the BBM polypeptide and, hence, retain their ability to induce maternal haploid induction in a female gametophyte when expressed by a promoter that expresses in an egg cell. In some examples, the BBM polypeptide fragments include one or more of following: the A motif, the B motif, or the AP2 DNA binding domain. Non-limiting examples of BBM B motifs are set forth in FIG. 1 and FIG. 2, including in SEQ ID NO: 108 and 110. A non-limiting example of the BBM AP2 DNA binding domain is set forth in FIG. 1 and FIG. 2, including in SEQ ID NO: 111.

[0026] In some embodiments, the BBM polynucleotide is altered to create a nucleotide sequence that encodes a variant BBM polypeptide. In some aspects, the variant BBM polypeptide is derived from SEQ ID NO: 2, 4, 6, 8, 10, 12, 14,

21, 23, 25, 27, 29, 42, 44, 46, or 48 by alteration of one or more amino acid residues, for example, by deletion, substitution, addition insertion of one or more nucleotides. In some examples, amino acid sequences are maintained in one or more conserved regions of BBM polypeptides, such as the A motif, the B motif, or the AP2 DNA binding domain, and altered in non-conserved regions. See, for example, FIGS. 1 and 2. Non-limiting examples of A motifs are set forth in FIG. 1 and FIG. 2, including in SEQ ID NO: 107 and 109. Non-limiting examples of B motifs are set forth in FIG. 1 and FIG. 2, including in SEQ ID NO: 108 and 110. A non-limiting example of the AP2 DNA binding domain is set forth in FIG. 1 and FIG. 2, including in SEQ ID NO: 111. Variants of a BBM polypeptide useful in the methods and compositions described herein retain their ability to induce maternal haploid induction in a female gametophyte when expressed by a promoter that expresses in an egg cell. Fragments and variants may be obtained via methods including but not limited to site-directed mutagenesis and synthetic construction.

[0027] Any of the BBM polynucleotides described herein may be introduced into a female gametophyte, including the gametophyte's genome, using any suitable techniques, including but not limited to gene editing, or transformation. In some examples, the BBM polynucleotides are part of recombinant nucleic acids, such as heterologous polynucleotides and DNA constructs. As used herein, "heterologous" refers to a nucleic acid that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. For example, a promoter operably linked to a heterologous structural gene that is from a species different from that from which the structural gene was derived, or, if from the same species, one or both are substantially modified from their original form and/or genomic location.

[0028] Because BBM is not naturally expressed in a female gametophyte, its expression is modified by operatively linking the polynucleotide to a promoter that drives expression in an egg cell. As used herein, "egg cell" means the female gamete giving rise to the embryo.

[0029] As used herein "promoter" is an exemplary regulatory element and generally refers to a nucleotide sequence capable of controlling the expression of a coding sequence or functional RNA. In general, a coding sequence is located 3' to a promoter sequence. The promoter sequence comprises proximal and more distal upstream elements, the latter elements are often referred to as enhancers.

[0030] An "enhancer" is a nucleotide sequence that may stimulate promoter activity and may be an innate element of the promoter or a heterologous element inserted to enhance the level or tissue-specificity of a promoter. An example of an enhancer is an EME described elsewhere herein.

[0031] A promoter generally includes a core promoter (also known as minimal promoter) sequence that includes a minimal regulatory region to initiate transcription, that is a transcription start site. Generally, a core promoter includes a TATA box. A core promoter is a minimal sequence required to direct transcription initiation and generally may not include enhancers or other UTRs. Promoters may be derived in their entirety from a native gene, or be composed of different elements derived from different promoters found in nature, or even comprise synthetic DNA segments.

[0032] A "plant promoter" is a promoter capable of initiating transcription in plant cells. Any suitable promoter that is capable of initiating transcription of a polynucleotide in a plant egg cell, i.e. expressing the polynucleotide in the egg cell, may be utilized in the methods and compositions of the present disclosure, including but not limited to egg cell promoters, egg cell-specific promoters, or egg cell-preferred promoters, and promoters active during female gamete development.

[0033] Exemplary promoters that express in a plant egg cell include but are not limited to SEQ ID NOs: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, and 86. Also included in the embodiments are any of their fragments or variants.

[0034] Any egg cell promoters and/or egg cell-specific promoters known to one skilled in the art may be used, for example, the egg cell-specific EC1.1 and EC1.2 promoters disclosed in Sprunck et al., (2012) *Science*, 338, 1093-1097 and Steffen et al., (2007) *Plant J.*, 51:281-92) and those disclosed in US2015/0152430 and US2018/0094273, each of which is incorporated herein by reference in its entirety.

[0035] In one embodiment, the present disclosure provides a method of inducing parthenogenesis that includes expressing, in a female gametophyte, a BBM polypeptide or a fragment thereof encoded by a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell. As shown herein, expression of a BBM polypeptide in a female gametophyte of a plant is useful for inducing maternal haploid induction, resulting the production of a haploid embryo without the use of pollination.

[0036] The egg cell promoter nucleotide sequences and methods disclosed herein are useful in regulating expression of any heterologous nucleotide sequences (such as but not limited to BBM sequences) in egg cells of a plant. In some embodiments, the promoter that expresses in an egg cell is an egg cell-preferred promoter. In some embodiments, the promoter that expresses in an egg cell is an egg cell-specific promoter.

[0037] In one embodiment, this disclosure provides a method of expressing a gene product in a plant egg cell. In some aspects, the egg cell is in the plant. In other aspects, the egg cell is isolated from the plant. As used herein, gene product refers to a polynucleotide or a polypeptide. In some aspects, the method includes introducing into an egg cell a promoter comprising a polynucleotide sequence set forth in SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86.

[0038] In some aspects, the egg cell promoters comprise a nucleotide sequence having at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleotide sequence set forth in SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, when aligned across the full length of the sequences.

[0039] Fragments of the egg cell promoters are also encompassed by the embodiments. "Fragment" as used herein with reference to an egg cell promoter refers to a portion of the nucleotide sequence of the egg cell promoter. Nucleic acid molecules that are fragments of a nucleotide sequence of the egg cell promoters comprise at least about 50, 100, 150, 180, 210, 240, 270, 300, 330, 360, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1050, 1100, 1150, 1200, 1250, 1300, 1350, 1400, 1450,

1500, 1550, 1600, 1650, 1700, 1750, 1800, 1850, 1900, 1950, or 2000 contiguous nucleotides or up to the number of nucleotides present in a full-length nucleic acid promoter sequence disclosed herein, for example, those in SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, and 86. “Contiguous nucleotides” is used herein to refer to nucleotide residues that are immediately adjacent to one another. Fragments of the nucleic acid sequences of egg cell promoters that retain the ability to initiate transcription in an egg cell and, hence, retain their ability to express a heterologous polynucleotide in an egg cell.

[0040] In some embodiments, the polynucleotide of the egg cell promoter is altered to create a nucleotide sequence that encodes a variant. In some aspects, the egg cell promoter variant is derived from SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86 by alteration of one or more nucleotides by deletion, substitution, addition insertion, or combinations thereof.

[0041] A “variant” of an egg cell promoter may include changes in its nucleotide sequence in which one or more nucleotides of the original sequence is deleted, added, and/or substituted, while substantially maintaining promoter function in an egg cell. One or more base pairs can be inserted, deleted, or substituted internally to a promoter. In the case of a promoter fragment, variant promoters can include changes affecting the transcription of a minimal promoter to which it is operably linked. Variant promoters can be produced, for example, by standard DNA mutagenesis techniques or by chemically synthesizing the variant promoter or a portion thereof.

[0042] In some embodiments, the egg cell promoter may be operably linked to a heterologous polynucleotide sequence so that the polypeptide encoded by the heterologous polynucleotide sequence is expressed in a female reproductive tissue or cell of a plant, e.g. an egg cell.

[0043] Any of the egg cell promoter polynucleotides described herein may be introduced into a female gametophyte, including the gametophyte’s genome, using any suitable techniques, including but not limited to, for example, gene editing, transformation. In some embodiments, the egg cell promoter polynucleotides may be part of chimeric polynucleotides, such as a heterologous polynucleotide and DNA constructs. Accordingly, provided herein is a recombinant DNA construct comprising a nucleotide sequence comprising any of the sequences set forth in SEQ ID NOS: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86; a nucleotide sequence having at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleotide sequence set forth in SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86; or a functional fragment thereof, or variant thereof, operably linked to at least one heterologous sequence. Also provided herein are embryos, plants, and parts thereof comprising a nucleotide sequence comprising any of the sequences set forth in SEQ ID NOS: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86; a nucleotide sequence having at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity with the nucleotide sequence set forth in SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86; or a

functional fragment thereof, or variant thereof, operably linked to at least one heterologous sequence. In some examples, the operably linked heterologous polynucleotide sequence encodes a BBM polypeptide, or functional BBM fragment thereof, or BBM variant. The egg cell promoter sequence and BBM polynucleotide sequence are heterologous with respect to one another as this combination is not naturally occurring. In another example, the egg cell promoter or BBM polynucleotide may be homologous, or native, or heterologous, or foreign, to the plant cell, for example, a maize egg cell promoter is heterologous to a wheat female gametophyte. Fragments and variants can be obtained via any suitable methods including but not limited to site-directed mutagenesis and synthetic construction.

[0044] The promoter sequences disclosed herein may be comprised in a recombinant DNA construct or in a chimeric polynucleotide, e.g. combined with a heterologous polynucleotide, for example, as part of an engineered genomic locus.

[0045] Also provided herein is a method of modulating expression of a polynucleotide encoding a polypeptide of interest in an egg cell. In some aspects, the method includes expressing the polynucleotide by operably linking the polynucleotide with a promoter nucleotide sequence that expresses in an egg cell, wherein the egg cell promoter includes one of more expression modulating/modulation elements (EMEs). “Expression modulating/modulation element” or “EME” as used herein refers to a nucleotide sequence that up or down-regulates the expression of one or more plant genes. EME may have one or more copies of the same sequence arranged head-to-head, tail-to-head, or head-to-tail or any combination of these configurations. EMEs may be derived from plant sequences, or from bacterial or viral enhancer elements. Exemplary EMEs for use in the methods and compositions include but are not limited to SEQ ID NO: 30 and those described in Publication No. WO 2018/183878, which is incorporated herein in its entirety. In some aspects, the EME is heterologous to the egg cell promoter.

[0046] In another aspect, provided herein is a recombinant DNA construct or chimeric polynucleotides comprising an egg cell promoter and an EME, wherein the EME is heterologous to the egg cell promoter. Also provided is a method of modulating the expression of a polynucleotide sequence of interest in a plant egg cell, the method comprising expressing the polynucleotide sequence, where expression of the polynucleotide sequence is regulated by a promoter that expresses in an egg cell, where the promoter comprises one of more EMEs, where the EME is heterologous to the polynucleotide sequence of interest and/or the egg cell promoter. In some aspects, the polynucleotide sequence of interest is a BBM polynucleotide, which is described elsewhere herein. The methods and compositions of the present disclosure may use different promoters with or without the EME sequences and/or enhancers.

[0047] In one embodiment, a method of modulating expression of an endogenous BBM polynucleotide in a plant cell is provided. The method may include altering one or more nucleotides in a regulatory region of the genomic locus of an endogenous BBM polynucleotide, e.g. a BBM gene, to create a modified regulatory region. The creation of the modified regulatory region results in expression of the BBM polypeptide in a female gametophyte. In some embodiments, a nucleotide sequence of a promoter that expresses in

an egg cell is introduced into the BBM genomic locus so that the promoter is operably linked to the nucleotide sequence encoding the BBM polypeptide. In some embodiments, the egg cell promoter may be replaced for the endogenous BBM promoter sequence. In some embodiments, the endogenous BBM promoter nucleotide sequence may be altered to that of an egg cell promoter nucleotide sequence. Also provided herein is the resulting modified BBM genomic locus.

[0048] In one embodiment, a method of modulating expression of a coding or genomic sequence in an endogenous genomic egg cell locus in a plant cell is provided. The method may include altering one or more nucleotides in a genomic or coding region of the genomic locus of an egg cell polynucleotide, e.g. to create a modified genomic or coding nucleotide sequence that encodes a BBM polypeptide. The creation of the modification results in expression of the BBM polypeptide in a female gametophyte. In some embodiments, a heterologous polynucleotide encoding a BBM polypeptide is introduced into the egg cell genomic locus so that the heterologous polynucleotide encoding a BBM polypeptide is operably linked to the regulatory region or promoter of the egg cell locus and expressed. In some embodiments, the heterologous polynucleotide encoding a BBM polypeptide replaces the endogenous genomic or coding nucleotide sequence in the genomic egg cell locus. In some embodiments, the endogenous genomic or coding nucleotide sequence in the genomic egg cell locus is altered to that of the heterologous polynucleotide encoding a BBM polypeptide. Also provided herein is the resulting modified genomic egg cell locus.

[0049] One embodiment of the disclosure includes a method of obtaining a wheat plant that produces clonal, non-reduced, non-recombined male and/or female gametes. In certain embodiments, wheat plants that have suppressed expression and/or activity with respect to its endogenous Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotides and polypeptides are either heterozygous or homozygous for the suppression.

[0050] The terms “suppress”, “suppressed”, “suppression”, “suppressing” and “silencing”, are used interchangeably herein and include lowering, reducing, declining, decreasing, inhibiting, eliminating or preventing. “Silencing” or “gene silencing” does not specify mechanism and is inclusive, and not limited to, anti-sense, cosuppression, viral-suppression, hairpin suppression, stem-loop suppression, RNAi-based approaches, and small RNA-based approaches and the like. “Silencing,” as used herein with respect to a target gene, such as Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1, refers generally to the suppression of levels of mRNA or protein expressed by the target gene, and/or the level of protein functionality.

[0051] In certain embodiments, the methods include suppressing endogenous wheat Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotides and polypeptides or combinations thereof in a plant or plant cell. In certain embodiments, the plant cell is a microspore mother cell (which gives rise to haploid microspores), megaspore mother cell (which gives rise to a haploid megaspore) or any other plant cell where the genes, e.g. Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1, are expressed.

[0052] Methods are provided to suppress the activity of endogenous Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotides or polypeptides in a wheat plant cell. In some aspects, suppressing activity also includes suppressing

the expression level of the Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotides or polypeptides. Any suitable method or technique may be used. In certain embodiments, Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotides or polypeptides are suppressed using anti-sense, cosuppression, viral-suppression, hairpin suppression, stem-loop suppression, RNAi-based approaches, and small RNA-based approaches. In certain examples, genome editing approaches, including but not limited to, Cas endonuclease and guide RNA, are employed to introduce into a plant cell's genome polynucleotides that suppress Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 expression and/or activity. In certain examples, genome editing approaches, including but not limited to, Cas endonuclease and guide RNA, are employed to introduce, into a plant cell's genome, nucleotide deletions or modifications, e.g. additions or substitutions, that suppress Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 expression and/or activity. In one embodiment, Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 expression and/or activity may be suppressed using any combinations of RNA-based and gene editing approaches.

[0053] One of ordinary skill in the art would readily recognize a suitable control or reference to be utilized when assessing or measuring expression level or activity of Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotides or polypeptides in any embodiment of the present disclosure. For example, by way of non-limiting illustrations, a plant or plant cell comprising a modified Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotide or polypeptide; or Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 suppression polynucleotide would be typically measured relative to a plant or plant cell not comprising the modified Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotide or polypeptide, or the Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 suppression polynucleotide as the control or reference plant or plant cell. In some examples, the control is a wild type plant or cell. One skilled in the art will be able to determine Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 expression level or activity using assays, such as PCR, Northern, and Western blot assays.

[0054] In certain embodiments, wheat plants that have suppressed activity with respect to endogenous Spo11 or Prd1, Prd2, or PRD3; Rec8; and OSD1 polynucleotides and polypeptides are either heterozygous or homozygous for the suppression. These plants may be crossed, intercrossed, or selfed until a wheat plant is obtained that comprises suppressed activity for each and all of these endogenous genes: (Spo11 or Prd1, Prd2, or PRD3); Rec8; and OSD1 activity. A female gametophyte derived from this wheat plant is modified so that a Babyboom (BBM) polynucleotide encoding a BBM polypeptide, or a fragment thereof, is expressed in the female gametophyte to produce an embryo.

[0055] The modification may be performed using any suitable approach. In one example, the method includes introducing a BBM polynucleotide encoding a BBM polypeptide into the coding or genomic nucleotide sequence of a genomic egg cell locus. In another example, the method includes introducing an egg cell promoter or regulatory region into a genomic BBM locus. In another example, a chimeric polynucleotide comprising a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell, where the BBM polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) poly-

peptide or a fragment thereof is introduced into the female gametophyte. In another example, a recombinant DNA construct comprising a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell, where the BBM polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof is introduced into the female gametophyte.

[0056] Using methods described herein, a wheat plant derived from this embryo may be used to produce progeny that are non-reduced, non-recombined and clonal with respect to the parent wheat plant. Plants produced from this method allow for the clonal reproduction of the wheat plant through seed. The resulting seeds from the wheat plant may be grown into plants and the seed harvested from those plants, allowing for the clonal reproduction of the parent plant through seeds. In certain embodiments, the wheat plant is a hybrid. In certain embodiments, the wheat plant is an inbred or a variety.

[0057] As used herein, the term “wheat” refers to any species of the genus *Triticum*, including progenitors thereof, as well as progeny thereof produced by crosses with other species. Wheat includes “hexaploid wheat” which has genome organization of AABBDD, comprised of 42 chromosomes, and “tetraploid wheat” which has genome organization of AABB, comprised of 28 chromosomes. Hexaploid wheat includes *T. aestivum*, *T. spelta*, *T. mocha*, *T. compactum*, *T. sphaerococcum*, *T. vavilovii*, *Triticum boeoticum*, or to the domesticated form, *Triticum monococcum*, and interspecies cross thereof. Tetraploid wheat includes *T. durum* (also referred to as durum wheat or *Triticum turgidum* ssp. *durum*), *T. dicoccoides*, *T. dicoccum*, *T. polonicum*, and interspecies cross thereof. In addition, the term “wheat” includes possible progenitors of hexaploid or tetraploid *Triticum* sp. such as *T. urartu*, *T. monococcum* or *T. boeoticum* for the A genome, *Aegilops speltoides* for the B genome, and *T. tauschii* (also known as *Aegilops squarrosa* or *Aegilops tauschii*) for the D genome. A wheat cultivar for use in the present disclosure may belong to, but is not limited to, any of the above-listed species. Also encompassed are plants that are produced by conventional techniques using *Triticum* sp. as a parent in a sexual cross with a non-*Triticum* species, such as rye (*Secale cereale*), including but not limited to Triticale. In some embodiments, the wheat plant is suitable for commercial production of grain, such as commercial varieties of hexaploid wheat or durum wheat, having suitable agronomic characteristics which are known to those skilled in the art.

[0058] The present disclosure also includes plants obtained by any of the disclosed methods or compositions herein. The present disclosure also includes seeds from a plant obtained by any of the methods or compositions disclosed herein. As used herein, the term “plant” refers to whole plants, plant organs (e.g., leaves, stems, roots, etc.), plant tissues, plant cells, plant parts, seeds, propagules, embryos and progeny of the same. As used herein, the term “plant” refers to whole plants, plant organs (e.g., leaves, stems, roots, etc.), plant tissues, plant cells, plant parts, seeds, propagules, embryos and progeny of the same. Plant cells can be differentiated or undifferentiated (e.g. callus, undifferentiated callus, immature and mature embryos, immature zygotic embryo, immature cotyledon, embryonic axis, suspension culture cells, protoplasts, leaf, leaf cells, root cells, phloem cells and pollen).

[0059] Grain is intended to mean the mature seed produced by commercial growers for purposes other than growing or reproducing the species.

[0060] The compositions and methods of the present disclosure are applicable to a broad range of plant species, including dicotyledonous plants and monocotyledonous plants. Representative examples of plants that can be treated in accordance with the methods disclosed herein include, but are not limited to, wheat, cotton, sunflower, safflower, tobacco, *Arabidopsis*, barley, oats, rice, maize, triticale, sorghum, rye, millet, flax, sugarcane, banana, cassava, common bean, cowpea, tomato, potato, beet, grape, *Eucalyptus*, wheat grasses, turf grasses, alfalfa, clover, soybean, peanuts, citrus, *papaya*, *Setaria* sp, cacao, cucumber, apple, *Capsicum*, bamboo, melon, *cannabis*, ornamentals including commercial garden and flower bulb species, fruit trees, vegetable species, *Brassica* species, as well as interspecies hybrids. In one embodiment, the compositions and methods of the disclosure are applied to wheat plants.

[0061] The methods of the disclosure may include introducing a polypeptide, polynucleotide (i.e., DNA or RNA), or nucleotide construct (i.e., DNA or RNA) into a plant. As used herein, “introducing” means presenting to the plant the polynucleotide, polypeptide, or nucleotide construct in such a manner that the polynucleotide, polypeptide, or nucleotide construct gains access to the interior of a cell of the plant. The methods of the disclosure do not depend on a particular method for introducing the polynucleotide, polypeptide, or nucleotide construct into a plant, only that the polynucleotide, polypeptide, or nucleotide construct gains access to the interior of at least one cell of the plant. Methods for introducing polynucleotides, polypeptides, or nucleotide constructs into plants are known in the art including, but not limited to, gene editing, stable or transient transformation methods, and virus-mediated methods.

[0062] The compositions and methods of the present disclosure optionally include producing doubled haploid plants from haploid embryos.

[0063] The present disclosure also provides methods of contacting haploid cells with an amount of a chromosome doubling agent before, during, after, or overlapping with any portion of the isolation and embryogenesis induction process used for generating a paternal gamete (androgenic) or a maternal gamete (gynogenic) doubled haploid population.

[0064] The terms “polypeptide,” “peptide” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers.

[0065] In some embodiments, parthenogenesis inducing methods may be used to improve maternal haploid embryo regeneration productivity and enable gene editing to provide regenerated gene-edited maternal haploids.

[0066] In an aspect, haploid cells may be contacted with an amount of a chromosome doubling agent to promote chromosome doubling followed by regenerating homozygous diploid plants from the treated haploid cells. The haploid microspore cells can be in contact with the doubling agent before, during, or after initiation of microspore embryogenesis or embryo maturation. After chromosome doubling, the doubled haploid embryo will contain 2 copies of paternally derived chromosomes. The efficiency of the

process for obtaining doubled haploid plants from haploid embryos may be greater than 10%, 20%, 30%, 50%, 60%, 70%, 80%, or 90%. The duration of contact between the haploid cells and the chromosomal doubling agent may vary. Contact may be from less than 24 hours, for example 4-12 hours, to about a week. The duration of contact is generally from about 8 hours to 2 days.

[0067] Methods of chromosome doubling are disclosed in Antoine-Michard, S. et al., Plant cell, tissue organ cult., Cordrecht, the Netherlands, Kluwer Academic Publishers, 1997, 48 (3): 203-207; Kato, A., Maize Genetics Cooperation Newsletter 1997, 36-37; and Wan, Y. et al., TAG, 1989, 77:889-892. Wan, Y. et al., TAG, 1991, 81:205-211. The disclosures of which are incorporated herein by reference.

Typical doubling methods involve contacting the cells with colchicine, anti-microtubule agents or anti-microtubule herbicides, pronamide, nitrous oxide, or any mitotic inhibitor to create homozygous doubled haploid cells. The amount of colchicine used in medium is generally 0.01%-0.2% or approximately 0.05% of amiprophos-methyl (APM) (5-225 μ M) may be used. The amount of colchicine can range from approximately 400-600 mg/L or approximately 500 mg/L. The amount of pronamide in medium is approximately 0.5-20 μ M. Examples of mitotic inhibitors are included in Table B. Other agents may be used with the mitotic inhibitors to improve doubling efficiency. Such agents include dimethyl sulfoxide (DMSO), adjuvants, surfactants, and the like.

TABLE B

Chemical chromosome doubling agents		
Common Name/ Trade name	CAS	IUPAC
Colchicine and Colchicine Derivatives		
colchicine/acetyltrimethylcolchicinic acid		(S)-N-(5,6,7,9-tetrahydro-1,2,3,10-tetramethoxy-9-oxobenzo (a) heptalen-7-yl) acetamide
colchicine derivatives		
Carbamates		
Carbetamide	(R)-1-(ethylcarbamoyl)ethyl carbanilate	(2R)-N-ethyl-2-[[[(phenylamino)carbonyl]oxy]propanamide
chloropropham Propham		
Benzamides		
Pronamide/ propyzamide Tebutam	3,5-dichloro-N-(1,1-dimethylpropynyl)benzamide	3,5-dichloro-N-(1,1-dimethyl-2-propynyl)benzamide
Benzoic Acids		
Chlorthal dimethyl (DCPA), Dicamba/dianat/ disugran (dicamba-methyl) (BANVEL, CLARITY)	3,6-dichloro-o-anisic acid	3,6-dichloro-2-methoxybenzoic acid
Dinitroaniline chromosome doubling agents		
benfluralin/benefin/ (BALAN)	N-butyl-N-ethyl- α,α,α -trifluoro-2,6-dinitro-p-toluidine	N-butyl-N-ethyl-2,6-dinitro-4-(trifluoromethyl)benzenamine
Butralin	(RS)-N-sec-butyl-4-tert-butyl-2,6-dinitroaniline	4-(1,1-dimethylethyl)-N-(1-methylpropyl)-2,6-dinitrobenzenamine
Chloralin dinitramine	N1,N1-diethyl-2,6-dinitro-4-trifluoromethyl-m-phenylenediamine	N3,N3-diethyl-2,4-dinitro-6-(trifluoromethyl)-1,3-benzenediamine
ethalfluralin (Sonalan)	N-ethyl- α,α,α -trifluoro-N-(2-methylallyl)-2,6-dinitro-p-toluidine	N-ethyl-N-(2-methyl-2-propenyl)-2,6-dinitro-4-(trifluoromethyl)benzenamine
fluchloralin	N-(2-chloroethyl)-2,6-dinitro-N-propyl-4-(trifluoromethyl)aniline or N-(2-chloroethyl)- α,α,α -trifluoro-2,6-dinitro-N-propyl-p-toluidine	N-(2-chloroethyl)-2,6-dinitro-N-propyl-4-(trifluoromethyl)benzenamine
isopropalin	4-isopropyl-2,6-dinitro-N,N-dipropylaniline	4-(1-methylethyl)-2,6-dinitro-N,N-dipropylbenzenamine
methalpropalin	α,α,α -trifluoro-N-(2-methylallyl)-2,6-dinitro-N-propyl-p-toluidine	N-(2-methyl-2-propenyl)-2,6-dinitro-N-propyl-4-(trifluoromethyl)benzenamine
nitralin	4-methylsulfonyl-2,6-dinitro-N,N-dipropylaniline	4-(methylsulfonyl)-2,6-dinitro-N,N-dipropylbenzenamine
oryzalin (SURFLAN)	3,5-dinitro-N4,N4-dipropylsulfanilamide	4-(dipropylamino)-3,5-dinitrobenzenesulfonamide

TABLE B-continued

Chemical chromosome doubling agents		
Common Name/ Trade name	CAS	IUPAC
pendimethalin (PROWL)	N-(1-ethylpropyl)-2,6-dinitro-3,4-xylydine	N-(1-ethylpropyl)-3,4-dimethyl-2,6-dinitrobenzenamine
prodiamine	5-dipropylamino- α,α,α -trifluoro-4,6-dinitro-o-toluidine or 2,6-dinitro-N1,N1-dipropyl-4-trifluoromethyl-m-phenylenediamine	2,4-dinitro-N3,N3-dipropyl-6-(trifluoromethyl)-1,3-benzenediamine
profluralin	N-cyclopropylmethyl- α,α,α -trifluoro-2,6-dinitro-N-propyl-p-toluidine or N-cyclopropylmethyl-2,6-dinitro-N-propyl-4-trifluoromethylaniline	N-(cyclopropylmethyl)-2,6-dinitro-N-propyl-4-(trifluoromethyl)benzenamine
trifluralin (TREFLAN, TRIFIC, TRILLIN)	α,α,α -trifluoro-2,6-dinitro-N,N-dipropyl-p-toluidine	2,6-dinitro-N,N-dipropyl-4-(trifluoromethyl)benzenamine
Phosphoramidates		
APM (Amiprophos methyl); amiprophos-methyl		
Butamifos	O-ethyl O-6-nitro-m-tolyl (RS)-sec-butylphosphoramidothioate	O-ethyl O-(5-methyl-2-nitrophenyl) (1-methylpropyl)phosphoramidothioate
Pyridines		
Dithiopyr Thiazopyr	methyl 2-difluoromethyl-5-(4,5-dihydro-1,3-thiazol-2-yl)-4-isobutyl-6-trifluoromethylnicotinate	methyl 2-(difluoromethyl)-5-(4,5-dihydro-2-thiazolyl)-4-(2-methylpropyl)-6-(trifluoromethyl)-3-pyridinecarboxylate

[0068] As an alternative to using chemical chromosome doubling agents, modulating expression of genes known to impact the plant cell cycle (genetic chromosome doubling protein), either through stimulation of the cell cycle (and cell division) or through stimulation of endoreduplication, can be used to double the chromosome complement in an embryo. Increasing ploidy level in plant cells can be achieved by modulating expression of genes that stimulate key control points in the cell cycle cell. It is expected that other plant genes known to simulate the cell cycle (or cell division) in plants may be used to produce a similar doubling of the chromosome number in the forming maternal haploid embryos. Examples of plant genes whose over-expression stimulates the cell cycle include cyclin-A in tobacco (Yu et al., 2003), cyclin-D in tobacco (Cockcroft et al., 2000, Nature 405:575-79; Schnittger et al., 2002, PNAS 99:6410-6415; Dewitte et al., 2003, Plant Cell 15:79-92), E2FA in *Arabidopsis* (De Veylder et al., 2002, EMBO J 21:1360-1368), E2FB in *Arabidopsis* (Magyar et al., 2005, Plant Cell 17:2527-2541). Similarly, over-expression of viral genes known to modulate plant cell cycle machinery can be used, such as when over-expression of the Wheat Dwarf Virus RepA gene stimulates cell cycle progression (G1/S transition) and cell division in maize (Gordon-Kamm et al., 2002, PNAS 99:11975-11980). Conversely, plant genes whose encoded products are known to inhibit the cell cycle have been shown to result in increased cell division when the gene, such as Cyclin-Dependent Kinase Inhibitor (ICK1/KRP), is down-regulated in *Arabidopsis* (Cheng et al 2013, Plant J 75:642-655). Thus, down-regulation of the KRP gene using an egg cell specific promoter to drive expression may have a similar effect as over-expression of DZ470, resulting

in chromosome doubling. Methods of down-regulation of a gene such as KRP are known in the art and include expression of an artificial micro-RNA targeted to the KRP mRNA, or expression of a dCas9-repressor fusion that is targeted to the KRP promoter by a gRNA to that sequence. Finally, there are plant genes that are known to specifically impact the process of endoreduplication. When using such genes, such as for example the *ccs52* gene or the *Dell* gene, over-expression of *ccs52* may result in an increased ploidy level as observed in *Medicago sativa* (Cebolla et al., 1999, EMBO J 18:4476-4484), and down-regulation of *Dell* may result in an increased ploidy level as observed in *Arabidopsis* (Vlieghe et al., 2005, Current Biol 15:59-63). It is expected that other genes that are known to stimulate the cell cycle, the G1/S transition, or endoreduplication may be used in the methods disclosed herein to increase ploidy level.

[0069] Repressor motifs are known in the art, for example see Kagale and Rozwadowski (Epigenetics. 2011. 6:141-146). Ethylene-responsive element binding factor-associated Amphiphilic Repression (EAR) motif-mediated transcriptional repression is known in plants, including EAR motifs defined by the consensus sequence patterns of either LxLxL and DLNxxP (see Hiratsu et al., 2003, Plant J. 35:177-192). Of interest to the present disclosure are peptides including the amphiphilic repression motif disclosed in WO 2013/109754 A1 and all references cited therein and the Dr1/DRAP1 global repressor complex (see U.S. Pat. No. 7,288,695 B2 and all references cited therein), including the Dr1 motif that is similar to the motif found in *Arabidopsis thaliana* MYBL2 (see Matsui K, Umemura Y, Ohme-Takagi M. 2008. Plant J. 55:954-967).

[0070] Methods are known in the art for the targeted insertion of a polynucleotide at a specific location in the plant genome. The insertion of the polynucleotide at a desired genomic location may be achieved using a site-specific recombination system. See, for example, WO99/25821, WO99/25854, WO99/25840, WO99/25855 and WO99/25853, all of which are herein incorporated by reference in their entirety. Briefly, a polynucleotide of interest, flanked by two non-identical recombination sites, can be contained in a T-DNA transfer cassette. The T-DNA transfer cassette may be introduced into a plant having stably incorporated into its genome a target site which is flanked by two non-identical recombination sites that correspond to the sites of the transfer cassette. An appropriate recombinase is provided, and the transfer cassette is integrated at the target site. The polynucleotide of interest is thereby integrated at a specific chromosomal position in the plant genome.

[0071] Any suitable technique may be used to introduce, into any plant cell, polynucleotides that are useful to target a specific site for modification in the genome of a plant derived from the plant cell. Site specific modifications that can be introduced include those produced using any method for introducing site specific modification, including, but not limited to, through the use of gene repair oligonucleotides (e.g. US Publication 2013/0019349), or through the use of double-stranded break technologies such as TALENs, meganucleases, zinc finger nucleases, CRISPR-Cas, other endonucleases, and the like. For example, the disclosed methods can be used to introduce a CRISPR-Cas system into a plant cell or plant, for the purpose of genome modification of a target sequence in the genome of a plant or plant cell, for selecting plants, for deleting a base or a sequence, for gene editing, and for inserting a polynucleotide of interest into the genome of a plant or plant cell. Thus, the disclosed methods can be used together with a CRISPR-Cas system to provide for an effective system for modifying or altering target sites and nucleotides of interest within the genome of a plant, plant cell or seed. Any suitable Cas endonuclease may be used in the methods and to create the compositions described herein, for example, any suitable Cas endonuclease that is capable of binding to and creating a double strand break in a genomic target sequence of the plant genome.

[0072] In some embodiments, polynucleotides, such as egg cell or BBM polynucleotides, may be introduced into the genome of a plant using genome editing technologies, or polynucleotides in the genome of a plant, such as egg cell or BBM polynucleotides, may be edited using genome editing technologies. For example, the polynucleotides may be introduced into a desired location in the genome of a plant through the use of a genome editing system such as TALENs, meganucleases, zinc finger nucleases, CRISPR-Cas, and the like. For example, the disclosed polynucleotides may be introduced into a desired location in a genome using a CRISPR-Cas system, for the purpose of site-specific insertion. The desired location in a plant genome can be any desired target site for insertion, such as a genomic region.

[0073] In some embodiments, where the polynucleotide has previously been introduced into a genome, genome editing or genome engineering technologies may be used to alter or modify the introduced polynucleotide sequence, including the flanking chromosomal genomic sequences. Site-specific modifications that can be introduced into the disclosed compositions include those produced using any

method for introducing site-specific modification, including, but not limited to, through the use of gene repair oligonucleotides, or through the use of site-directed genome modification tools such as TALENs, meganucleases, zinc finger nucleases, CRISPR-Cas, and the like. Site-specific modifications to the disclosed polynucleotides (including genomic flanking and junction sequences) may include, but are not limited to, changes to codon usage, changes to regulatory elements such as promoters, introns, terminators, enhancers, 5' or 3' untranslated regions (UTRs), or other noncoding sequences, and other regions of the polynucleotide, where the modifications do not adversely affect the phenotypic characteristics of the resulting maize plant.

[0074] Cas polypeptides suitable for introducing site-specific modifications include, for example, Cas9, Cas 12f (Cas-alpha, Cas14), Cas121 (Cas-beta), Cas12a (Cpf1), Cas12b (a C2c1 protein), Cas13 (a C2c2 prot cin), Cas12c (a C2c3 protein), Cas12d, Cas12e, Cas12g, Cas12h, Cas12i, Cas12j, Cas12k, Cas3, Cas3-HD, Cas 5, Cas6, Cas7, Cas8, Cas10, or combinations or complexes of these. In some aspects, transposon-associated TnpB, a programmable RNA-guided DNA endonuclease can be used.

[0075] In some aspects, a genome editing system comprises a Cas-alpha (e.g., Cas12f) endonuclease and one or more guide polynucleotides that introduce one or more site-specific modifications in a target polynucleotide sequence, resulting in a modified target sequence. As used herein, "altered target site", "altered target sequence", "modified target site," and "modified target sequence" are used interchangeably and refer to a target sequence as disclosed herein that comprises at least one alteration or modification when compared to a non-altered target sequence. Such alterations or modifications include, for example: (i) replacement or substitution of at least one nucleotide, (ii) deletion of at least one nucleotide, (iii) insertion of at least one nucleotide, or (iv) any combination of (i)-(iii).

[0076] In some aspects, a genome editing system comprises a Cas-alpha endonuclease, one or more guide polynucleotides, and optionally a donor DNA. Some exemplary Cas-alpha endonucleases are described, for example, in WO2020123887.

[0077] In some aspects, a genome editing system comprises a Cas polypeptide, one or more guide polynucleotides, and optionally donor DNA, and editing a target polynucleotide sequence comprises nonhomologous end-joining (NHEJ) or homologous recombination (HR) following a Cas polypeptide-mediated double-strand break. Once a double-strand break is induced in the DNA, the cell's DNA repair mechanism is activated to repair the break. The most common repair mechanism to bring the broken ends together is the nonhomologous end-joining pathway (Bleuyard et al., (2006) DNA Repair 5:1-12). The structural integrity of chromosomes is typically preserved by the repair, but deletions, insertions, or other rearrangements are possible (Siebert and Puchta, (2002) Plant Cell 14:1121-31; Pacher et al., (2007) Genetics 175:21-9). Alternatively, the double-strand break can be repaired by homologous recombination between homologous DNA sequences. Once the sequence around the double-strand break is altered, for example, by exonuclease activities involved in the maturation of double-strand breaks, gene conversion pathways can restore the original structure if a homologous sequence is available, such as a homologous chromosome in non-dividing somatic

cells, or a sister chromatid after DNA replication (Molinier et al., (2004) *Plant Cell* 16:342-52). Ectopic and/or epigenic DNA sequences may also serve as a DNA repair template for homologous recombination (Puchta, (1999) *Genetics* 152: 1173-81).

[0078] In some aspects, the genome editing system comprises a Cas polypeptide, one or more guide polynucleotides, and a donor DNA. As used herein, “donor DNA” is a DNA construct that comprises a polynucleotide of interest to be inserted into the target site of a Cas polypeptide. Once a double-strand break is introduced in the target site by the endonuclease, the first and second regions of homology of the donor DNA can undergo homologous recombination with their corresponding genomic regions of homology resulting in exchange of DNA between the donor and the target genome. As such, the provided methods result in the integration of the polynucleotide of interest of the donor DNA into the double-strand break in the target site in the plant genome, thereby altering the original target site and producing an altered genomic target site.

[0079] In some aspects, a genome editing system comprises a base editing agent and a plurality of guide polynucleotides and editing a target polynucleotide sequence comprises introducing a plurality of nucleobase edits in the target polynucleotide sequence resulting in a variant nucleotide sequence.

[0080] One or more nucleobases of a target polynucleotide can be chemically altered, in some cases to change the base from one type to another, for example from a Cytosine to a Thymine, or an Adenine to a Guanine. In some aspects, a plurality of bases, for example 2 or more, 5 or more, 10 or more, 20 or more, 30 or more, 40 or more, 50 or more, 60 or more, 70 or more, 80 or more 90 or more, 100 or more, or even greater than 100, 200 or more, up to thousands of bases may be modified or altered, to produce a plant with a plurality of modified bases.

[0081] Any base editing complex, such as a base editing agent associated with an RNA-guided protein, may be used to target and bind to a desired locus in the genome of an organism and chemically modify one or more components of a target polynucleotide.

[0082] Site-specific base conversions can be achieved to engineer one or more nucleotide changes to create one or more edits into the genome. These include for example, a site-specific base edit mediated by an C-G to T-A or an A-T to G-C base editing deaminase enzymes (Gaudelli et al., “Programmable base editing of A-T to G-C in genomic DNA without DNA cleavage.” *Nature* (2017); Nishida et al. “Targeted nucleotide editing using hybrid prokaryotic and vertebrate adaptive immune systems.” *Science* 353 (6305) (2016); Komor et al. “Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage.” *Nature* 533 (7603) (2016): 420-4. A catalytically “dead” or inactive Cas polypeptide, for example an inactive Cas9 (dCas9), Cas12f (dCas12f), or another Cas polypeptide disclosed herein, fused to a cytidine deaminase or an adenine deaminase protein becomes a specific base editor that can alter DNA bases without inducing a DNA break. Base editors convert C->T (or G->A on the opposite strand) or an adenine base editor that would convert adenine to inosine, resulting in an A->G change within an editing window specified by the gRNA. Any molecule that effects a change in a nucleobase is a “base editing agent”. The dCas forms a functional complex with a guide polynucleotide that shares

homology with a polynucleotide sequence at the target site, and is further complexed with the deaminase molecule. The guided Cas polypeptide recognizes and binds to a double-stranded target sequence, opening the double-strand to expose individual bases. In the case of a cytidine deaminase, the deaminase deaminates the cytosine base and creates a uracil. Uracil glycosylase inhibitor (UGI) is provided to prevent the conversion of U back to C. DNA replication or repair mechanisms then convert the Uracil to a thymine (U to T), and subsequent repair of the opposing base (formerly G in the original G-C pair) to an Adenine, creating a T-A pair.

[0083] The Cas endonuclease is guided by the guide nucleotide to recognize and optionally introduce a double strand break at a specific target site into the genome of a cell. The CRISPR-Cas system provides for an effective system for modifying target sites within the genome of a plant, plant cell or seed. Further provided are methods employing a guide polynucleotide/Cas endonuclease system to provide an effective system for modifying target sites within the genome of a cell and for editing a nucleotide sequence in the genome of a cell. Once a genomic target site is identified, a variety of methods may be employed to further modify the target site. The disclosed methods can be used to introduce a CRISPR-Cas system for editing a nucleotide sequence in the genome of a cell. The nucleotide sequence to be edited (the nucleotide sequence of interest) can be located within or outside a target site that is recognized by a Cas endonuclease.

[0084] The term “Cas polypeptide” or “Cas endonuclease” refers to a polypeptide encoded by a Cas (CRISPR-associated) gene.

[0085] A Cas polypeptide includes but is not limited to: Cas9, Cas 12f (Cas-alpha, Cas14), Cas12i (Cas-beta), Cas12a (Cpf1), Cas12b (a C2c1 protein), Cas13 (a C2c2 protein), Cas12c (a C2c3 protein), Cas12d, Cas12e, Cas12g, Cas12h, Cas12i, Cas12j, Cas12k, Cas3, Cas3-HD, Cas 5, Cas6, Cas7, Cas8, Cas10, or combinations or complexes of these.

[0086] Endonucleases are enzymes that cleave the phosphodiester bond within a polynucleotide chain and include restriction endonucleases that cleave DNA at specific sites without damaging the bases. Restriction endonucleases include Type I, Type II, Type III, and Type IV endonucleases, which further include subtypes. In the Type I and Type III systems, both the methylase and restriction activities are contained in a single complex. Endonucleases also include meganucleases, also known as homing endonucleases (HEases), which like restriction endonucleases, bind and cut at a specific recognition site, however the recognition sites for meganucleases are typically longer, about 18 bp or more (Patent application PCT/US 12/30061 filed on Mar. 22, 2012). Meganucleases have been classified into four families based on conserved sequence motifs. These motifs participate in the coordination of metal ions and hydrolysis of phosphodiester bonds. Meganucleases are notable for their long recognition sites, and for tolerating some sequence polymorphisms in their DNA substrates. The naming convention for meganuclease is similar to the convention for other restriction endonuclease. Meganucleases are also characterized by prefix F-, I-, or PI- for enzymes encoded by free-standing ORFs, introns, and inteins, respectively. One step in the recombination process involves polynucleotide cleavage at or near the recognition site. This cleaving

activity can be used to produce a double-strand break. For reviews of site-specific recombinases and their recognition sites, see, Sauer (1994) *Curr Op Biotechnol* 5:521-7; and Sadowski (1993) *FASEB* 7:760-7. In some examples the recombinase is from the Integrase or Resolvase families. TAL effector nucleases are a new class of sequence-specific nucleases that can be used to make double-strand breaks at specific target sequences in the genome of a plant or other organism. (Miller, et al. (2011) *Nature Biotechnology* 29:143-148). Zinc finger nucleases (ZFNs) are engineered double-strand break inducing agents comprised of a zinc finger DNA binding domain and a double-strand-break-inducing agent domain. Recognition site specificity is conferred by the zinc finger domain, which typically comprising two, three, or four zinc fingers, for example having a C2H2 structure, however other zinc finger structures are known and have been engineered. Zinc finger domains are amenable for designing polypeptides which specifically bind a selected polynucleotide recognition sequence. ZFNs include an engineered DNA-binding zinc finger domain linked to a nonspecific endonuclease domain, for example nuclease domain from a Type II endonuclease such as FokI.

[0087] A catalytically active and/or inactive Cas endonuclease can be fused to a heterologous sequence (US20140068797 published 6 Mar. 2014). A “Dead-CAS9” (dCAS9) as used herein, is used to supply a transcriptional repressor domain. The dCAS9 has been mutated so that can no longer cut DNA. The dCAS9 can still bind when guided to a sequence by the gRNA and can also be fused to repressor elements. The dCAS9 fused to the repressor element, as described herein, is abbreviated to dCAS9-REP, where the repressor element (REP) can be any of the known repressor motifs that have been characterized in plants. An expressed guide RNA (gRNA) binds to the dCAS9-REP protein and targets the binding of the dCAS9-REP fusion protein to a specific predetermined nucleotide sequence within a promoter (a promoter within the T-DNA). For example, if this is expressed beyond-the border using a ZM-UBI PRO::dCAS9-REP::PINII TERM cassette along with a U6-POL PRO::gRNA::U6 TERM cassette and the gRNA is designed to guide the dCAS9-REP protein to bind the SB-UBI promoter in the expression cassette SB-UBI PRO::moPAT::PINII TERM within the T-DNA, any event that has integrated the beyond-the-border sequence would be bialaphos sensitive. Transgenic events that integrate only the T-DNA would express moPAT and be bialaphos resistant. The advantage of using a dCAS9 protein fused to a repressor (as opposed to a TETR or ESR) is the ability to target these repressors to any promoter within the T-DNA. TETR and ESR are restricted to cognate operator binding sequences. Alternatively, a synthetic Zinc-Finger Nuclease fused to a repressor domain can be used in place of the gRNA and dCAS9-REP (Urritia et al., 2003, *Genome Biol.* 4:231) as described above.

[0088] The type II CRISPR/Cas system from bacteria employs a crRNA and tracrRNA to guide the Cas endonuclease to its DNA target. The crRNA (CRISPR RNA) contains the region complementary to one strand of the double strand DNA target and base pairs with the tracrRNA (trans-activating CRISPR RNA) forming a RNA duplex that directs the Cas endonuclease to cleave the DNA target. As used herein, the term “guide nucleotide” relates to a synthetic fusion of two RNA molecules, a crRNA (CRISPR RNA) comprising a variable targeting domain, and a

tracrRNA. In an aspect, the guide nucleotide comprises a variable targeting domain of 12 to 30 nucleotide sequences and a RNA fragment that can interact with a Cas endonuclease.

[0089] As used herein, the term “guide polynucleotide” relates to a polynucleotide sequence that can form a complex with a Cas endonuclease and enables the Cas endonuclease to recognize and optionally cleave a DNA target site. The guide polynucleotide can be a single molecule or a double molecule. The guide polynucleotide sequence can be a RNA sequence, a DNA sequence, or a combination thereof (a RNA-DNA combination sequence).

[0090] In an aspect, the guide nucleotide and Cas endonuclease are capable of forming a complex that enables the Cas endonuclease to introduce a double strand break at a DNA target site. Cas endonucleases, either as single effector proteins or in an effector complex with other components, unwind the DNA duplex at the target sequence and optionally cleave at least one DNA strand, as mediated by recognition of the target sequence by a polynucleotide (such as, but not limited to, a crRNA or guide RNA) that is in complex with the Cas effector protein. Such recognition and cutting of a target sequence by a Cas endonuclease typically occurs if the correct protospacer-adjacent motif (PAM) is located at or adjacent to the 3' end of the DNA target sequence. Alternatively, a Cas endonuclease herein may lack DNA cleavage or nicking activity, but can still specifically bind to a DNA target sequence when complexed with a suitable RNA component. (See also U.S. Patent Application US20150082478 published 19 Mar. 2015 and US20150059010 published 26 Feb. 2015).

[0091] Cas endonucleases may occur as individual effectors (Class 2 CRISPR systems) or as part of larger effector complexes (Class 1 CRISPR systems).

[0092] Cas endonucleases that have been described include, but are not limited to, for example: Cas3 (a feature of Class 1 type I systems), Cas9 (a feature of Class 2 type II systems) and Cas12 (Cpf1) (a feature of Class 2 type V systems).

[0093] Cas endonucleases and effector proteins can be used for targeted genome editing (via simplex and multiplex double-strand breaks and nicks) and targeted genome regulation (via tethering of epigenetic effector domains to either the Cas protein or sgRNA. A Cas endonuclease can also be engineered to function as an RNA-guided recombinase, and via RNA tethers could serve as a scaffold for the assembly of multiprotein and nucleic acid complexes (Mali et al., 2013, *Nature Methods* Vol. 10:957-963).

[0094] In some examples, Cas endonucleases, when complexed with a cognate guide RNA, recognize, bind to, and optionally nick or cleave a target polynucleotide.

[0095] In an aspect of the present disclosure the variable target domain is 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 nucleotides in length.

[0096] The guide polynucleotide enables target recognition, binding, and optionally cleavage by the Cas endonuclease, and can be a single molecule or a double molecule. The guide polynucleotide sequence can be a RNA sequence, a DNA sequence, or a combination thereof (a RNA-DNA combination sequence). Optionally, the guide polynucleotide can comprise at least one nucleotide, phosphodiester bond or linkage modification such as, but not limited, to Locked Nucleic Acid (LNA), 5-methyl dC, 2,6-Diaminopurine, 2'-Fluoro A, 2'-Fluoro U, 2'-O-Methyl RNA, phosphorothioate bond, linkage to a cholesterol molecule, linkage to a polyethylene glycol molecule, linkage to a spacer 18 (hexaethylene glycol chain) molecule, or 5' to 3' covalent linkage resulting in circularization. A guide polynucleotide

that solely comprises ribonucleic acids is also referred to as a “guide RNA” or “gRNA” (US20150082478 published 19 Mar. 2015 and US20150059010 published 26 Feb. 2015). A guide polynucleotide may be engineered or synthetic.

[0097] The guide polynucleotide includes a chimeric non-naturally occurring guide RNA comprising regions that are not found together in nature (i.e., they are heterologous with each other). For example, a chimeric non-naturally occurring guide RNA comprising a first nucleotide sequence domain (referred to as Variable Targeting domain or VT domain) that can hybridize to a nucleotide sequence in a target DNA, linked to a second nucleotide sequence that can recognize the Cas endonuclease, such that the first and second nucleotide sequence are not found linked together in nature.

[0098] The guide polynucleotide can be a double molecule (also referred to as duplex guide polynucleotide) comprising a crNucleotide sequence (such as a crRNA) and a tracrNucleotide (such as a tracrRNA) sequence. In some cases, there is a linker polynucleotide that connects the crRNA and tracrRNA to form a single guide, for example an sgRNA.

[0099] In an aspect of the present disclosure, the guide nucleotide comprises a crRNA (or crRNA fragment) and a tracrRNA (or tracrRNA fragment) of the type II CRISPR/Cas system that can form a complex with a type II Cas endonuclease, wherein the guide nucleotide Cas endonuclease complex can direct the Cas endonuclease to a plant genomic target site, enabling the Cas endonuclease to introduce a double strand break into the genomic target site. The guide nucleotide can be introduced into a plant or plant cell directly using any method known in the art such as, but not limited to, particle bombardment or topical applications.

[0100] In an aspect, the guide nucleotide can be introduced indirectly by introducing a recombinant DNA molecule comprising the corresponding guide DNA sequence operably linked to a plant specific promoter that is capable of transcribing the guide nucleotide in the plant cell. The term “corresponding guide DNA” includes a DNA molecule that is identical to the RNA molecule but has a “T” substituted for each “U” of the RNA molecule.

[0101] In an aspect, the guide nucleotide is introduced via particle bombardment or using the disclosed methods for *Agrobacterium* transformation of a recombinant DNA construct comprising the corresponding guide DNA operably linked to a plant U6 polymerase III promoter.

[0102] In an aspect, the RNA that guides the RNA Cas9 endonuclease complex, is a duplexed RNA comprising a duplex crRNA-tracrRNA. One advantage of using a guide nucleotide versus a duplexed crRNA-tracrRNA is that only one expression cassette needs to be made to express the fused guide nucleotide.

[0103] The terms “target site,” “target sequence,” “target DNA,” “target locus,” “genomic target site,” “genomic target sequence,” and “genomic target locus” are used interchangeably herein and refer to a polynucleotide sequence in the genome (including chloroplast and mitochondrial DNA) of a plant cell at which a double-strand break is induced in the plant cell genome by a Cas endonuclease. The target site can be an endogenous site in the plant genome, or alternatively, the target site can be heterologous to the plant and thereby not be naturally occurring in the genome, or the target site can be found in a heterologous genomic location compared to where it occurs in nature.

[0104] As used herein, terms “endogenous target sequence” and “native target sequence” are used interchangeably herein to refer to a target sequence that is endogenous or native to the genome of a plant and is at the endogenous or native position of that target sequence in the genome of the plant.

[0105] An “artificial target site” or “artificial target sequence” are used interchangeably herein and refer to a target sequence that has been introduced into the genome of a plant. Such an artificial target sequence can be identical in sequence to an endogenous or native target sequence in the genome of a plant but be located in a different position (i.e., a non-endogenous or non-native position) in the genome of a plant.

[0106] In an aspect, the disclosed methods can be used to introduce into plants polynucleotides useful for gene suppression of a target gene in a plant. Reduction of the activity of specific genes (also known as gene silencing, or gene suppression) is desirable for several aspects of genetic engineering in plants. Many techniques for gene silencing are well known to one of skill in the art, including but not limited to antisense technology, siRNA, microRNA, and the like. Other approaches may also be used, for example, using a dCas fused to a transcriptional repressor and/or chromatin modifying domain to induce a targeted reduction in transcript. See, for example, WO2022087616, incorporated herein by reference in its entirety.

[0107] The crNucleotide includes a first nucleotide sequence domain (referred to as Variable Targeting domain or VT domain) that can hybridize to a nucleotide sequence in a target DNA and a second nucleotide sequence (also referred to as a tracr mate sequence) that is part of a Cas endonuclease recognition (CER) domain. The tracr mate sequence can hybridize to a tracrNucleotide along a region of complementarity and together form the Cas endonuclease recognition domain or CER domain. The CER domain is capable of interacting with a Cas endonuclease polypeptide. The crNucleotide and the tracrNucleotide of the duplex guide polynucleotide can be RNA, DNA, and/or RNA-DNA-combination sequences. In some embodiments, the crNucleotide molecule of the duplex guide polynucleotide is referred to as “crDNA” (when composed of a contiguous stretch of DNA nucleotides) or “crRNA” (when composed of a contiguous stretch of RNA nucleotides), or “crDNA-RNA” (when composed of a combination of DNA and RNA nucleotides). The crNucleotide can comprise a fragment of the crRNA naturally occurring in Bacteria and Archaea. The size of the fragment of the crRNA naturally occurring in Bacteria and Archaea that can be present in a crNucleotide disclosed herein can range from, but is not limited to, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more nucleotides.

[0108] In some embodiments the tracrNucleotide is referred to as “tracrRNA” (when composed of a contiguous stretch of RNA nucleotides) or “tracrDNA” (when composed of a contiguous stretch of DNA nucleotides) or “tracrDNA-RNA” (when composed of a combination of DNA and RNA nucleotides). In one embodiment, the RNA that guides the RNA/Cas9 endonuclease complex is a duplexed RNA comprising a duplex crRNA-tracrRNA. The tracrRNA (trans-activating CRISPR RNA) comprises, in the 5'-to-3' direction, (i) a sequence that anneals with the repeat region of CRISPR type II crRNA and (ii) a stem loop-comprising portion (Deltcheva et al., Nature 471:602-607).

The duplex guide polynucleotide can form a complex with a Cas endonuclease, wherein said guide polynucleotide/Cas endonuclease complex (also referred to as a guide polynucleotide/Cas endonuclease system) can direct the Cas endonuclease to a genomic target site, enabling the Cas endonuclease to recognize, bind to, and optionally nick or cleave (introduce a single or double-strand break) into the target site. (US20150082478 published 19 Mar. 2015 and US20150059010 published 26 Feb. 2015).

[0109] The guide polynucleotide can also be a single molecule (also referred to as single guide polynucleotide) comprising a crNucleotide sequence linked to a tracrNucleotide sequence. The single guide polynucleotide comprises a first nucleotide sequence domain (referred to as Variable Targeting domain or VT domain) that can hybridize to a nucleotide sequence in a target DNA and a Cas endonuclease recognition domain (CER domain), that interacts with a Cas endonuclease polypeptide.

[0110] A “protospacer adjacent motif” (PAM) herein refers to a short nucleotide sequence adjacent to a target sequence (protospacer) that can be recognized (targeted) by a guide polynucleotide/Cas endonuclease system. The Cas endonuclease may not successfully recognize a target DNA sequence if the target DNA sequence is not followed by a PAM sequence. The sequence and length of a PAM herein can differ depending on the Cas protein or Cas protein complex used. The PAM sequence can be of any length but is typically 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 nucleotides long.

[0111] A “randomized PAM” and “randomized protospacer adjacent motif” are used interchangeably herein, and refer to a random DNA sequence adjacent to a target sequence (protospacer) that is recognized (targeted) by a guide polynucleotide/Cas endonuclease system. The randomized PAM sequence can be of any length but is typically 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 nucleotides long. A randomized nucleotide includes any one of the nucleotides A, C, G or T.

[0112] A guide polynucleotide/Cas endonuclease complex described herein is capable of recognizing, binding to, and optionally nicking, unwinding, or cleaving all or part of a target sequence.

[0113] Proteins may be altered in various ways including amino acid substitutions, deletions, truncations, and insertions. Methods for such manipulations are generally known. For example, amino acid sequence variants of the protein(s) can be prepared by mutations in the DNA. Methods for mutagenesis and nucleotide sequence alterations include, for example, Kunkel, (1985) *Proc. Natl. Acad. Sci. USA* 82:488-92; Kunkel et al., (1987) *Meth Enzymol* 154:367-82; U.S. Pat. No. 4,873,192; Walker and Gaastra, eds. (1983) *Techniques in Molecular Biology* (MacMillan Publishing Company, New York) and the references cited therein. Guidance regarding amino acid substitutions not likely to affect biological activity of the protein is found, for example, in the model of Dayhoff et al., (1978) *Atlas of Protein Sequence and Structure* (Natl Biomed Res Found, Washington, D.C.). Conservative substitutions, such as exchanging one amino acid with another having similar properties, may be preferable. Conservative deletions, insertions, and amino acid substitutions are not anticipated to produce radical changes in the characteristics of the protein, and the effect of any substitution, deletion, insertion, or combination thereof can be evaluated by routine screening assays. Assays for double-

strand-break-inducing activity are known and generally measure the overall activity and specificity of the agent on DNA substrates comprising target sites.

[0114] The nucleotide to be edited can be located within or outside a target site recognized and cleaved by a Cas endonuclease. In one embodiment, the at least one nucleotide modification is not a modification at a target site recognized and cleaved by a Cas endonuclease. In another embodiment, there are at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 30, 40, 50, 100, 200, 300, 400, 500, 600, 700, 900 or 1000 nucleotides between the at least one nucleotide to be edited and the genomic target site.

[0115] A knock-out may be produced by an indel (insertion or deletion of nucleotide bases in a target DNA sequence through NHEJ), or by specific removal of sequence that reduces or completely destroys the function of sequence at or near the targeting site.

[0116] A guide polynucleotide/Cas endonuclease induced targeted mutation can occur in a nucleotide sequence that is located within or outside a genomic target site that is recognized and cleaved by the Cas endonuclease.

[0117] The method for editing a nucleotide sequence in the genome of a cell can be a method without the use of an exogenous selectable marker by restoring function to a non-functional gene product.

[0118] In one embodiment, the invention describes a method for modifying a target site in the genome of a cell, the method comprising introducing into a cell at least one PGEN described herein and at least one donor DNA, wherein said donor DNA comprises a polynucleotide of interest, and optionally, further comprising identifying at least one cell that said polynucleotide of interest integrated in or near said target site.

[0119] In one aspect, the methods disclosed herein may employ homologous recombination (HR) to provide integration of the polynucleotide of interest at the target site.

[0120] Various methods and compositions can be employed to produce a cell or organism having a polynucleotide of interest inserted in a target site via activity of a CRISPR-Cas system component described herein. In one method described herein, a polynucleotide of interest is introduced into the organism cell via a donor DNA construct. The donor DNA construct further comprises a first and a second region of homology that flank the polynucleotide of interest. The first and second regions of homology of the donor DNA share homology to a first and a second genomic region, respectively, present in or flanking the target site of the cell or organism genome.

[0121] The donor DNA can be tethered to the guide polynucleotide. Tethered donor DNAs can allow for colocalizing target and donor DNA, useful in genome editing, gene insertion, and targeted genome regulation, and can also be useful in targeting post-mitotic cells where function of endogenous HR machinery is expected to be highly diminished (*Mali et al.*, 2013, *Nature Methods* Vol. 10:957-963).

[0122] In one embodiment, the disclosure comprises a method for editing a nucleotide sequence in the genome of a cell, the method comprising introducing into at least one PGEN described herein, and a polynucleotide modification template, wherein said polynucleotide modification template comprises at least one nucleotide modification of said

nucleotide sequence, and optionally further comprising selecting at least one cell that comprises the edited nucleotide sequence.

[0123] The guide polynucleotide/Cas endonuclease system can be used in combination with at least one polynucleotide modification template to allow for editing (modification) of a genomic nucleotide sequence of interest. (See also US20150082478, published 19 Mar. 2015 and WO2015026886 published 26 Feb. 2015).

[0124] Further uses for guide RNA/Cas endonuclease systems have been described (See for example: US20150082478 published 19 Mar. 2015, WO2015026886 published 26 Feb. 2015, US20150059010 published 26 Feb. 2015, WO2016007347 published 14 Jan. 2016, and PCT application WO2016025131 published 18 Feb. 2016) and include but are not limited to modifying or replacing nucleotide sequences of interest (such as a regulatory elements), insertion of polynucleotides of interest, gene knock-out, gene-knock in, modification of splicing sites and/or introducing alternate splicing sites, modifications of nucleotide sequences encoding a protein of interest, amino acid and/or protein fusions, and gene silencing by expressing an inverted repeat into a gene of interest.

[0125] The methods and compositions described herein do not depend on a particular method for introducing a sequence into an organism or cell, only that the polynucleotide or polypeptide gains access to the interior of at least one cell of the organism. Introducing includes reference to the incorporation of a nucleic acid into a eukaryotic or prokaryotic cell where the nucleic acid may be incorporated into the genome of the cell, and includes reference to the transient (direct) provision of a nucleic acid, protein or polynucleotide-protein complex (PGEN, RGEN) to the cell.

[0126] Methods for introducing polynucleotides or polypeptides or a polynucleotide-protein complex into cells or organisms are known in the art including, but not limited to, microinjection, electroporation, stable transformation methods, transient transformation methods, ballistic particle acceleration (particle bombardment), whiskers mediated transformation, *Agrobacterium*-mediated transformation, direct gene transfer, viral-mediated introduction, transfection, transduction, cell-penetrating peptides, mesoporous silica nanoparticle (MSN)-mediated direct protein delivery, topical applications, sexual crossing, sexual breeding, and any combination thereof.

[0127] The polynucleotide or recombinant DNA construct can be provided to or introduced into a prokaryotic and eukaryotic cell or organism using a variety of transient transformation methods. Such transient transformation methods include, but are not limited to, the introduction of the polynucleotide construct directly into the plant. Nucleic acids and proteins can be provided to a cell by any method including methods using molecules to facilitate the uptake of anyone or all components of a guided Cas system (protein and/or nucleic acids), such as cell-penetrating peptides and nanocarriers. See also US20110035836 published 10 Feb. 2011, and EP2821486A1 published 7 Jan. 2015.

[0128] Other methods of introducing polynucleotides into a prokaryotic and eukaryotic cell or organism or plant part can be used, including plastid transformation methods, and the methods for introducing polynucleotides into tissues from seedlings or mature seeds.

[0129] Stable transformation is intended to mean that the nucleotide construct introduced into an organism integrates

into a genome of the organism and is capable of being inherited by the progeny thereof. Transient transformation is intended to mean that a polynucleotide is introduced into the organism and does not integrate into a genome of the organism or a polypeptide is introduced into an organism. Transient transformation indicates that the introduced composition is only temporarily expressed or present in the organism.

[0130] A variety of methods are available to identify those cells having an altered genome at or near a target site without using a screenable marker phenotype. Such methods can be viewed as directly analyzing a target sequence to detect any change in the target sequence, including but not limited to PCR methods, sequencing methods, nuclease digestion, Southern blots, and any combination thereof.

[0131] In an aspect, the disclosed methods can be used to introduce into plants polynucleotides useful for the targeted integration of nucleotide sequences into a plant. For example, the disclosed methods can be used to introduce T-DNA expression cassettes comprising nucleotide sequences of interest flanked by non-identical recombination sites to transform a plant comprising a target site. See, for example, WO2022087616, incorporated herein by reference in its entirety.

[0132] In an aspect, the genomic target site contains at least a set of non-identical recombination sites corresponding to those on the T-DNA expression cassette. A recombinase mediates the exchange of the nucleotide sequences flanked by the recombination sites. Thus, the disclosed methods can be used for the introduction of T-DNA expression cassettes for targeted integration of nucleotide sequences, wherein the T-DNA expression cassettes flanked by non-identical recombination sites are recognized by a recombinase which implements recombination at the non-identical recombination sites. Accordingly, the disclosed methods and composition can be used to improve efficiency and speed of development of plants containing non-identical recombination sites.

[0133] Thus, the disclosed methods can further comprise methods for the directional, targeted integration of exogenous nucleotides into a plant cell. In an aspect, the disclosed methods use recombination sites in a gene targeting system which facilitates directional targeting of desired genes and nucleotide sequences into corresponding recombination sites previously introduced into the target plant genome.

[0134] By suitable non-identical recombination site implies that in the presence of active recombinase, excision of sequences between two non-identical recombination sites occurs, if at all, with an efficiency considerably lower than the recombinationally-mediated exchange targeting arrangement of nucleotide sequences into the plant genome. Thus, suitable non-identical sites for use in the present disclosure include those sites where the efficiency of recombination between the sites is low; for example, where the efficiency is less than about 30 to about 50%, preferably less than about 10 to about 30%, more preferably less than about 5 to about 10%.

[0135] As noted above, in one embodiment, the recombination sites in the targeting cassette correspond to those in the target site of the transformed plant. That is, if the target site of the transformed plant contains flanking non-identical recombination sites of FRT and a mutant FRT, the targeting cassette will contain the same FRT and mutant FRT non-

identical recombination sites. If FRT sites are utilized, the FLP recombinase will be needed. In the same manner, where lox sites are utilized, the Cre recombinase is required. If the non-identical recombination sites comprise both a FRT and a lox site, both the FLP and Cre recombinase will be required in the plant cell.

[0136] Where appropriate, the nucleotide sequences to be inserted in the plant genome may be optimized for increased expression. Where mammalian, yeast, or bacterial genes are used in the present disclosure, they can be synthesized using plant preferred codons for improved expression. It is recognized that for expression in monocots, dicot genes can also be synthesized using monocot preferred codons. It is further recognized that all or any part of the gene sequence may be optimized or synthetic. That is, fully optimized or partially optimized sequences may also be used.

[0137] Additional sequence modifications are known to enhance gene expression in a cellular host and can be used in the present disclosure. These include elimination of sequences encoding spurious polyadenylation signals, exon-intron splice site signals, transposon-like repeats, and other such well-characterized sequences, which may be deleterious to gene expression. The G-C content of the sequence may be adjusted to levels average for a given cellular host, as calculated by reference to known genes expressed in the host cell. When possible, the sequence is modified to avoid predicted hairpin secondary RNA structures.

[0138] The present disclosure also encompasses FLP recombination target sites (FRT). The FRT has been identified as a minimal sequence comprising two 13 base pair repeats, separated by an eight (8) base spacer. The nucleotides in the spacer region can be replaced with a combination of nucleotides, so long as the two 13-base repeats are separated by eight nucleotides.

[0139] FRT mutant sites can be used in the practice of the disclosed methods. Although mutant FRT sites are known (see SEQ ID NO: 2, 3, 4 and 5 of WO1999/025821), it is recognized that other mutant FRT sites may be used in the practice of the present disclosure. The present disclosure is not restricted to the use of a particular FRT or recombination site, but rather that non-identical recombination sites or FRT sites can be utilized for targeted insertion and expression of nucleotide sequences in a plant genome. Thus, other mutant FRT sites can be constructed and utilized based upon the present disclosure.

[0140] As discussed above, bringing genomic DNA containing a target site with non-identical recombination sites together with a vector containing a T-DNA expression cassette with corresponding non-identical recombination sites, in the presence of the recombinase, results in recombination. The nucleotide sequence of the T-DNA expression cassette located between the flanking recombination sites is exchanged with the nucleotide sequence of the target site located between the flanking recombination sites. In this manner, nucleotide sequences of interest may be precisely incorporated into the genome of the host.

[0141] It is recognized that many variations of the present disclosure can be practiced. For example, target sites can be constructed having multiple non-identical recombination sites. Thus, multiple genes or nucleotide sequences can be stacked or ordered at precise locations in the plant genome. Likewise, once a target site has been established within the genome, additional recombination sites may be introduced by incorporating such sites within the nucleotide sequence

of the T-DNA expression cassette and the transfer of the sites to the target sequence. Thus, once a target site has been established, it is possible to subsequently add sites, or alter sites through recombination.

[0142] Another variation includes providing a promoter or transcription initiation region operably linked with the target site in an organism. Preferably, the promoter will be 5' to the first recombination site. By transforming the organism with a T-DNA expression cassette comprising a coding region, expression of the coding region will occur upon integration of the T-DNA expression cassette into the target site. This aspect provides for a method to select transformed cells, particularly plant cells, by providing a selectable marker sequence as the coding sequence.

[0143] The present disclosure is further illustrated in the following embodiments. It should be understood that these embodiments are given by way of illustration only and not by way of limitation.

[0144] Embodiment 1. A method of inducing parthenogenesis, the method comprising:

[0145] expressing, in a female gametophyte, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in a plant egg cell, wherein the Babyboom (BBM) polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof, wherein the female gametophyte is rendered parthenogenic and forms a haploid embryo without pollination, and wherein the Babyboom (BBM) polynucleotide is not naturally expressed in a female gametophyte.

[0146] Embodiment 2. The method of embodiment 1, wherein the nucleotide sequence encoding the Babyboom (BBM) polypeptide or fragment thereof is selected from the group consisting of: BBM1, BBM2, BMN2, BMN3, ODP2, and BBML.

[0147] Embodiment 3. The method of embodiment 1, wherein the Babyboom (BBM) polynucleotide is selected from the group consisting of:

[0148] (a) a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0149] (b) a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0150] (c) a nucleotide sequence that has at least 85% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0151] (d) a fragment of any of the nucleotide sequences of (a), (b), or (c), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity;

[0152] (e) a fragment of any of the nucleotide sequences of (a), (b), (c), or (d), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity, wherein the amino acid fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

[0153] Embodiment 4. The method of embodiment 1, wherein the Babyboom (BBM) polypeptide is selected from the group consisting of:

- [0154] (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- [0155] (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- [0156] (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- [0157] (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and
- [0158] (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.
- [0159] Embodiment 5. The method of embodiment 1, wherein the promoter that expresses in the egg cell is an egg cell-specific promoter or an egg cell-preferred promoter.
- [0160] Embodiment 6. The method of embodiment 5, wherein the promoter that expresses in the egg cell comprises:
- [0161] (a) a polynucleotide sequence of SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;
- [0162] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;
- [0163] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragment retains its ability to drive expression of the operably linked Babyboom (BBM) polynucleotide encoding the Babyboom (BBM) polypeptide or the fragment thereof.
- [0164] Embodiment 7. The method of embodiment 1, wherein the promoter that expresses in the egg cell further comprises an expression modulation element.
- [0165] Embodiment 8. The method of embodiment 1, wherein the expression of the Babyboom (BBM) polynucleotide by an egg cell promoter comprising an expression modulation element increases haploid induction frequency of the female gametophyte, as compared to haploid induction frequency resulting from expression of a Babyboom (BBM) polynucleotide driven by an egg cell promoter without an expression modulation element.
- [0166] Embodiment 9. The method of embodiment 1, wherein the female gametophyte has not been fertilized prior to or during expression of the parthenogenesis factor, e.g. the BBM polynucleotide, in the egg cell.
- [0167] Embodiment 10. The method of embodiment 1, further comprising: modifying a regulatory region of an endogenous Babyboom (BBM) polynucleotide so that the Babyboom (BBM) polynucleotide expresses in a female gametophyte.
- [0168] Embodiment 11. The method of embodiment 1, the method comprising:
- [0169] expressing the Babyboom (BBM) polynucleotide from a modified endogenous genomic BBM locus, wherein the genomic BBM locus comprises a modified regulatory region of an endogenous polynucleotide encoding a BBM polypeptide, wherein one or more nucleotides in the regulatory region have been modified so that the BBM polypeptide expresses in a female gametophyte.
- [0170] Embodiment 12. The method of embodiment 1, the method comprising:
- [0171] expressing the Babyboom (BBM) polynucleotide from a modified endogenous genomic egg cell locus, wherein the egg cell's coding or genomic sequence has been modified so that it encodes a BBM polypeptide, wherein the BBM polypeptide expresses in a female gametophyte.
- [0172] Embodiment 13. The method of embodiment 1, wherein the Babyboom (BBM) polynucleotide has been modified from its native form using a gene editing technology.
- [0173] Embodiment 14. The method of embodiment 13, wherein the gene editing technology uses a DNA modification enzyme that is a site-directed nuclease selected from the group comprising meganucleases (MNs), zinc-finger nucleases (ZFN), transcription-activator like effector nucleases (TALENs), Cas polypeptides, such as, dCas9-FokI, dCpf1-FokI, chimeric Cas9-cytidine deaminase, chimeric Cas9 adenine deaminase, chimeric FEN1-FokI, Mega-TALs, a nickase Cas9 (nCas9), chimeric dCas9 non-FokI nuclease, dCpf1-non-FokI nuclease, Cas9, Cas12f (Cas-alpha, Cas14), Cas12i (Cas-beta), Cas12a (Cpf1), Cas12b (a C2c1 protein), Cas13 (a C2c2 protein), Cas12c (a C2c3 protein), Cas12d, Cas12e, Cas12g, Cas12h, Cas12i, Cas12j, Cas12k, Cas3, Cas3-HD, Cas 5, Cas6, Cas7, Cas8, Cas10, or combinations or complexes of these.
- [0174] Embodiment 15. The method of embodiment 1, wherein the female gametophyte is a monocot or dicot female gametophyte.
- [0175] Embodiment 16. The method of embodiment 1, wherein the female gametophyte is a wheat, cotton, sunflower, safflower, tobacco, *Arabidopsis*, *cannabis*, sugarcane, soy, barley, oats, rice, maize, triticale, sorghum, rye, millet, or a flax female gametophyte.
- [0176] Embodiment 17. The method of embodiment 1, further comprising:
- [0177] (a) contacting the haploid embryo with a chromosome doubling agent for a period sufficient to generate a doubled haploid embryo;
- [0178] (b) isolating the doubled haploid embryo; and
- [0179] (c) regenerating a doubled haploid plant from the doubled haploid embryo of step (b).
- [0180] Embodiment 18. The method of embodiment 1, further comprising:
- [0181] (a) regenerating a parthenogenic plant from a haploid embryo comprising the Babyboom (BBM) polynucleotide operably linked to the promoter that expresses in an egg cell;
- [0182] (b) pollinating the parthenogenic plant of (a) with pollen from a non-haploid inducer; and
- [0183] (c) rescuing a haploid embryo from the parthenogenic plant of (b).
- [0184] Embodiment 19. A method for obtaining a wheat plant producing clonal, non-reduced, non-recombined gametes, the method comprising:
- [0185] (a) suppressing in a wheat plant cell the activity of:
- [0186] (1) all endogenous Spo11 or Prd1, Prd2, or Prd3 polynucleotides or polypeptides;
- [0187] (2) all endogenous Rec8 polynucleotides or polypeptides;

- [0188] (3) all endogenous Osd1 polynucleotides or polypeptides;
- [0189] (b) expressing, in a female gametophyte derived from the wheat plant cell, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell, wherein the Babyboom (BBM) polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof, wherein the female gametophyte is rendered parthenogenic and forms an embryo;
- [0190] (c) obtaining a plant from the embryo, wherein the embryo comprises the egg cell expressed Babyboom (BBM) polynucleotide and suppressed Spo11, Rec8, Osd1, Prd1, Prd2, or Prd3 polynucleotides or polypeptides, thereby producing a wheat plant producing clonal, non-reduced, non-recombined gametes.
- [0191] Embodiment 20. The method of embodiment 19, wherein the plant is a first filial generation hybrid plant.
- [0192] Embodiment 21. The method of embodiment 19, wherein the activity of the endogenous Spo11 polynucleotides or polypeptides, Rec8 polynucleotides or polypeptides, Osd1 polynucleotides or polypeptides, Prd1, Prd2, or Prd3 polynucleotides or polypeptides, or combinations thereof is suppressed by introducing a nucleotide modification into its polynucleotide sequence or an amino acid modification into its polypeptide sequence.
- [0193] Embodiment 22. The method of embodiment 21, wherein the nucleotide modification is a deletion, addition, or substitution of one or more nucleotides, and wherein the amino acid modification is a deletion, addition, or substitution of one or more amino acids
- [0194] Embodiment 23. The method of embodiment 21, wherein said the nucleotide modification is introduced by a nuclease selected from the group consisting of: a TALEN, a meganuclease, a zinc finger nuclease, and a CRISPR-associated nuclease.
- [0195] Embodiment 24. The method of embodiment 21, wherein the nucleotide modification is introduced by a Cas endonuclease, e.g., Cas9, Cas12f (Cas-alpha, Cas14), Cas12i (Cas-beta), Cas12a (Cpf1), Cas12b (a C2c1 protein), Cas13 (a C2c2 protein), Cas12c (a C2c3 protein), Cas12d, Cas12e, Cas12g, Cas12h, Cas12i, Cas12j, Cas12k, Cas3, Cas3-HD, Cas 5, Cas6, Cas7, Cas8, Cas10, or combinations or complexes of these, guided by at least one guide RNA.
- [0196] Embodiment 25. The method of embodiment 21, wherein the promoter that expresses in the egg cell comprises:
- [0197] (a) a polynucleotide sequence of SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;
- [0198] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;
- [0199] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragment retains its ability to drive expression of the operably linked Babyboom (BBM) polynucleotide encoding the Babyboom (BBM) polypeptide or the fragment thereof.
- [0200] Embodiment 26. The method of embodiment 19, wherein the nucleotide sequence encoding the Babyboom (BBM) polypeptide or fragment thereof is selected from the group consisting of BBM1, BBM2, BMN2, BMN3, ODP2, and BBML.
- [0201] Embodiment 27. The method of embodiment 19, wherein the Babyboom (BBM) polynucleotide is selected from the group consisting of:
- [0202] (a) a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- [0203] (b) a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- [0204] (c) a nucleotide sequence that has at least 85% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- [0205] (d) a fragment of any of the nucleotide sequences of (a), (b), or (c), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity;
- [0206] (e) a fragment of any of the nucleotide sequences of (a), (b), (c), or (d), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity, wherein the amino acid fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.
- [0207] Embodiment 28. The method of embodiment 19, wherein the Babyboom (BBM) is selected from the group consisting of:
- [0208] (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- [0209] (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- [0210] (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- [0211] (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and
- [0212] (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.
- [0213] Embodiment 29. A method of obtaining a clonal apomictic plant from one or more gametophytic cells in a plant in the absence of egg cell fertilization comprising: (a) expressing, in one or more gametophytic cell, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell, wherein the Babyboom (BBM) polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof that retains haploid induction activity, wherein the Babyboom (BBM) polynucleotide is not naturally expressed in a female gametophyte;
- [0214] (b) developing an embryo from the gametophytic cell in the absence of egg cell fertilization; and
- [0215] (c) obtaining progeny plant from one or more gametophytic cells wherein the progeny plant contains the chromosomes from the gametophytic cell of (a),

thereby achieving propagation of a flowering plant in the absence of egg cell fertilization.

[0216] Embodiment 30. The method of embodiment 29, wherein the embryo is formed from an unreduced plant cell

[0217] Embodiment 31. The method of embodiment 30, wherein the unreduced plant cell is an egg cell or is formed from a somatic cell.

[0218] Embodiment 32. The method of embodiment 29, wherein the promoter that expresses in the egg cell comprises:

[0219] (a) a polynucleotide sequence of SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;

[0220] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;

[0221] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragment retains its ability to drive expression of the operably linked Babyboom (BBM) polynucleotide encoding the Babyboom (BBM) polypeptide or the fragment thereof.

[0222] Embodiment 33. The method of embodiment 29, wherein the nucleotide sequence encoding the Babyboom (BBM) polypeptide or fragment thereof is selected from the group consisting of BBM1, BBM2, BMN2, BMN3, ODP2, and BBML.

[0223] Embodiment 34. The method of embodiment 29, wherein the Babyboom (BBM) polynucleotide is selected from the group consisting of:

[0224] (a) a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0225] (b) a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0226] (c) a nucleotide sequence that has at least 85% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0227] (d) a fragment of any of the nucleotide sequences of (a), (b), or (c), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity;

[0228] (e) a fragment of any of the nucleotide sequences of (a), (b), (c), or (d), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity, wherein the amino acid fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

[0229] Embodiment 35. The method of embodiment 29, wherein the Babyboom (BBM) is selected from the group consisting of:

[0230] (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0231] (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0232] (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0233] (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and

[0234] (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

[0235] Embodiment 36. A method of modulating expression of an endogenous polynucleotide in a genomic locus of a plant cell, the method comprising:

[0236] modifying one or more nucleotides in a regulatory region of the genomic locus comprising an endogenous polynucleotide encoding a BBM polypeptide to create a modified regulatory region, wherein creating the modified regulatory region results in expression of the BBM polypeptide in a female gametophyte.

[0237] Embodiment 37. The method of embodiment 36, wherein the modification of one or more nucleotides is by introducing one or more nucleotides, altering one or more nucleotides, or deleting one or more nucleotides, or combinations thereof.

[0238] Embodiment 38. The method of embodiment 36, wherein the BBM polypeptide is selected from the group consisting of:

[0239] (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0240] (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0241] (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0242] (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and

[0243] (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

[0244] Embodiment 39. The method of embodiment 36, wherein the regulatory region has been modified to comprise egg cell-specific promoters or egg cell-specific sequence motifs.

[0245] Embodiment 40. The method of embodiment 36, wherein the regulatory region has been modified to comprise a promoter that expresses in the egg cell, wherein the promoter comprises:

[0246] (a) a polynucleotide sequence of SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;

[0247] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;

[0248] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragment retains its ability to drive expression of the operably linked Baby-

boom (BBM) polynucleotide encoding the Babyboom (BBM) polypeptide or the fragment thereof.

[0249] Embodiment 41. The method of embodiment 36, wherein the regulatory region has been modified to comprise an expression modulating element.

[0250] Embodiment 42. A modified genomic locus of a plant cell, wherein the genomic locus comprises a modified regulatory region of an endogenous polynucleotide encoding a BBM polypeptide, wherein one or more nucleotides in the regulatory region have been modified so that the BBM polypeptide expresses in a female gametophyte.

[0251] Embodiment 43. The modified genomic locus of embodiment 42, wherein the BBM polypeptide is selected from the group consisting of:

[0252] (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0253] (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0254] (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0255] (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and

[0256] (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

[0257] Embodiment 44. The modified genomic locus of embodiment 42, wherein the regulatory region has been modified to comprise egg cell-specific promoters or egg cell-specific sequence motifs.

[0258] Embodiment 45. The modified genomic locus of embodiment 42, wherein the regulatory region has been modified to comprise a promoter that expresses in the egg cell, wherein the promoter comprises:

[0259] (a) a polynucleotide sequence of SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;

[0260] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;

[0261] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragment retains its ability to drive expression of the operably linked Babyboom (BBM) polynucleotide encoding the Babyboom (BBM) polypeptide or the fragment thereof.

[0262] Embodiment 46. The modified genomic locus of embodiment 42, wherein the regulatory region has been modified to comprise an enhancer expressing element.

[0263] Embodiment 47. A method of modulating a genomic or coding sequence of an endogenous polynucleotide in a genomic locus of a plant cell, the method comprising: modifying a genomic or coding sequence of an endogenous genomic egg cell locus to create a modified genomic or coding sequence comprising a heterologous

polynucleotide encoding a BBM polypeptide, wherein the modification results in expression of the BBM polypeptide in a female gametophyte.

[0264] Embodiment 48. The method of embodiment 47, wherein the modification of the endogenous genomic or coding sequence is by

[0265] (a) replacing one or more nucleotides of the endogenous coding sequence;

[0266] (b) introducing one or more nucleotides into the endogenous coding sequence; (c) altering one or more nucleotides of the endogenous coding sequence; or

[0267] (d) deleting one or more nucleotides of the endogenous coding sequence;

[0268] or combinations thereof.

[0269] Embodiment 49. The method of embodiment 47, wherein the heterologous polynucleotide is selected from the group consisting of:

[0270] (a) a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0271] (b) a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0272] (c) a nucleotide sequence that has at least 85% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;

[0273] (d) a fragment of any of the nucleotide sequences of (a), (b), or (c), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity;

[0274] (e) a fragment of any of the nucleotide sequences of (a), (b), (c), or (d), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity, wherein the amino acid fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

[0275] Embodiment 50. The method of embodiment 47, wherein the BBM polypeptide is selected from the group consisting of:

[0276] (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0277] (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0278] (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0279] (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and

[0280] (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

[0281] Embodiment 51. The method of embodiment 47, wherein the genomic egg cell locus has been modified to comprise an enhancer expressing element so that the heterologous polynucleotide expresses in the female gametophyte and/or the genomic egg cell locus comprises an

endogenous promoter that expresses in the egg cell, wherein the endogenous promoter comprises:

- [0282] (a) a polynucleotide sequence of SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;
- [0283] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86, or variants thereof;
- [0284] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragment retains its ability to drive expression of the operably linked Babyboom (BBM) polynucleotide encoding the Babyboom (BBM) polypeptide or the fragment thereof.
- [0285] Embodiment 52. A modified genomic locus of a plant cell, wherein the genomic locus comprises a modified coding sequence of an endogenous genomic egg cell locus, wherein one or more nucleotides in the endogenous egg cell's coding sequence have been modified to express a BBM polypeptide, wherein the BBM polypeptide expresses in a female gametophyte. See, for example, SEQ ID NOs: 114 and 115.
- [0286] Embodiment 53. The modified genomic locus of embodiment 52, wherein the heterologous polynucleotide is selected from the group consisting of:
 - [0287] (a) a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
 - [0288] (b) a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
 - [0289] (c) a nucleotide sequence that has at least 85% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
 - [0290] (d) a fragment of any of the nucleotide sequences of (a), (b), or (c), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity;
 - [0291] (e) a fragment of any of the nucleotide sequences of (a), (b), (c), or (d), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity, wherein the amino acid fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.
- [0292] Embodiment 54. The modified genomic locus of embodiment 52, wherein the BBM polypeptide is selected from the group consisting of:
 - [0293] (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
 - [0294] (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
 - [0295] (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;

[0296] (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and

[0297] (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

[0298] Embodiment 55. The modified genomic locus of embodiment 52, wherein the endogenous genomic egg cell locus has been modified to comprise an enhancer expressing element.

[0299] Embodiment 56. A plant comprising a promoter operably linked to a heterologous polynucleotide sequence, wherein the promoter is selected from the group consisting of:

- [0300] (a) a polynucleotide sequence of SEQ ID NO: 37, 38, or 39;
- [0301] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 37, 38, or 39;
- [0302] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragments retains its ability to drive expression of the heterologous polynucleotide sequence in an egg cell of a plant.

[0303] Embodiment 57. A recombinant DNA construct comprising a promoter operably linked to a heterologous polynucleotide sequence, wherein the promoter is selected from the group consisting of:

- [0304] (a) a polynucleotide sequence of SEQ ID NO: 37, 38, or 39;
- [0305] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 37, 38, or 39;
- [0306] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragment retains its ability to drive expression of the heterologous polynucleotide sequence in an egg cell of a plant.

[0307] Embodiment 58. A method of expressing a gene product in an egg cell of a plant, comprising:

[0308] introducing into the plant or plant cell or egg cell of the plant a polynucleotide comprising a polynucleotide sequence selected from the group consisting of:

- [0309] (a) a polynucleotide sequence of SEQ ID NO: 37, 38, or 39;
- [0310] (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 37, 38, or 39;
- [0311] (c) a fragment of the polynucleotide of (a) or (b), e.g., a fragment of at least 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, or 2000 contiguous nucleotides of (a) or (b), wherein the fragment retains its ability to drive expression of a heterologous polynucleotide sequence in an egg cell;
- [0312] wherein said polynucleotide sequence drives expression of an operably linked heterologous polynucleotide sequence.

[0313] Embodiment 59. The method of embodiment 56, 57, or 58, wherein the operably linked sequence is a Baby-

boom (BBM) polynucleotide that encodes a polypeptide that, when expressed in a female gametophyte, induces haploid induction.

[0314] Embodiment 60. The method of embodiment 60, wherein the operably linked sequence encodes a BBM polypeptide or functional fragment thereof.

[0315] Embodiment 61. The method of embodiment 56, 57, or 58, wherein the plant is a monocot or dicot plant.

[0316] Embodiment 62. The method of embodiment 56, 57, or 58, wherein the plant is a wheat, cotton, sunflower, safflower, tobacco, *Arabidopsis*, soy, barley, oats, rice, maize, triticale, sorghum, *cannabis*, sugarcane, rye, millet, or a flax plant.

[0317] The following examples are offered by way of illustration and not by way of limitation.

EXAMPLES

[0318] The aspects of the disclosure are further defined in the following Examples, in which parts and percentages are by weight and degrees are Celsius, unless otherwise stated. These Examples, while indicating aspects of the disclosure, are given by way of illustration only. From the above discussion and these Examples, one skilled in the art can ascertain the essential characteristics of the aspects of the disclosure, and without departing from the spirit and scope thereof, can make various changes and modifications of them to adapt to various usages and conditions. Thus, various modifications in addition to those shown and described herein will be apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims.

Example 1: Methods of Inducing Parthenogenesis in Wheat

[0319] Polynucleotides encoding BBM peptides, linked to regulatory elements active in maternal tissues or haploid cells, were evaluated for their ability to induce parthenogenesis in the egg cells of wheat.

[0320] Maternal Haploid Induction In Response to Wheat BBM Peptides Polynucleotides encoding BBM type transcription factor peptides were identified in the wheat genome on homocologous chromosome 3 (chromosome 3A (SEQ ID NO: 1, 2, 74, 75), 3B (SEQ ID NO: 3, 4, 76, 77) and 3D (SEQ ID NO: 5, 6, 78, 79)) and homocologous chromosome 6 (chromosome 6A (SEQ ID NO: 7, 8, 80, 81), 6B (SEQ ID NO: 9, 10, 82, 83) and 6D (SEQ ID NO: 11, 12, 84, 85)). An alignment of wheat BBM peptides with BBM peptides from additional monocot species is shown in FIG. 1. DNA polynucleotide sequences encoding TA-BBM peptides were operably linked to various different promoters active during female gamete development.

[0321] Expression cassettes were prepared to test parthenogenesis using the wheat TA-BBM3A sequence, an ortholog of the ZM-ODP2 (SEQ ID NO: 13, 14). In one example, a ZM-DD45 PRO: TA-BBM3A expression cassette contained a polynucleotide encoding a TA-BBM3A peptide (SEQ ID NO: 1) operably linked to regulatory elements ZM-DD45 PRO (SEQ ID NO: 15). In another example, a PvEC1 PRO: TA-BBM3A expression cassette contained a polynucleotide encoding a TA-BBM3A peptide (SEQ ID NO: 1) operably linked to regulatory elements comprising PV-EGG CELL PRO (TR1) (SEQ ID NO: 16), EGG MIN PRO (SEQ ID NO: 17), and PV-PRO31696.1

5UTR (SEQ ID NO: 18), this combination of regulatory elements SEQ ID NO: 16, SEQ ID NO: 17, and SEQ ID NO: 18, is called the “PvEC1 promoter” (SEQ ID NO: 19).

[0322] Similar expression cassettes were prepared to evaluate parthenogenesis using the wheat TA-BBM6A sequence, an ortholog of the ZM-BBM2 (SEQ ID NO: 20, 21) sequence in wheat. In one example, the ZM-DD45 PRO: TA-BBM6A expression cassette contained a polynucleotide encoding a TA-BBM6A peptide (SEQ ID NO: 7) operably linked to regulatory elements ZM-DD45 PRO (SEQ ID NO: 15). In another example, a PvEC1 PRO: TA-BBM6A expression cassette contained a polynucleotide encoding a TA-BBM6A peptide (SEQ ID NO: 7) was operably linked to regulatory elements comprising PV-EGG CELL PRO (TR1) (SEQ ID NO: 16), EGG MIN PRO (SEQ ID NO: 17), and PV-PRO31696.1 5UTR (SEQ ID NO: 18), this combination of regulatory elements SEQ ID NO: 16, SEQ ID NO: 17, and SEQ ID NO: 18, is called the “PvEC1 promoter” (SEQ ID NO: 19).

[0323] These expression cassettes were designed and used as a parthenogenesis factor to induce egg cells to differentiate into embryo cells and to initiate embryogenesis. These expression cassettes were included in constructs for *Agrobacterium*-mediated transformation of the wheat line SBC0456D, resulting in random integration of T-DNA within the wheat genome. Leaf samples were collected from TO seedlings to determine copy number and transgene integrity of multiple different elements within the left border and right border of the T-DNA integrated into the plant genome. Only plants determined as single copy for the multiple different elements were characterized further. After transformation, each regenerated hemizygous TO plant that contained a single, stably inserted T-DNA was considered a unique event, was grown to maturity, allowed to self-pollinate, and mature T1 seed was harvested. In plants containing a hemizygous BBM expression cassette, the transgene is expected to segregate as a 1:1 ratio (transgene positive: transgene negative) in the female gametes, and thus 50% is the expected maximum frequency of haploid progeny if fully penetrant and expressive for the parthenogenetic phenotype. If parthenogenesis occurred, pollination may only entail the fertilization of the central cell, and initiate the normal development of endosperm, but not fertilization of the egg cell which has already initiated embryogenesis. During female gametogenesis, expression of the wheat BBM polynucleotides in or near the embryo sac cell, particularly the egg cell, driven by the egg-cell specific promoter stimulates embryogenesis and parthenogenesis of the egg cell. In bread wheat, progeny derived from parthenogenetic egg cells will be haploid ($In=3x$, ABD) instead of diploid ($2n=6x$, AABBDD), and contain half the number of chromosomes and relative DNA content, as compared to the parental plant. The ploidy of individual plants in the T1 progeny was determined using flow cytometry analysis of nuclei isolated from leaf tissue.

[0324] Evaluation of haploid progeny demonstrated the ability of wheat BBM polynucleotide sequences to induce parthenogenesis. The percent of haploid progeny (%) were scored for each unique event using the number of haploid progeny divided by the total number of progeny examined to determine the haploid induction (HI) frequency. Progeny from transgenic events containing expression cassettes ZM-DD45 PRO::TA-BBM3A, Pv-EC1 PRO: TA-BBM3A, were evaluated by flow cytometry to determine the HI frequency.

For the ZM-DD45 PRO::TA-BBM3A expression cassette, individual events showed a range of 0% to 10% HI frequency, and showed an average rate of 1.94% HI frequency with a standard deviation of 4%, as shown in Table 1. For the Pv-EC1 PRO: TA-BBM3A expression cassette, individual events showed a range of 31.03% to 48.57% HI frequency, and showed an average rate of 41.56% HI frequency with a standard deviation of 6.97%, as shown in Table 1. Similarly, progeny from transgenic events containing expression cassettes ZM-DD45 PRO::TA-BBM6A, Pv-EC1 PRO: TA-BBM6A, were evaluated by flow cytometry to determine the frequency of haploid progeny. For the ZM-DD45 PRO::TA-BBM6A expression cassette, individual events showed a range of 0% to 6.5% HI frequency, and showed an average rate of 0.89% HI frequency with a standard deviation of 2.22%, as shown in Table 1. For the Pv-EC1 PRO: TA-BBM6A expression cassette, individual events showed a range of 16.7% to 56.3% HI frequency, and showed an average rate of 38.1% HI frequency with a standard deviation of 12.91%, as shown in Table 1.

TABLE 1

Haploid Induction (HI) in T1 progeny of transgenic events with wheat BBM peptides				
Expression cassette	Number of independent transgenic events	Number of T1 seeds screened	Mean (% HI)	SD (% HI)
ZM-DD45 PRO::TA-BBM6A	11	328	0.89	2.22
PvEC1 PRO::TA-BBM6A	8	493	38.1	12.91
ZM-DD45 PRO::TA-BBM3A	6	352	1.94	4
PvEC1 PRO::TA-BBM3A	6	225	41.56	6.97

[0325] As shown above, a regulatory element expressed in the egg cell of the plant is useful for regulating TA-BBM peptide activity to induce maternal haploid induction, resulting in a percentage of the progeny produced being haploid (having half the number of chromosomes compared to the parent). In addition, alternative regulatory elements can be used to further optimize parthenogenetic maternal haploid induction levels. For example, regulatory elements such as those disclosed in US2015/0152430 (promoters including, but not limited to the AT-DD5 promoter, the AT-DD31 promoter, and the AT-DD65 promoter) and those disclosed in US2018/0094273 (*Zea mays* egg cell promoters) can be used in the methods of the present disclosure (See, for example, US2015/0152430 and US2018/0094273 incorporated herein by reference in their entireties).

[0326] In summary, wheat BBM polynucleotides were demonstrated to induce parthenogenesis and produce viable haploid progeny when expressed in the egg cell of the wheat female gametophyte. In addition, it was unexpected that the egg cell-specific regulatory element PvEC1 promoter improved the parthenogenetic haploid frequency in comparison to the egg cell-specific regulatory elements ZM-DD45 PRO (Table 1).

[0327] A. Maternal Haploid Induction In Response to Non-Wheat BBM Peptides

[0328] BBM type transcription factors such as those disclosed in Publication No. US20210180077, incorporated

herein by reference in its entirety, and polynucleotides encoding BBM type transcription factors identified in additional species were used in this example. Haploid progeny assays were performed to evaluate whether BBM genes from various species would initiate parthenogenesis in wheat. Expression cassettes of DNA polynucleotide sequences encoding additional BBM peptides (SEQ ID NO: 13-14, 20-29) were operably linked to the PvEC1 promoter (SEQ ID NO: 19) active during female gamete development described in Table 2.

TABLE 2

Summary of BBM polynucleotides and peptides.			
DNA Polynucleotide Name	DNA Polynucleotide SEQ ID NO:	Peptide Name	Protein SEQ ID NO:
TA-BBM3A	1	TA-BBM3A	2
TA-BBM6A	7	TA-BBM6A	8
ZM-ODP2	13	ZM-ODP2	14
ZM-BBM2	20	ZM-BBM2	21
OS-ODP2	22	OS-ODP2	23
OS-BBM2	24	OS-BBM2	25
SV-BBM1	26	SV-BBM1	27
SV-BBM2	28	SV-BBM2	29

[0329] Progeny from transgenic events containing *Zea mays* BBM type sequences, with expression cassettes PvEC1 PRO: ZM-ODP2 and Pv-EC1 PRO: ZM-BBM2, were evaluated by flow cytometry to determine the frequency of haploid progeny in wheat. The PvEC1 PRO::ZM-ODP2 expression cassette showed an average rate of 43.1% HI frequency with a standard deviation of 12.2%, and the Pv-EC1 PRO: ZM-BBM2 expression cassette showed an average rate of 42.8% HI frequency with a standard deviation of 10.4%, as shown in Table 3. Additionally, progeny from transgenic events containing *Oryza sativa* and *Setaria viridis* BBM type sequences, with expression cassettes PvEC1 PRO::OS-ODP2, Pv-EC1 PRO: OS-BBM2, PvEC1 PRO:SV-BBM1 and Pv-EC1 PRO: SV-BBM2, were evaluated by flow cytometry to determine the frequency of haploid progeny in wheat. The PvEC1 PRO::OS-ODP2 expression cassette showed an average rate of 46.8% HI frequency with a standard deviation of 8.5%, and the Pv-EC1 PRO: OS-BBM2 expression cassette showed an average rate of 49.7% HI frequency with a standard deviation of 18.6%, as shown in Table 3. The PvEC1 PRO::SV-BBM1 expression cassette showed an average rate of 36.5% HI frequency with a standard deviation of 11.0%, and the Pv-EC1 PRO: SV-BBM2 expression cassette showed an average rate of 44.1% HI frequency with a standard deviation of 10.3%, as shown in Table 3.

TABLE 3

Haploid Induction (HI) in T1 progeny of transgenic events with BBM polynucleotides				
Expression cassette	Number of independent transgenic events	Number of T1 seeds screened	Mean (% HI)	SD (% HI)
PvEC1 PRO::ZM-ODP2	8	336	43.1	12.2
PvEC1 PRO::ZM-BBM2	10	824	42.8	10.4
PvEC1 PRO::OS-ODP2	10	505	46.8	8.5

TABLE 3-continued

Haploid Induction (HI) in T1 progeny of transgenic events with BBM polynucleotides				
Expression cassette	Number of independent transgenic events	Number of T1 seeds screened	Mean (% HI)	SD (% HI)
PvEC1 PRO::OS-BBM2	10	326	49.7	18.6
PvEC1 PRO::SV-BBM1	10	234	36.5	11.0
PvEC1 PRO::SV-BBM2	10	658	44.1	10.3

[0330] In summary, BBM polynucleotides from *Zea mays*, *Oryza sativa* and *Setaria viridis* were demonstrated to induce parthenogenesis and produce viable haploid progeny when expressed in the egg cell of the wheat female gametophyte.

Example 2: Methods of Inducing Parthenogenesis in Wheat With Modified Enhanced Promoters

[0331] As shown in Example 1, the wheat TA-BBM3A polynucleotide sequence was operably linked to the ZM-DD45 PRO regulatory element, and demonstrated a low

netic maternal haploid induction levels. For example, expression modulating elements such as those disclosed in Publication No. WO2018/183878, incorporated herein by reference in its entirety, may be used in the methods of the present disclosure.

[0332] Expression cassettes encoding a TA-BBM3A operably linked to a regulatory element containing at least one EME are useful in the methods of the present disclosure. A modified ZM-DD45 PRO including at least one EME is used to alter mRNA transcription levels during female gametogenesis, thereby further improving and/or optimizing parthenogenic maternal haploid induction in comparison to the ZM-DD45 PRO: TA-BBM3A expression cassette as shown in Example 1.

[0333] For example, exemplary promoters are shown in Table 4. As shown in Table 4, the ZM-DD45 PRO (SEQ ID NO: 15) is modified using the ZM-AS2 EME (SEQ ID NO: 30) having one, two or three copies, shown as 1X ZM-AS2, 2X ZM-AS2, or 3X ZM-AS2, respectively, at varying positions. The EME location indicates the number of DNA base pairs upstream of the TATA box where each respective EME sequence is inserted. The TATA sequence used for each promoter is shown in the TATA box column.

TABLE 4

Exemplary promoters that include EME modification					
SEQ ID NO:	Regulatory Element	Promoter Source	EME	EME location	TATA box
15	ZM-DD45 PRO	ZM-DD45 PRO	not applicable	not applicable	TATTTATA
31	ZM-DD45 PRO: Z (1XM-AS2-20)	ZM-DD45 PRO	1X ZM-AS2	-20	TATTTATA
32	ZM-DD45 PRO: (2XZM-AS2-20)	ZM-DD45 PRO	2X ZM-AS2	-20	TATTTATA
33	ZM-DD45 PRO: (3XZM-AS2-20)	ZM-DD45 PRO	3X ZM-AS2	-20	TATTTATA
34	ZM-DD45 PRO: (2XZM-AS2-20): TATATATA	ZM-DD45 PRO	2X ZM-AS2	-20	TATATATA
35	ZM-DD45 PRO: (1XZM-AS2-20): TATAGCGC	ZM-DD45 PRO	1X ZM-AS2	-20	TATAGCGC
36	ZM-DD45 PRO: (2XZM-AS2-100)	ZM-DD45 PRO	2X ZM-AS2	-100	TATTTATA

frequency of parthenogenesis as measured by frequency of haploid progeny. In addition, alternative regulatory elements are used to further optimize parthenogenic maternal haploid induction levels. For example, regulatory elements such as those disclosed in US2015/0152430 (promoters including, but not limited to the AT-DD5 promoter, the AT-DD31 promoter, and the AT-DD65 promoter) and those disclosed in US2018/0094273 (*Zea mays* egg cell promoters) are used in the methods of the present disclosure (US2015/0152430 and US2018/0094273 incorporated herein by reference in their entireties). In addition, the further modification of regulatory elements to include an expression modulating elements (EME) are used to further optimize parthenoge-

[0334] The modified regulatory elements shown in Table 4 were used to alter mRNA transcription levels during female gametogenesis, thereby further improving and/or optimizing parthenogenetic maternal haploid induction in comparison to TA-BBM3A polynucleotide (SEQ ID NO:1) operably linked to a regulatory element comprising DNA fragments named ZM-DD45 PRO (SEQ ID NO: 15) as described in Example 1. Expression cassettes were prepared to evaluate parthenogenesis using the wheat TA-BBM3A polynucleotide sequence with the modified ZM-DD45 PRO regulatory sequences. In one example, a ZM-DD45 PRO: (1X ZM-AS2-20)::TA-BBM3A expression cassette contained a polynucleotide encoding a TA-BBM3A peptide

(SEQ ID NO: 1) operably linked to regulatory element ZM-DD45 PRO: (1X ZM-AS2-20) (SEQ ID NO: 31). In another example, a ZM-DD45 PRO: (2X ZM-AS2-100)::TA-BBM3A expression cassette contained a polynucleotide encoding a TA-BBM3A peptide (SEQ ID NO: 1) operably linked to regulatory element ZM-DD45 PRO: 2X ZM-AS2-100) (SEQ ID NO: 36).

[0335] Evaluation of haploid progeny demonstrate the ability of modified regulatory elements operably linked to the wheat TA-BBM3A polynucleotide sequence to induce parthenogenesis. The percent of haploid progeny (%) were scored for each unique event using the number of haploid progeny divided by the total number of progeny examined to determine the haploid induction (HI) frequency. Progeny from transgenic events containing expression cassettes ZM-DD45 PRO::TA-BBM3A, ZM-DD45 PRO: (1X ZM-AS2-20)::TA-BBM3A, and ZM-DD45 PRO: (2X ZM-AS2-100)::TA-BBM3A were evaluated by flow cytometry to determine the frequency of haploid progeny. As shown in Table 5, transgenic events showed an average rate of 28.5% HI frequency for the ZM-DD45 PRO: (1X ZM-AS2-20)::TA-BBM3A expression cassette, and an average rate of 4.2% HI frequency for ZM-DD45 PRO (2X ZM-AS2-100) PRO::TA-BBM3A expression cassette, relative to the average rate of 1.9% HI frequency for the unmodified ZM-DD45 PRO::TA-BBM3A (as shown in Example 1). Thus, modification to the regulatory elements operably linked to the BBM polynucleotide sequence can improve rate of parthenogenesis and haploid progeny.

TABLE 5

Haploid Induction (HI) in T1 progeny of transgenic events with modified regulatory element ZM-DD45 PRO			
Expression cassette	Number of independent transgenic events	Mean (% HI)	SD (% HI)
ZM-DD45 PRO::TA-BBM3A	6	1.94	4
ZM-DD45 PRO:(1X ZM-AS2-20):: TA-BBM3A	9	28.5	15.5
ZM-DD45 PRO:(2X ZM-AS2-100):: TA-BBM3A	9	4.2	6.7

[0336] The methods of the present disclosure may also use different promoters with or without the EME sequences, to improve haploid parthenogenesis as described herein.

Example 3: Identification of a Wheat Egg Cell Preferred Promoter

[0337] An egg cell-specific or -preferred expressed gene was identified in the wheat *Triticum aestivum* genomic DNA reference. Polynucleotides encoding regulatory element sequence were identified in the wheat genome on homoeologous chromosome 4 (SEQ ID NO: 37, 38, 39). The spatiotemporal pattern of gene expression can be visualized by operably linking DNA polynucleotide sequences encoding a TA-EGG regulatory sequences with a fluorescent protein, and observing fluorescent signal during female gamete development.

Example 4: Methods of Inducing Parthenogenesis with Wheat Egg Cell Promoters

[0338] As shown in Example 1, the wheat TA-BBM3A polynucleotide sequence was operably linked to the PvEC1

PRO regulatory element and demonstrated a high frequency of parthenogenesis as measured by frequency of haploid progeny. In addition, alternative regulatory elements are used to further optimize parthenogenic maternal haploid induction levels. DNA polynucleotide sequences encoding a TA-BBM3A and TA-BBM6A peptides were operably linked to native wheat promoters active during female gamete development, TA-EGG PRO and TA-EC PRO.

[0339] Expression cassettes were prepared to evaluate parthenogenesis using the wheat TA-EGG PRO (SEQ ID NO: 37) and TA-EC PRO (SEQ ID NO: 40) regulatory element sequences. In one example, the TA-EGG PRO: TA-BBM3A and TA-EGG PRO: TA-BBM6A expression cassettes contained a polynucleotide encoding either a TA-BBM3A or TA-BBM6A peptide (SEQ ID NO: 1, 7) operably linked to regulatory elements TA-EGG PRO (SEQ ID NO: 37). In another example, the TA-EC PRO: TA-BBM3A and TA-EC PRO: TA-BBM6A expression cassettes contained a polynucleotide encoding either a TA-BBM3A or TA-BBM6A peptide (SEQ ID NO: 1, 7) operably linked to regulatory elements TA-EC PRO (SEQ ID NO: 40).

[0340] These expression cassettes were designed and used as a parthenogenesis factor to induce egg cells to differentiate into embryo cells and to initiate embryogenesis. These expression cassettes were included in constructs for *Agrobacterium*-mediated transformation of the wheat line SBC0456D, resulting in random integration of T-DNA within the wheat genome. Leaf samples were collected from TO seedlings to determine copy number and transgene integrity of multiple different elements within the left border and right border of the T-DNA integrated into the plant genome.

[0341] Only plants determined as single copy (hemizygous) for the multiple different elements were characterized further. After transformation, each regenerated plant that was hemizygous TO, containing a single stably inserted T-DNA, was considered a unique event, grown to maturity, allowed to self-pollinate, and mature T1 seed was harvested. If parthenogenesis occurred, pollination may only entail the fertilization of the central cell, and initiate the normal development of endosperm, but not fertilization of the egg cell which has already initiated embryogenesis. During female gametogenesis, expression of the wheat BBM polynucleotides in or near the embryo sac cell, particularly the egg cell, driven by the egg-cell specific promoter stimulates embryogenesis and parthenogenesis of the egg cell. In bread wheat, progeny derived from parthenogenetic egg cells will be haploid (1n=3x, ABD) instead of diploid (2n=6x, AABBDD), and contain half the number of chromosomes and relative DNA content, as compared to the parental plant. The ploidy of individual plants in the T1 progeny was determined using flow cytometry analysis of nuclei isolated from leaf tissue.

[0342] Evaluation of haploid progeny demonstrate the ability of native wheat egg cell promoters combined with native wheat BBM polynucleotide sequences to induce parthenogenesis. HI frequency was determined. Evaluation of transgenic of events containing expression cassette TA-EGG PRO: TA-BBM6A had an average of 45.3% HI with a standard deviation of 12.4%, and transgenic events containing expression cassette TA-EGG PRO::TA-BBM3A had an average of 35.1% HI with a standard deviation of 9.5% (Table 6). Evaluation of transgenic of events containing

expression cassette TA-EC PRO::TA-BBM6A had an average of 46.6% HI with a standard deviation of 10.0%, and transgenic events containing expression cassette TA-EC PRO::TA-BBM3A had an average of 40.6% HI with a standard deviation of 18.7% (Table 6). Furthermore, multiple hemizygous T1 plants from four individual events containing expression cassette TA-EGG PRO::TA-BBM6A were grown to produce the T2 generation of progeny seed, and evaluated by flow cytometry. Stability of the parthenogenesis phenotype was observed as each individual event maintained a comparable HI frequency in the T1 and T2 generations.

TABLE 6				
Haploid Induction (HI) in T1 progeny of transgenic events with wheat egg cell promoters				
Expression cassette	Number of independent transgenic events	Number of T1 seeds screened	Mean (% HI)	SD (% HI)
TA-EGG PRO::TA-BBM3A	10	578	35.1%	9.5%
TA-EGG PRO::TA-BBM6A	10	568	45.3%	12.4%
TA-EC PRO::TA-BBM3A	8	539	40.6%	18.7%
TA-EC PRO::TA-BBM6A	7	614	46.6%	10.0%

[0343] Thus, expression cassettes containing native wheat promoter and BBM related polynucleotide sequences are sufficient to drive a high HI rate in a wheat system. Furthermore, the penetrance of the parthenogenesis phenotype shows that the wheat TA-EGG and TA-EC promoter sequences contain regulatory elements that are capable of driving expression in the egg cell. Though TA-BBM3A and TA-BBM6A only share 34.9% sequence homology, they were both able to drive a high HI rate in a wheat system.

Example 5: Methods of Inducing Parthenogenesis in Wheat With Modified BBM Peptides

[0344] Wheat polynucleotides encoding BBM peptides are linked to regulatory elements active in a haploid cell or tissue, for example a promoter active during female gamete development. To increase activity of BBM, the polynucleotide and peptide sequences of TA-BBM3A (SEQ ID NO: 1, 2) and TA-BBM6A (SEQ ID NO: 7, 8) are modified. An alignment of native and modified wheat BBM peptides is shown in FIG. 2. The PvEC1 promoter was operably linked to these modified wheat TA-BBM3A and TA-BBM6A sequences and screened for induction of parthenogenesis. [0345] TA-BBM3A (ALT1) (SEQ ID NO: 41, 42) is created by truncating the N- and C-terminal domains of TA-BBM3A (SEQ ID NO: 1, 2) to only contain a start codon, base pairs 154 through 2010, and a stop codon. It retains a start codon, and conserved motifs B, A, and AP2 DNA binding domains. In another variation, TA-BBM3A (ALT2) (SEQ ID NO: 43, 44) is created by truncating TA-BBM3A (SEQ ID NO: 1, 2) to leave a start codon, and base pairs 154-231, 442-504, and 684-2010, and a stop codon. TA-BBM6A (ALT1) (SEQ ID NO: 45, 46) is created by truncating the N- and C-domains of TA-BBM6A (SEQ ID NO: 7, 8) to only contain a start codon, and base pairs 328 through 1980, and a stop codon. In another variation,

TA-BBM6A (ALT2) (SEQ ID NO: 47, 48) is created by truncating TA-BBM6A (SEQ ID NO: 7, 8) to leave a start codon, base pairs 328-1908, and a stop codon.

TABLE 7					
Summary of Wheat TA-BBM3A and TA-BBM6A polynucleotide and peptide truncations					
DNA Polynucleotide	DNA SEQ ID NO:	DNA length (bp)	Peptide Name	PROTEIN SEQ ID NO:	Peptide length (aa)
TA-BBM3A	1	2109	TA-BBM3A	2	702
TA-BBM3A (154-2010)	41	1863	TA-BBM3A (ALT1)	42	620
TA-BBM3A (154-231)-(442-504)-(684-2010)	43	1473	TA-BBM3A (ALT2)	44	490
TA-BBM6A	7	1980	TA-BBM6A	8	659
TA-BBM6A (328-1980)	45	1656	TA-BBM6A (ALT1)	46	551
TA-BBM6A (328-1908)	47	1587	TA-BBM6A (ALT2)	48	528

[0346] Expression cassettes were prepared to evaluate parthenogenesis using the modified wheat TA-BBM3A sequence. In one example, a PvEC1 PRO: TA-BBM3A (ALT1) expression cassette contained a polynucleotide encoding a TA-BBM3A (ALT1) peptide (SEQ ID NO: 41) operably linked to regulatory elements PvEC1 PRO (SEQ ID NO: 19). In another example, a PvEC1 PRO: TA-BBM3A (ALT2) expression cassette contained a polynucleotide encoding a TA-BBM3A (ALT2) peptide (SEQ ID NO: 43) operably linked to regulatory elements PvEC1 PRO (SEQ ID NO: 19). Similarly, expression cassettes were prepared to evaluate parthenogenesis using the modified wheat TA-BBM6A sequence. In one example, a PvEC1 PRO: TA-BBM6A (ALT1) expression cassette contained a polynucleotide encoding a TA-BBM6A (ALT1) peptide (SEQ ID NO: 45) operably linked to regulatory elements PvEC1 PRO (SEQ ID NO: 19). In another example, a PvEC1 PRO: TA-BBM6A (ALT2) expression cassette contained a polynucleotide encoding a TA-BBM6A (ALT2) peptide (SEQ ID NO: 47) operably linked to regulatory elements PvEC1 PRO (SEQ ID NO: 19).

[0347] Evaluation of haploid progeny demonstrate the ability of modified BBM peptides, when operably linked to PvEC1 PRO regulatory element, to induce parthenogenesis. HI frequency was determined as previously described. Evaluation of transgenic events containing expression cassette PvEC1 PRO::TA-BBM3A (ALT1) had an average of 27.2% HI with a standard deviation of 9.8%, and transgenic events containing PvEC1 PRO::TA-BBM3A (ALT2) had an average of 0.0% HI with a standard deviation of 0.0% (Table 8). In addition, evaluation of transgenic events containing expression cassette PvEC1 PRO::TA-BBM6A (ALT1) had an average of 2.6% HI with a standard deviation of 3.1%, and transgenic events containing PvEC1 PRO::TA-BBM6A (ALT2) had an average of 0.0% HI with a standard deviation of 0.0% (Table 8).

TABLE 8

Haploid Induction (HI) in T1 progeny of transgenic events with wheat BBM polynucleotide and peptide truncations				
Expression cassette	Number of independent transgenic events	Number of T1 seeds screened	Mean (% HI)	SD (% HI)
PvEC1 PRO::TA-BBM3A (ALT1)	7	439	27.2%	9.8%
PvEC1 PRO::TA-BBM3A (ALT2)	7	474	0.0%	0.0%
PvEC1 PRO::TA-BBM6A (ALT1)	9	409	2.6%	3.1%
PvEC1 PRO::TA-BBM6A (ALT1)	9	491	0.0%	0.0%

[0348] In summary, minimal N- and C-terminal truncations of the wheat BBM polypeptides (ALT1 sequences) were sufficient to induce parthenogenesis and produce viable haploid progeny, but were less efficacious than the full length BBM polynucleotide sequence, when expressed in the egg cell of the wheat female gametophyte. However, further truncations of the N- or C-termini of the wheat BBM polypeptide between motifs B and A were not sufficient to induce parthenogenesis or haploid progeny.

Example 6: Method for Engineering a Native Parthenogenesis Locus in Wheat

[0349] To achieve optimal parthenogenesis phenotypes, promoters have been evaluated for driving TA-BBM egg cell-specific or -preferred expression operably linked to TA-BBM polynucleotides in transgenic wheat plants, as shown in Example 4.

A. Modifying an Endogenous Wheat BBM locus

[0350] Here we present a gene editing method to modify the promoter region of the endogenous BBM locus to attain the desired expression pattern for parthenogenesis phenotypes. In wheat, it is expected that modification of the promoter of a single homoeologous TA-BBM locus to include alternative egg cell-specific/preferred promoters or egg cell-specific/preferred sequence motifs, in addition to the native promoter sequence or in place of the native promoter sequence, would modify the expression pattern. This modified egg cell expression pattern of a single TA-BBM homeolog is expected to induce parthenogenesis in wheat.

[0351] The DNA reagents used during the modification of the native locus, including but not limited to the guideRNA, the Cas polypeptide, such as, Cas9, Cas12f (Cas-alpha, Cas14), Cas12i (Cas-beta), Cas12a (Cpf1), Cas12b (a C2c1 protein), Cas13 (a C2c2 protein), Cas12c (a C2c3 protein), Cas12d, Cas12e, Cas12g, Cas12h, Cas12i, Cas12j, Cas12k, Cas3, Cas3-HD, Cas 5, Cas6, Cas7, Cas8, Cas10, or combinations or complexes of these, and transformation selectable marker are not required for the function of the newly generated BBM allele and can be eliminated from the genome by segregation through breeding methods known to one skilled in the art. Because the promoter TA-EGG (SEQ ID NO: 37), having the same sequence as the endogenous TA-EGG locus, is inserted into the TA-BBM locus, for example the TraesCS3A02G395500 or the TraesCS6A02G229500 as described in the IWGSCv1.0 Chinese Spring wheat reference sequence assembly and anno-

tation, through homologous recombination, this TA-BBM allele is indistinguishable from natural mutant alleles.

[0352] To substitute the native promoter of TA-BBM with the TA-EGG promoter into the 5'UTR of the wheat TA-BBM gene, two guide RNAs (gRNA) can be prepared using the maize U6 promoter and terminator. One of the gRNAs can include the 5'-end of the gRNA containing a 20-bp variable targeting domain targeting the genomic sequence in the 5' upstream region of the TA-BBM promoter, and one of the gRNAs can include the 5'-end of the gRNA containing a 20-bp variable targeting domain targeting the genomic sequence in the 3' upstream region of the TA-BBM promoter, preferably near the ATG start codon. For example, potential gRNA targeting domain sequences for the TA-BBM3A promoter and 5'UTR sequence are described but not limited to sequences in Table 7 (SEQ ID NO: 49-56). A polynucleotide modification template containing the TA-EGG PRO will be flanked by two genomic DNA fragments (HR1 and HR2) derived and homologous to the upstream and downstream region of the genomic target sequence. Transformation can be initiated with constructs that include the gRNA, polynucleotide modification template, Cas cassette, such as a Cas9 cassette, and a transformation selectable marker. Transgenic plants that are produced by this method may be evaluated by PCR analysis to confirm the promoter swap by identifying PCR amplification products that include both the TA-EGG promoter sequence and the TA-BBM genomic coding sequence, but do not include amplification products for the of the TA-BBM promoter and the TA-BBM genomic coding sequence. Successful promoter swap events can be further characterized by sequencing the amplification products described above. It is expected that successful TA-EGG PRO: TA-BBM promoter swap events will induce parthenogenesis and result in haploid progeny as described in Example 4.

TABLE 9

Example of gRNA targeting domain sequences for the TA-BBM3A promoter region		
DNA Polynucleotide	SEQ ID NO:	Sequence
TA-BBM3A gRNA 1 (Nucleotide sequence of gRNA that targets wheat genomic BBM3A)	49	GTAATCCTAAAA TAATCCGTGGG
TA-BBM3A gRNA 2 (Nucleotide sequence of gRNA that targets wheat genomic BBM3A)	50	GAGTAAGTGGAC TCTCTGCTTGG
TA-BBM3A gRNA 3 (Nucleotide sequence of gRNA that targets wheat genomic BBM3A)	51	GCACGACATGTC CAAAGCGTAGG
TA-BBM3A gRNA 4 (Nucleotide sequence of gRNA that targets wheat genomic BBM3A)	52	GTAGTGTGCGACA CCAGTGTATGG
TA-BBM3A gRNA 5 (Nucleotide sequence of gRNA that targets wheat genomic BBM3A)	53	GGGCTAGTACAT CCAAATGTAGG

TABLE 9-continued

Example of gRNA targeting domain sequences for the TA-BBM3A promoter region			
DNA Polynucleotide	SEQ ID NO:	Sequence	
TA-BBM3A gRNA 6 (Nucleotide sequence of gRNA that targets wheat genomic BBM3A)	54	GGCTAGTACATC CAATGTAGGG	
TA-BBM3A gRNA 7 (Nucleotide sequence of gRNA that targets wheat genomic BBM3A)	55	GTTAAACCTTGT CAGGGATTCGG	
TA-BBM3A gRNA 8 (Nucleotide sequence of gRNA that targets wheat genomic BBM3A)	56	GTCCGACATAGG AGTGTTAATGG	

B. Modifying an Endogenous Wheat EGG CELL Locus

[0353] Here we present a gene editing method to modify the coding region of the endogenous EGG CELL locus to attain the desired expression pattern for parthenogenesis phenotypes. In wheat, it is expected that modification of the coding sequence of a single homoeologous TA-EGG locus to include alternative BBM coding sequence in place of the native coding sequence, would modify the expression pattern of a BBM polynucleotide. The introduction of a BBM coding sequence downstream of an egg cell-specific/preferred promoter is expected to induce parthenogenesis in wheat.

[0354] To substitute the native coding sequence of TA-EGG with the TA-BBM coding sequence into the 5'UTR of the wheat TA-BBM gene, two guide RNAs (gRNA) can be prepared using the maize U6 promoter and terminator. One of the gRNAs can include the 5'-end of the gRNA containing a 20-bp variable targeting domain targeting the genomic sequence in the 3' upstream region of the TA-EGG promoter, preferably near the ATG start codon, and one of the gRNAs can include the 5'-end of the gRNA containing a 20-bp variable targeting domain targeting the genomic sequence in the 3' region of the TA-EGG locus, preferably near the stop codon. A polynucleotide modification template containing the TA-BBM coding sequence will be flanked by two genomic DNA fragments (HR1 and HR2) derived and homologous to the upstream and downstream region of the genomic target sequence. Transformation can be initiated with constructs that include the gRNA, polynucleotide modification template, Cas cassette, such as a Cas 9 cassette, and a transformation selectable marker. Transgenic plants that are produced by this method can be evaluated by PCR analysis to confirm the coding sequence swap by identifying PCR amplification products that include both the TA-EGG promoter sequence and the TA-BBM genomic coding sequence, but do not include amplification products of the TA-EGG promoter and the TA-EGG genomic coding sequence. Successful promoter swap events can be further

characterized by sequencing the amplification products described above. It is expected that successful TA-EGG PRO: TA-BBM coding sequence swap events will induce parthenogenesis and result in haploid progeny as described in Example 4.

Example 7: Method for Producing Apomictic Plants

[0355] Apomixis is asexual reproduction resulting in progeny that are genetically identical to the parent. Methods disclosed herein are used to obtain an apomictic plant having an apomixis component with inhibited or mutated gene products that induce mitosis instead of meiosis, the MiMe phenotype, and containing a BBM parthenogenesis component as described in Examples 1, 2, 4-6.

[0356] In hexaploid wheat it is expected that MiMe phenotypes may require inhibition or mutation in each of the three homocologous gene loci. Genomic sequence for the coding region of the SPO11-1 homocologous gene loci (SEQ ID NO: 57, 58, 59) was identified from the IWGSCv1.0 Chinese Spring reference sequence and assembly. Genomic sequence for the coding region of the REC8 homocologous gene loci (SEQ ID NO: 60, 61, 62) was identified from the IWGSCv1.0 Chinese Spring reference sequence and assembly. Genomic sequence for the coding region of the OSD homocologous gene loci (SEQ ID NO: 63, 64, 65) was identified from the IWGSCv1.0 Chinese Spring reference sequence and assembly.

[0357] Natural mutations, or mutations induced by chemicals or radiation, in the individual homocologous loci of the MiMe genes may be identified, and then crossed together to create individual MiMe gene knockout lines. The individual MiMe gene knockout lines for Spo11, Rec8, Prd1, Prd2, Prd3 are expected to be both male and female sterile, and individual MiMe gene knockout lines for Osd1 are expected to increase the ploidy of the gametes. However, the combination of mutations to create a MiMe phenotype may be created by crossing plants that are heterozygous for alleles of the MiMe loci. In this example, mutations with knockout or knockdown alleles of MiMe loci may be combined with BBM parthenogenesis components to achieve an apomictic plant, e.g. a flowering plant, in the absence of egg cell fertilization comprising with clonal gametes and progeny.

[0358] Another example to improve the process to create knockout or knockdown alleles in the MiMe loci is to utilize gene editing components. Methods of the present disclosure use a gene editing trait comprising a first expression cassette encoding a CRISPR-Cas9 gene editing polynucleotide (or other Cas polypeptide) and a second expression cassette encoding gRNA molecules having sequence homology to the MiMe genes. For example, gRNA targeting domain sequences (SEQ ID NO: 66-73) for the TA-SPO11-1, TA-REC8 and TA-OSD genomic coding sequences are described but not limited to sequences shown in Table 8. It is expected that mutations at the MiMe gene target sites provide methods of obtaining a non-recombination and non-reduction. Alternatively, the methods of the present disclosure use a gene editing trait comprising a first expression cassette encoding a Cas alpha gene editing polynucleotide and a second expression cassette encoding gRNA molecules having sequence homology to the MiMe genes. It is expected that mutations at the MiMe gene target sites provide methods of obtaining a non-recombined and non-reduced gamete.

TABLE 10

Example of gRNA targeting domain sequences for the TA-SPO11-1, TA-REC8 and TA-OSD genomic coding region		
DNA Polynucleotide	SEQ ID NO:	Sequence
TA-SPO11-1 gRNA 1 (Nucleotide sequence of gRNA that targets wheat genomic SPO11-1)	66	GCCCCAAGCCATCGTATCTC
TA-SPO11-1 gRNA 2 (Nucleotide sequence of gRNA that targets wheat genomic SPO11-1)	67	GTGTTCTTTTACCTCCATTC
TA-REC8 gRNA 1 (Nucleotide sequence of gRNA that targets wheat genomic REC8)	68	GAGACTCAAATTATCGACC
TA-REC8 gRNA 2 (Nucleotide sequence of gRNA that targets wheat genomic REC8)	69	GTCGAGAGTAATATGGACTG
TA-REC8 gRNA 3 (Nucleotide sequence of gRNA that targets wheat genomic REC8)	70	GATGACTGATATACATTTCC
TA-REC8 gRNA 4 (Nucleotide sequence of gRNA that targets wheat genomic REC8)	71	GCTCTAAAAGGCGTCAAGTC
TA-OSD gRNA 1 (Nucleotide sequence of gRNA that targets wheat genomic OSD)	72	TGCTGGATCCTCAGGCTCCG
TA-OSD gRNA 2 (Nucleotide sequence of gRNA that targets wheat genomic OSD)	73	GAATCCACCGCCAGAGCGAT

[0359] Constructs used for gene editing processes may include an additional cassette containing a BBM parthenogenesis component to achieve an apomictic plant with clonal gametes and progeny. Constructs used for gene editing or parthenogenesis processes may also be transformed into lines that are previously transformed or gene edited to achieve an apomictic plant with clonal gametes and progeny. Alternatively, gene edits with knockout or knockdown alleles of the MiMe loci may be combined with BBM parthenogenesis components by crossing to achieve an apomictic plant with clonal gametes and progeny. In this example, gene edits with knockout or knockdown alleles of MiMe loci may be combined with BBM parthenogenesis components to achieve an apomictic plant with clonal gametes and progeny.

Example 8: Method for Engineering a Native Parthenogenesis Locus in Maize

[0360] To achieve optimal parthenogenesis phenotypes, promoters are evaluated for driving ZM-BBM or ZM-ODP2 egg cell-specific or -preferred expression operably linked to ZM-BBM2 or ZM-ODP2 polynucleotides in transgenic maize plants.

A. Modifying an Endogenous Maize BBM Locus

[0361] Here we present a gene editing method to modify the promoter region of the endogenous maize BBM2 locus or ODP2 locus to obtain the desired expression pattern for parthenogenesis phenotypes. In maize, it is expected that modification of the promoter of a single homologous ZM-BBM2 or ZM-ODP2 locus to include alternative egg cell-specific/preferred promoters or egg cell-specific/preferred sequence motifs, in addition to the native promoter sequence or in place of the native promoter sequence, would modify the expression pattern. This modified egg cell expression pattern of a single ZM-BBM2 or ZM-ODP2 homeolog is expected to induce parthenogenesis in maize.

[0362] The DNA reagents used during the modification of the native locus, including but not limited to the guideRNAs, a Cas endonuclease (such as Cas9, Cas12f (Cas-alpha, Cas14), Cas12i (Cas-beta), Cas12a (Cpf1), Cas12b (a C2c1 protein), Cas13 (a C2c2 protein), Cas12c (a C2c3 protein), Cas12d, Cas12e, Cas12f, Cas12g, Cas12h, Cas12i, Cas12j, Cas12k, Cas3, Cas3-HD, Cas 5, Cas6, Cas7, Cas8, Cas10, or combinations or complexes of these), and transformation selectable marker are not required for the function of the newly generated BBM allele and can be eliminated from the genome by segregation through standard breeding methods. For example, the promoter ZM-EGG PRO (SEQ ID NO: 86) is inserted upstream of the start codon of the endogenous ZM-BBM2 genomic locus or endogenous ZM-ODP2 genomic locus. In another example, the promoter ZM-DD45 (SEQ ID NO: 15) is inserted upstream of the start codon of the endogenous Zm-BBM2 or ZM-ODP2 genomic locus. In either of these cases, one gRNA could be used near the start codon or in the 5'UTR to insert the promoter sequence, or two gRNAs could be used to first excise out the native ZM-BBM2 or ZM-ODP2 promoter before insertion of the egg cell promoter. Because the promoter ZM-EGG PRO, having the same sequence as the endogenous ZM-EGG locus, is inserted into the ZM-BBM2 or ZM-ODP2 locus for example, through homologous recombination, this allele is indistinguishable from natural mutant alleles.

[0363] To substitute the native promoter of ZM-BBM2 with the ZM-EGG promoter (SEQ ID NO: 86) into the 5'UTR of the maize ZM-BBM2 gene, two guide RNAs (gRNA) can be prepared using the maize U6 promoter and terminator. One of the gRNAs can include the 5'-end of the gRNA containing a 20-bp variable targeting domain targeting the genomic sequence in the 5' upstream region of the ZM-BBM2 promoter, and one of the gRNAs can include the 5'-end of the gRNA containing a 20-bp variable targeting domain targeting the genomic sequence in the 3' upstream region of the ZM-BBM2 promoter, preferably near the ATG start codon. For example, potential gRNA targeting domain sequences for the ZM-BBM2 promoter and 5'UTR sequence are described but not limited to sequences in Table 9 (SEQ ID NO:91, 92, 93, 94, 95, 96, 97, and 98).

TABLE 11

Example of gRNA targeting domain sequences for the ZM-ODP2 and ZM-BBM2 promoter regions.		
DNA Polynucleotide	SEQ ID NO:	Sequence
ZM-ODP2 PRO gRNA 1 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> ODP2 promoter)	91	TGTTTCATACT TACGCGGTGA

TABLE 11-continued

Example of gRNA targeting domain sequences for the ZM-ODP2 and ZM-BBM2 promoter regions.		
DNA Polynucleotide	SEQ ID NO:	Sequence
ZM-ODP2 PRO gRNA 2 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> ODP2 promoter)	92	GTTACCTGTT CATACTTACG
ZM-ODP2 PRO gRNA 3 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> ODP2 promoter)	93	CCAGTTGTTC ACAGTGGCCA
ZM-ODP2 PRO gRNA 4 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> ODP2 promoter)	94	AGCGAGCCAG TGTTTCACAG
ZM-BBM2 PRO gRNA 1 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> BBM2 promoter)	95	TATGCACTTG TCGGCGCACA
ZM-BBM2 PRO gRNA 2 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> BBM2 promoter)	96	CACTTGTCGG CGCACACGGC
ZM-BBM2 PRO gRNA 3 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> BBM2 promoter)	97	TGCCTTGCTT GTGATCAGAG
ZM-BBM2 PRO gRNA 4 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> BBM2 promoter)	98	GGGCGTTGAT AGAACAGTGA

[0364] A polynucleotide modification template containing the ZM-EGG PRO will be flanked by two genomic DNA fragments (HR1 and HR2) derived and homologous to the upstream and downstream region of the genomic target sequence. Transformation can be initiated with constructs that include the gRNA, polynucleotide modification template, Cas cassette and a transformation selectable marker. Transgenic plants that are produced by this method may be evaluated by PCR analysis to confirm the promoter swap by identifying PCR amplification products that include both the ZM-EGG promoter sequence and the ZM-BBM2 genomic coding sequence. Successful promoter swap events can be further characterized by sequencing the amplification products described above. It is expected that successful ZM-EGG PRO: ZM-BBM2 promoter swap events will induce parthenogenesis and result in haploid progeny.

[0365] In another example, a similar strategy can be applied for inserting ZM-DD45 PRO or ZM-EGG PRO into the ZM-ODP2 genomic locus.

TABLE 2

Example of gRNA targeting domain sequences for the ZM-EGG and ZM-DD45 regions.		
DNA Polynucleotide	SEQ ID NO:	Sequence
ZM-EGG gRNA 1 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> Egg promoter)	99	GGAGCCCTTC CTTCCCGCCA
ZM-EGG gRNA 2 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> Egg promoter)	100	GCGTCATCAG CACC GCGAGG
ZM-EGG gRNA 3 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> Egg promoter)	101	ATCTAAGACC AGTGAAGACT
ZM-EGG gRNA 4 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> Egg promoter)	102	TGGGATGATG AATGGAGCTC
ZM-DD45 gRNA 1 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> DD45 promoter)	103	CGTACGTACA GGAGCACACC
ZM-DD45 gRNA 2 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> DD45 promoter)	104	AGCACACCAG GCACACGGCAA
ZM-DD45 gRNA 3 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> DD45 promoter)	105	GGTACCAGCT CACCCCTCA
ZM-DD45 gRNA 4 (Nucleotide sequence of gRNA that targets <i>Zea mays</i> DD45 promoter)	106	AGCTGGCTGC CGTAGCCGTG

B. Modifying an Endogenous Maize EGG CELL Locus

[0366] Here we present a gene editing method to modify the coding region of the endogenous EGG CELL locus to attain the desired expression pattern for parthenogenesis phenotypes in maize. In maize, it is expected that modification of the coding sequence of a single homoeologous ZM-EGG locus to include alternative BBM2 or ODP2 coding sequence in place of the native coding sequence, would modify the expression pattern of a BBM polynucleotide. The introduction of a BBM2 or ODP2 coding sequence downstream of an egg cell-specific/preferred promoter is expected to induce parthenogenesis in maize.

[0367] To substitute the native coding sequence of ZM-EGG with the ZM-BBM coding sequence into the 5'UTR of the maize ZM-BBM2 gene, two guide RNAs (gRNA) can be prepared using the maize U6 promoter and terminator. One of the gRNAs can include the 5'-end of the gRNA containing a 20-bp variable targeting domain targeting the genomic sequence in the 3' upstream region of the ZM-EGG promoter, preferably near the ATG start codon, and one of the gRNAs can include the 5'-end of the gRNA containing a 20-bp variable targeting domain targeting the genomic sequence in the 3' region of the ZM-EGG locus, preferably

near the stop codon. A polynucleotide modification template containing the ZM-BBM coding sequence will be flanked by two genomic DNA fragments (HR1 and HR2) derived and homologous to the upstream and downstream region of the genomic target sequence. Transformation can be initiated with constructs that include the gRNAs, polynucleotide modification template, Cas endonuclease cassette and a transformation selectable marker. Transgenic plants that are produced by this method can be evaluated by PCR analysis to confirm the coding sequence swap by identifying PCR amplification products that include both the ZM-EGG promoter sequence and the ZM-BBM2 coding sequence. Successful coding sequence swap events can be further characterized by sequencing the amplification products described above. It is expected that successful ZM-EGG PRO: ZM-BBM2 coding sequence swap events will induce parthenogenesis and result in haploid progeny.

[0368] In another example, similar to above, but a single gRNA is used to create a DNA break between the promoter

of an egg promoter such as ZM-EGG or ZM-DD45 and the coding sequence of ZM-EGG or DD45. A coding sequence, either genomic or CDS of ZM-ODP2 or ZM-BBM2 with their terminators are introduced and integrates into the cut site. In these examples, the coding sequence of the ZM-EGG or ZM-DD45 is not eliminated from the genome but is separated from its functional promoter. Instead, the functional egg promoter is driving expression of either ZM-ODP2 or ZM-BBM2. Transgenic plants that are produced by this method can be evaluated by PCR analysis to confirm the coding sequence swap by identifying PCR amplification products that include both the ZM-EGG promoter sequence and the ZM-BBM2/ZM-ODP2 coding sequence. Successful coding sequence swap events can be further characterized by sequencing the amplification products described above. It is expected that successful ZM-EGG PRO: ZM-BBM2, ZM-DD45: ZM-BBM2, ZM-EGG: ZM-ODP2, and ZM-EGG: ZM-ODP2 coding sequence swap events will induce parthenogenesis and result in haploid progeny.

SEQUENCE LISTING

Sequence total quantity: 122

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 mol_type = other DNA
 organism = Triticum aestivum

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 organism = Triticum aestivum

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 organism = Triticum aestivum

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 organism = Triticum aestivum

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 organism = Triticum aestivum

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gcggaggcct	acgacatcgc	ggcgatcaag	ttccgtgggc	tgaacgccc	caccaacttc	1320
gacatgagtc	gctacgagct	caagaccatc	ctcgagagca	gcaccctgcc	gggtgggtggc	1380
gcggcgaggc	gccccaaagga	ggctcgctgat	catccagagg	ctggcgccac	gatctggcgg	1440
gcgggcattg	acggcgcgct	catctcccag	ctgaccgacg	tcgggatggg	cacgtacgcg	1500
tcgtaccacc	acggctggcc	aaccatcgca	ttccagcagc	cctcgccgct	gtcgggtgca	1560
taccatacgc	tcgcccagcc	gtctcgccgg	tgggtgcaagc	ccgagcagga	cgccagcgct	1620
gccacgcaaa	gcctgcagga	cctccagcag	ctgcacctcg	ggaccgccc	tcacaactat	1680
ttccaggcat	cgtcgagctc	cacggtctac	aacggcgccg	ggtagccagca	aggcaccagt	1740
ggcaacgcct	tcttgatgcc	ggctagcacc	gtggtggatg	aacaggggca	cagcagcact	1800
gccaccaacc	aggggaagcag	atgcagctac	ggggacgaag	aggggaagct	caatctcggg	1860
tacgatgcca	tggcgatggc	gagcacccgc	gcgacccat	acgcgcgccg	gccaactgtg	1920
agcattgcga	gggcgaacgg	gtactccaac	aactggagct	cgccgttcaa	tggcatgtga	1980

SEQ ID NO: 8 moltype = AA length = 659
 FEATURE Location/Qualifiers
 source 1..659
 mol_type = protein
 organism = Triticum aestivum

SEQUENCE: 8

MASGNDWLGF	SLSGQENPQP	HQDSSPPADI	DIAGASGYFG	LPTQPAPDAQ	LGVPGHHPQS	60
YGITTEAFNRG	AHEIHDWMMR	GDLDYNGGAS	ELSMVLVGSSA	VGGKRAAMEA	ETEPKLEDFL	120
VGNSFVSEQD	QAGGFLFSGV	PMATSTNSNS	RSNTMELSMI	KSWLRDNQVP	QAPHPQSQTG	180
APAQQPQPHE	EMGTDASSFD	PLGRNGALVV	AGSSQSLSLS	MSTGSGSHLP	MAVLGGSPSG	240
GVSESTSSEN	KRASGAMDSP	GGAVEAVARK	SIDTFGQRTS	IYRGVTRHRW	TGRYEHLWD	300
NSCRREGQSR	KGRQVYLGYY	DKEDKAARAY	DLAALKYWGT	TTTTNFPIINT	YEKEVDEMKG	360
MTRQEIYIAYL	RRNSSGFSRG	ASKYRGVTRH	HQHGRWQARI	GRVAGNKDLY	LGTFSSTEEEA	420
AEAYDIAAIK	FRGLNAVTFN	DMSRYDVKTI	LESSTLPVGG	AARRPKKEVAD	HPEAGATIWR	480
AGMDGGVISQ	LTDVGMGTIA	SYHHGWPTIA	FQQPSPLSVH	YPYVAQPSRG	WCKPEQDASV	540
ATQSLQDLQQ	LHLGTAAHNY	FQASSSSTVY	NGGGYQQGTS	GNAFLMPAST	VVDEQGHST	600
ATNQGSTCSY	GDEEGKLNLG	YDAMAMASTG	ADPYAAAPT	SIARANGYSN	NWSSPFNGM	659

SEQ ID NO: 9 moltype = DNA length = 1980
 FEATURE Location/Qualifiers
 source 1..1980
 mol_type = other DNA
 organism = Triticum aestivum

SEQUENCE: 9

atggcttctg	gaaacgactg	gttgggggttc	tcgctctccg	gccaaagaaa	cccacagccg	60
catcaggata	gctcgccgcc	cgcggcgatc	gacatctccg	gcgccagtga	cttctatggc	120
ctaccaccgc	agccagcccc	cgacgctcaa	cttggcgtgc	caggccatca	gccttcgtac	180
gggatcacgg	aggccttcac	tcatgaaacc	cacgattgga	acatgagggg	tgatctggac	240
tataacggcg	gtgcgtccga	gctctcgatg	ctcgtcgggt	cgagcgcagt	cgccggcaag	300
agggcgccga	tggagggccga	gaccgagccc	aagcttgagg	acttctccg	tggaactcgc	360
ttcgtctccg	agcaagatca	ggcggcgccg	ttcttgttct	ccggagtcgc	gatggcgacc	420
agcggcaaca	gcaacagcgg	gagcaacacc	atggagctct	ctatgatcaa	gagctggctc	480
cgtaacaacc	aggtacccca	agcggccgac	ccgcagtcgc	ataccggggc	gcccgtgcag	540
cagccgcagc	cccacgagga	gatgagcacc	gacgcaagcg	cgagcagctt	cgatgcgctg	600
gggagaaacg	gcgcgttggt	ggtggcgggg	agctcgacga	gcctggcgct	ctcgatgagc	660
acgggctcag	gatcgacttt	gccgatggct	gcggtcggcg	cgagtgcacc	tgggggagtc	720
tcagagagca	ctcctcgga	gaacaagcgg	gccagcgcca	ccatggattc	gccggcggtg	780
gccgtagaag	ctgtcgcgag	gaagtccatc	gacacattcg	ggcaaaaggc	gtcgatatat	840
cgaggtgtta	caagacatag	atggacaggg	cgttatgagg	ctcatctctg	ggacaatagc	900
tgcaagaagag	aagggcagag	tcgcaagggt	aggcaagggtg	gctatgacaa	ggaggacaag	960
gcagcaaggg	cttatgattt	ggcagcactc	aagtattggg	gtacaacaac	aacaacaaat	1020
ttcccaatta	atacctacga	aaaagagggt	gatgaaatga	aacatagac	taggcaggaa	1080
tacatcgcat	acctaagaag	gaatagcagc	ggattttctc	gtggtgcgtc	aaaatatcgt	1140
gggtgacta	ggcaccatca	gcatgggaga	tggcaagcaa	gaattggcg	gggtgcagga	1200
aacaaggatc	tctacttagg	caccttcagc	accgaggagg	aggcgccgga	ggcctacgac	1260

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atcgcgccga	tcaagttccg	tgggctcaat	gccgtcacca	acttcgacat	gagtcgctac	1320
gacgtcaaga	ccatocctcca	gagcagcacc	ctaccgggtg	cggtgagtg	cgcgccgagg	1380
cgctcaag	aggccgctga	tcattccagag	gctggcgcca	cgatctggcg	ggccgcatg	1440
gatggcgccg	tcattctccc	gctgaccgac	gtcgggatgg	gcgcttacgc	gtcgtaccac	1500
cacggctggc	cgacaatcgc	attccagcag	ccgtcgccgc	tctcggtgca	ctaccatac	1560
gtcgcccgag	cgtctcgccg	gtgggtgcaag	cccgagcagg	acgccaccgt	cgccacgcaa	1620
aaacctgcagg	acctccagga	gctgcacctc	gggatcgccg	ctcacaacta	tttccaggca	1680
tcgtcgagct	ccacgggtcta	caacggcgcc	ggcgccggcg	ggtaccagca	gggcacgggt	1740
ggcaacgcct	tcctgatgcc	ggctagcgcc	gtcgtggacg	aacaggggca	cagcagcact	1800
gccaccaacc	aggggaagcag	ctgcagctac	ggggacgaag	aagggaagct	caatattggg	1860
tacgacgcca	tgccgagctgc	gagcaccggc	gccgacccat	acgcgccggc	gtcaacgggt	1920
agcatcgcca	gggcgaacgg	gtactccaac	aactggagct	cgccgttcaa	tggcatgtga	1980

SEQ ID NO: 10 moltype = AA length = 659
 FEATURE Location/Qualifiers
 source 1..659
 mol_type = protein
 organism = Triticum aestivum

SEQUENCE: 10

MASGNDWLGF	SLSGQENPQP	HQDSSPPAGI	DISGASDFYG	LPTQPAPDAQ	LGVPGHQPSY	60
GITEAFTHET	HDWMMRGDLD	YNGGASELSM	LVGSSAVGGK	RAAMEAETEP	KLEDPLGGNS	120
FVSEQDQAGG	FLFSGVPMAT	SANSNSGNT	MELSMIKSWL	RNNQVPQAPH	PQSHTGAPVQ	180
QPQPHEEMST	DASASSFDAL	GRNGALVVAG	SSQSLALSMS	TGSGSHLPMA	AVGGSATGGV	240
SESTSENENR	ASGTMDSFGG	AVEAVARKSI	DTFGQRTSIY	RGVTRHRWTG	RYEAHLWDNS	300
CRREGQSRKG	RQGGYDKEDK	AARAYDLAAL	KYWGTTTNTN	FPINTYEKEV	DEMKHMTREQ	360
YIAYLRNNS	GFSRGASKYR	GVTRHHQHGR	WQARIGRVAG	NKDLYLGTFS	TEEEAAEAYD	420
IAAIKFRGLN	AVTNFDMRSY	DVKTIQSST	LPVAVSGAAR	RLKEAADHPE	AGATIWRAGM	480
DGGVISQLTD	VGMGAYASYH	HGWPTIAFQQ	PSPLSVHYPY	VAQPSRGWCK	PEQDATVATQ	540
NLQDLQELHL	GIAAHNYFQA	SSSSTVYNGG	GGGGYQQGTG	GNAFLMPASA	VVDEQGHSSST	600
ATNQGSTCSY	GDEEGKLNIG	YDAMAMASTG	ADPYAAASTV	SIARANGYSN	NWSSPFNGM	659

SEQ ID NO: 11 moltype = DNA length = 1986
 FEATURE Location/Qualifiers
 source 1..1986
 mol_type = other DNA
 organism = Triticum aestivum

SEQUENCE: 11

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gtaccacgcg	gcagcgtccc	cgacgctcaa	ctcgcggtgc	caggctcatca	gccttcatac	180
gggatcacgc	aggacttcaa	taggggagct	catgaaaccc	acgattggaa	catgaggggt	240
gatctggact	ataacggcgg	tcgctccgag	ctctcgatgc	tcgtcggtgc	gagcgcagtc	300
ggcgccaaaga	ggcgccgcat	ggaggccgag	accgagccca	agcttgagga	cttctcgggt	360
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atgaccacca	gcagcaaacg	caacacgggg	agcaaacacca	tggagctctc	tatgatcaag	480
agctgggtcc	gtgacaacca	gctaccccaa	gcgcgcgacc	cgagctcgca	gaccggggcg	540
cccgcgcagc	agccgcaacc	ccacgaggag	atgggcaccc	acgcgagcag	cttcgatccg	600
cgggggagaa	acggcgcggt	ggtggtggcg	gggagctcgc	agagcctggc	gctctcgatg	660
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ggtcggttag	aagctgtcgc	gaggaagtcc	atcgacacat	tcgggcaaa	gacgtcgata	840
tatcgaggtg	taacaagaca	tagatggaca	ggcggttatg	aggctcatct	ctgggacaat	900
agctgcagaa	gagaagggca	gagtcgcaag	ggccgacaag	tttatcttgg	tggctatgac	960
aaggaggaca	aggcagcaag	ggcttatgat	ttggcagcac	tcaagtattg	gggtacaaca	1020
acaaactacaa	atttcccaat	taatacctac	gaaaaagagg	tggatgaaat	gaaacatatg	1080
actaggcagg	aatatatcgc	atacttaaga	aggaatagca	gcggattttc	tcgtggtgcg	1140
tcaaaatc	gtggtgtaac	taggcacccat	cagcatggga	gatggcaagc	aagaattggg	1200
cgggttgacg	gaaacaaaga	tctctactta	ggcaccttca	gcaccgagga	ggaggcgccg	1260
gaggcctacg	acatcgcgcc	gatcaagttc	cgcggggtca	acgcgctcac	caacttcgac	1320
atgagtcgct	acgacgtcaa	gaccatcctc	gagagcagca	ccctgcccgt	gggtggcgcg	1380
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taccaccacg	gctggccgac	catcgatttc	cagcagccgt	cgccgctctc	ggtgcactac	1560
ccatagctcg	cccagccgcc	ccgcccgggtg	tgcaagcccg	agcaggagcg	caccgtcgcc	1620
acgcaaaacc	tcaggagctg	ccagcagctg	ccctcgggga	ccgcgcgtca	caactatttc	1680
caggcatcgt	cgagctcaac	ggtctacaac	ggcgccggcg	tcggccggga	ccagcagggc	1740
accggtggca	acgcctcttc	gatgcgggct	agcacccgtg	tggatgaaca	ggggcacagc	1800
agcactggcg	ccaaccaggg	agctacgggg	acgaagaggg	gaagctcaat		1860
cttgggtacg	atgccatggc	gatggcgagc	accggcgccg	accatacgc	ggcgcggtca	1920
actgtgagca	tcgcgagggc	gaacgggtac	tccaacaact	ggagctcgcc	gttcagtggc	1980
atgtga						1986

SEQ ID NO: 12 moltype = AA length = 661
 FEATURE Location/Qualifiers
 source 1..661
 mol_type = protein

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organism = Triticum aestivum

SEQUENCE: 12
MASGNDWLGF SLSGQENPQP HQDSPPPAGI DISGASDFYG VPTQPAPDAQ LGVPGHQPSY 60
GITEDFNRRGA HETHDWNMRG DLDYNGGASE LSMLVGSSAV GKKRAAMEAE TEPKLEDFLG 120
CNSFVSEQDQ AGGFLFSGVP MTTSSNSNTG SNTMELSMIK SWLRDNQLPQ APHPQSQTGA 180
PAQQQPQPHEE MGTDASSFDP RGRNGALVVA GSSQSLALSM STGSGSHLPM AVVSGSASGG 240
VSESTSENK RASGAMDSPG GAVEAVARKS IDTFGQRTSI YRGVTRHRWT GRYEAHLWDN 300
SCRREGQSRK GRQVYLGGYD KEDKAARAYD LAALKYWGTT TTTNFPINTY EKEVDEMCKHM 360
TRQEIYIAYLR RNSSGFSRGA SKYRGVTRHH QHGRWQARIG RVAGNKDLYL GTFSTEEEA 420
EAYDIAAIKF RGLNAVTNFD MSRYDVKITL ESSTLPVGGG ARRLKEADH PEAGATIWRA 480
GMDGAIISQL TDVGMGAYAS YHHGWPTIAF QQPSPLSVHY PYVAQPPRGW CKPEQDATVA 540
TQSLQDLQQL HLGTAAHNYF QASSSSTVYN GGGVGGYQQG TGGNAFLMPA STVVDEQGH 600
STANQGSTC SYGDEEGKLN LGYDAMAMAS TGADPYAAS TVSIARANGY SNNWSPPFSG 660
M

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SEQ ID NO: 13      moltype = DNA length = 2133
FEATURE           Location/Qualifiers
source            1..2133
                  mol_type = other DNA
                  organism = Zea mays

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SEQUENCE: 13
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tgcttcaaca tcccccaaga ttggagcatg aggggacag agctttcgcc gctcgtcgcg 180
gagccgaagc tggaggactt cctcgccggc atctccttct cgcagcagca tcacaaggcc 240
aactgcaaca tgataccacg cactagcagc acagtgtgtc acgcgagctc aggtgctagc 300
accggctacc atcaccagct gtaccaccag cccaccagct cagcgctcca cttcgcgga 360
tcgtaaatgg tggcctcttc ggcgggtgtc cagcagcgcg gtgcatgtc cagcgcgggc 420
gccgtcaacg cgctcgctgg cgctgccagt gccaacggcg gcggcatcgg gctgtccatg 480
attaagaact ggtcgcgagg ccaaccggcg cccatgcagc cgaggggtgg gccggctgag 540
ggcgcgacgg ggctctcttt gtccatgaac atggcgggga cgaccaagg cgctgctggc 600
atgccacttc tcgctggaga gcgcgcacgg gcgcccgaga gtgtatcgac gtcagcacag 660
ggtggagccg tcgctgctac ggcgcggaag gaggatagcg gtggcagcgg tgttgccggc 720
gctctagtag ccgtgagcag ggcacgggtt ggcagcgcg cgcgctcgcc tgacaacacg 780
gcaaggaaga cgggtggcac gttcgggcag cgcacgtcga tttaccgttg cgtgacaagg 840
catagatgga ctggggagata tgaggcacat ctttgggata acagtgtcag aaggggaagg 900
caaacctcgt aaggtcgtca agtctattta ggtggtatg ataagagga gaaagctgct 960
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gtgagtaact acgaaaagga gctcgaggac atgaagcaca tgacaaggca ggagtttgta 1080
gcgtctctga gaaggaagag cagtggtttc tccagaggtg catccattta caggggagtg 1140
actaggcatc accaacatgg aagatggcaa gcacggattg gacgagttgc agggaaacaag 1200
gatctttact tgggcacctt cagcacccag gaggaggcag cggaggcgta cgacatcgcg 1260
gcgatcaagt tcgcggcgct caacgccgtc acatgagccg ctacgacgtg 1320
aagagcatcc tggacagcag cgcctctccc atcggcagcg ccgccaagcg cctcaaggag 1380
gccgagggcg cagcgctcgc gcagcaccac cagccggcgg ttggtgagcta cgacgtcgcc 1440
cgcatcgctc cctcgctcgg cagcggcgga gccctggcgg cgcggtacgg cgcgcactac 1500
cacggcgccg cctggcgagc catcgcggtc cagccggggc ccgccaagcag aggcctgtac 1560
caccgctacg cgcagcagcc aatgcgcggc ggcgggtggt gcaagcagga gcaggaccac 1620
gcggtgtagc agcgccgcga cagcctgcag gacctccacc acctgaacct gggcgcgccc 1680
ggcgcgacag actttttctc ggcagggcag caggcccgcc cgctgcgat gcacggcctg 1740
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ggcggggtcg gcgacagcaa gcgcgccagc gccgtcgcg gcagtggcgg tggctacatg 1860
atgccgatga gcgctgcggg agcaaccact acatcggcaa tgggtgagcca cgagcaggtg 1920
catgcacggg cctacgacga agccaagcag gctgctcaga tggggtagca gactacactg 1980
gtgaacggcg agaacaatgg tggcggaagg atgtctgcat gggggactgt cgtgctgca 2040
gcgcggggcg cagcagcaag aacaaacgac aacatggccg ccgacgtcgg ccatggcgcg 2100
gcgcagctct tcagtgtctg gaacgacact taa 2133

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SEQ ID NO: 14      moltype = AA length = 710
FEATURE           Location/Qualifiers
source            1..710
                  mol_type = protein
                  organism = Zea mays

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SEQUENCE: 14
MATVNNWLLAF SLSPQELPPS QTTDSTLISA ATADHVSGDV CFNIPQDWSM RGSELSALVA 60
EPKLEDFLGG ISFSEQHHKA NCNMIPTSS TVCYASSGAS TGYHHQLYHQ PTSSALHFAD 120
SVMVASSAGV HDGGAMLASAA AANGVAGAAS ANGGGIGLSM IKNWLRSQPA PMQPRVAAA 180
GAQGLSLSMN MAGTTQGAAG MPLLAGERAR APESVSTSAQ GGAVVVTAPK EDGGSGVAG 240
ALVAVSTD TG GSGGASADNT ARKTVDTFGQ RTSIYRGVTR HRWTGRYEAH LWDNSCRREG 300
QTRKGRQVYL GGYDKEEKAA RAYDLAALKY WGATTTTNFP VSNYEKELED MKHMTREFFV 360
ASLRKSSGF SRGASIVRGV TRHHQHGRWQ ARIGRVAGNK DLYLGTFTSQ EEAAEAYDIA 420
AIKFRGLNAV TNFDMRYDV KSILDSSALP IGSAAKRLKE AEAASQAHH HAGVVSVDVG 480
RIASQLGDGG ALAAAYGAHY HGAAWPTIAF QPGAASTGLY HPYAQQPMRG GGWCKQE QDH 540
AVIAAAHSLQ DLHHLNLGAA GAHDFFSAGQ QAAAAAMHGL GSIDSASLEH STGSNSVVYN 600
GGVDSNGAS AVGGSGGSGYM MPMSAAGATT TSAMVSHQV HARAYDEAKQ AAQMGYESYL 660
VNAENNGGGR MSAWGTTVSA AAAAAASSND NMAADVGHGG AQLFSVWNDT 710

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SEQ ID NO: 15      moltype = DNA length = 1358
FEATURE           Location/Qualifiers
source            1..1358
                  mol_type = genomic DNA
                  organism = Zea mays

SEQUENCE: 15
acacaggacc aagaacttga agatgcattt gaaggccttt atcttgttga ctcccaaggg 60
ccctagactt tgtaatcttg cattttgtgct ctgctgatct ggtctgatac tgatgtaact 120
gatcaatgaa ctaattgtat tagaactgga ttgtactctt ttttcccttt atatggtttt 180
ctcataaggg gagtttttac ctagaagggt ttttaataag acagccattg cacaacacgc 240
tataatattt tatttaaggt ctatgagact gactccgtgt gtgctactgc ctactggcta 300
ctactatctg tgaatttgtg acctgtgaac tttgaaatgt gaaatttgtg acttgagaac 360
tatgatttta tgacatatga agttgtgaac tgtgtatttg atacctgtgt gaatttatga 420
cctatttagg ccttggtcgt ttacaccaat ccagctctgg attgacatgg attggaatta 480
aatacatgtc acaatctatg tcccaaaata atccaagcct actcattttt ttatttggtt 540
aaacccatca tagattataa cccaaggatt taggaaattt ttaactatg gaagacatga 600
attctattca tagcttatta ggtatggaat aaatccatga atatttgca caagtttata 660
ttagaattca tgaatcaaaa gaataactag ttttgagaga tacatggatt aaatggtaga 720
tttaactcga ctatgggatt gagtgtgata tatggattta ttcaatccaa atccggatta 780
aatccatggt ggatctatat atattggtgt gctcttagct cggttgtgta ggtgggcat 840
gtttgacgtg ccgagctggc acgatcggac cttttaccgc tgcctgtctc gtgcaagggg 900
tgttgccgtg caggagcgac cgtgagttaa tcggactcaa ttggaccgga ctctcggat 960
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tgccgaatgac attcttgcat cctcctcac caatcaaggc ggcaacatac cgccctttg 1080
gccttccatg aacatgaacg cggcggaacg ccacgcccgc gtgcactact cacttgcagt 1140
aattcgccgc ccaactcacg cgccaacca acttgattgc acgcactacc atcaattcgc 1200
cgcccgccgc atcccttcac ccagctgcta ttatacgcc tcgcccgcgt ccagctcag 1260
cagaaccacc agtccctcac tccatcttct actccgacca caaccacagc gaccacgacc 1320
gtgcacgtac gtacatcagc acaccaggca acggcacc 1358

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SEQ ID NO: 16      moltype = DNA length = 1299
FEATURE           Location/Qualifiers
source            1..1299
                  mol_type = genomic DNA
                  organism = Panicum virgatum

SEQUENCE: 16
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ttggcaacag tgtataaatg aaagcagcag tgtgcagggt acttctaata ttgaagtgtc 120
caatcaaaat catgaggaag gcaaccaaaa tagtgaaggc caaattactc ctattcgaag 180
catttgtgtc gaagatgcag atactgaaag atcagaaaagg gaaataggtc tgaccctcc 240
tgaggagacc tcagcattgc cggatgagca gcttaacacg aaggtggagt gccacgaagc 300
tgaaacgaac ctgtgctatg atgaggacac ttcgagctta tttgatactg aatcgctgta 360
cagcaaaagca tccagtttgc acgaagacac tcgtaaggat acttctccaa gtgatctatc 420
tgctccgaga tctcccgagg aatctacagt tttctgaac tctctgttcc cggtatcagt 480
aggatcattg gctgattctg attttgttca cgttgctctg acataatgct ttgttcaatc 540
ttttattgta gtatctgtga gagcctcctg actacatctt tcattttgct ttttcattag 600
gaatctgtcg aagcagcagg agaagaggta tagacgaact ccgcccgaag ctgcagagtt 660
tcaaatgtct cagcactgta aaaggaaagt acattgctat gagtgcacct cgcccgaagc 720
aaggtgacaa tctgagccaa tctgcaatcg cgttgctccg gaacagcgag aacgcacctg 780
ctgttaaagt aggccatcct ggcaagccgg accctgatct tccggctgac aagcgtgca 840
gccatcagtg ggaagatcaa gggatcactg ataggcagta gggcgtaacg cgtagtttat 900
tattgtggac agttagcgtt tgtgctcaac attagtgtgt ggtgggtggg ttcaaccttt 960
ggctgaaggt tattaggtgt gcatctgacc ggaactatc cgtgtgtttg cgcagcgtgt 1020
agtgcgaagc ctgctctgtg catttgttat tagcagttaa catattgttg gtgtgggtgc 1080
cgtgggtgaa atgttaaagca actggcagcc gtacagttag ttaattgggt atgcaggcgt 1140
ttggtttttg cttttttgta atcctgttat cgtgaacaga tccgttgtgc ccctgtctgt 1200
tattcgtaca cgtgctcact tgtttaatct cttgcaatgg cttaccatg tactgggtcc 1260
agaagacagg cgaggagcat ggctattcgg aacagaaga 1299

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SEQ ID NO: 17      moltype = DNA length = 100
FEATURE           Location/Qualifiers
source            1..100
                  mol_type = genomic DNA
                  organism = Panicum virgatum

SEQUENCE: 17
gccgcaggtc gtttggcggt ccatgagcag acactgcagt gagtggatag caatcaatag 60
cgatgctctt tggccttcca tgaaacgcac gcactcatgt 100

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SEQ ID NO: 18      moltype = DNA length = 163
FEATURE           Location/Qualifiers
source            1..163
                  mol_type = genomic DNA
                  organism = Panicum virgatum

SEQUENCE: 18
cgccgccac aatccaagtc ggtgtacccc cgaatcgctg atcatattaa ttccggccgc 60

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ctagcgctgc	tatatataga	gagacctcgc	ggcactccac	tcggatcctg	caaggetgca	120
acgcacacac	ttccggcgga	ctcgatccgt	ccaagtcgca	gca		163

SEQ ID NO: 19 moltype = DNA length = 1398
 FEATURE Location/Qualifiers
 source 1..1398
 mol_type = genomic DNA
 organism = Panicum virgatum

SEQUENCE: 19

cctttatgta	ttgtgaattg	cctcatgact	ctgtttttaca	tcaggacctt	gtcagactag	60
ttggcaacag	tgtataaatg	aaagcagcag	tgtgcagggtg	acttctaata	ttgaagtgtc	120
caatcaaaat	catgaggaag	gcaaccaaaa	tagtgaaggc	caaattactc	ctattcgaag	180
caattgtgtc	gaagatgcag	atactgaaag	atcagaaaag	gaaataggtc	tgacccctcc	240
tcgaggacgc	tcagcattgc	cggatgagca	gcttaacacg	aaggtggagt	gccacgaagc	300
tgaacgcaac	tggtgctatg	atgaggacac	ttcgagctta	tttgatactg	aatcgctgta	360
cagcaagaca	tccagtttgc	acgaagacac	tcgtaaggat	acttctccaa	gtgatctatc	420
tgctccgaga	tctcccgagg	aatctacagt	ttttctgaac	tcttctgttc	cgggatcagt	480
aggatcattt	gctgattctg	attttgttca	cgttgctctg	acataatgtc	ttgttcaatc	540
ttttattgta	gtatctgtga	gagcctcctg	actacatctt	tcattttgct	tttctattag	600
gaatctgtcg	aagcagcagg	agaagaggta	tagacgaact	ccgcgcgaag	ctgcagagtt	660
tcaaggtctc	cagcactgta	aaaggaagct	acattgctat	gagtgccctc	cgcccggaagc	720
aaggtgacaa	tctgagccaa	ctcgcaatcg	cgttgctccg	gaacagcgag	aacgcacctg	780
ctgttaaaat	aggccatcct	ggcgaagccg	accctgatct	tccgggtcgac	aagcgctgca	840
gcccacagtc	ggaagatcaa	gggatcctga	taggcagtag	ggcgtagcgc	gtagtttatt	900
attgtggaca	gttagcgttt	gtgctcaaca	ttagttgggt	gtggtgggtg	tcaacctttg	960
gctgaaggtt	attaggtgtg	catctgaccg	gaaactatcc	gtgtgtttgc	gcagcgtgta	1020
gtgacgaagc	tgctctgtgc	atttgttatt	agcagttaac	atattgtggg	tggtggtggc	1080
gtggtggaaa	gtttaagcaa	ctggcagccg	tacagttggg	taattggtta	tgacggcggt	1140
tggttttggc	ttttttgtaa	tctgtttatc	gtgaacagat	ccgtttgtgc	ccgtgctggt	1200
attcgtacac	gtgctcactt	gtttaacttc	ttgcaatggc	ttacccatgt	actgggtcca	1260
gaagacaggg	gaggagcagc	gctattcgga	acagaagagc	cgcaggtcgt	ttggcggtcc	1320
atgagcagac	actgcagtga	gtggatagca	atcaatagcg	atgctctttg	gccttccatg	1380
aaacgcacgc	actcatgt					1398

SEQ ID NO: 20 moltype = DNA length = 2040
 FEATURE Location/Qualifiers
 source 1..2040
 mol_type = other DNA
 organism = Zea mays

SEQUENCE: 20

atggcttcag	cgaacaactg	gctgggcttc	tcgctctcgg	gccaggataa	cccgagcct	60
aaccaggaca	gctcgctgc	cgcgggtatc	gacatctccg	gcgcccagca	cttctatggc	120
ctgcccacgc	agcagggttc	cagcgggcat	ctcggcgctg	cgggcctcgc	ggacgatcac	180
gcttcttatg	gtatcatgga	ggcctacaac	agggttcctc	aagaaaccca	agattggaac	240
atgaggggct	tggactacaa	cggcggtggc	tcggagctct	cgatgcttgt	gggggtccagc	300
ggcgcgcgcg	ggggcaacgc	caagaggggc	gtggaaagaca	gcgagcccaa	gctcgaagat	360
tccctcgccg	gcaactcggt	cgtctccgat	caagatcagt	ccggcggtta	cctgttctct	420
ggagtcocga	tagccagcag	cgcacaatgc	aacagcggga	gcaacaccat	ggagctctcc	480
atgatcaaga	cttggctacg	gaacaaccag	gtggcccagc	cccagccgcg	agctccacat	540
cagccgcagc	ctgaggaat	gagcaccgac	gccagcggca	gcagctttgg	atgctcggat	600
tcgatgggaa	ggaacagcat	ggtggcggtc	ggtgggagct	gcgagagcct	ggcgctctcg	660
atgagcacgg	gctcgcaact	gcccattggt	gtgcccagcg	gcgcccagcg	cggagcggcc	720
tcggagagca	catcgctcga	gaacaagcga	gcgagcgggt	ccatggattc	gcccggcagc	780
gcggtagaag	ccgtaccgag	gaagtccatc	gacacgttcg	ggcaaaaggac	ctctatata	840
cgaggtgtta	caaggcatag	atggacaggg	cggatagagg	ctcatctatg	ggataaatag	900
tgtagaaggg	aagggcagag	tcgcaagggg	aggcaagtat	acttaggtgg	ctatgacaag	960
gaggacaagg	cagcaagggc	ttacgatttg	gcagctctca	agtattgggg	cactacgaca	1020
acaaacaaat	tccctataag	caactacgaa	aaagagctag	aagaaatgaa	acatatgact	1080
agacaggagt	acattgcata	cctaagaaga	aatagcagtg	gattttctcg	tggggcatca	1140
aagtatcggt	gagtaactag	acatcatcag	catgggagat	ggcaagcaag	gatagggaga	1200
gttgacagaa	acaaggatct	ttacttgggc	acattcagca	ccgagaggga	ggcgccggag	1260
gcctacgaca	tcgcccgcgt	caagttccgc	ggtctcaacg	ccgtcaccac	cttcgacatg	1320
agccgctacg	acgtgaagag	catcctcgag	agcagcacac	tgctctgctg	cgggtcgggc	1380
aggcgctcca	aggacgctg	ggaccacgtg	gaggccggcg	ccaccatctg	gcgcgcccag	1440
atggacggcg	ccgtgatctc	ccagctggcc	gaagccggga	tggcgcgcta	cgctcgtac	1500
ggccaccacg	gctggccgac	catcgcgctc	cagcagccgt	cgccgctctc	cgtccactac	1560
cgtacggccc	agcgtcccg	cgggtgggtg	aaacccgagc	aggacgcggc	cgccgcccgc	1620
gcgcacagcc	tcgaggacct	ccagcagctg	cacctcgcca	gcgcggccca	caactctctc	1680
caggcgctcgt	cgagctccac	agtctacaac	ggcgggcgccg	gcgccaagtg	tggctaccag	1740
ggcctcggtg	gtggcagctc	ttctctcatg	ccgtcgagca	ctgtcgtggc	ggcgccggac	1800
caaggggcaca	gcagcagggc	agcacgtgca	gctacgggga	cgaccaccag		1860
gaggggaagc	tcatcggtta	cgacgcggcc	ccgcagctgg	tggagaccgc		1920
tacgctcgcg	cgaggaacgg	ctaccagttc	tcgcagggct	cgggttccac	ggtgagcatc	1980
gcgagggcga	acgggtacgc	taacaactgg	agctctcctt	tcaacaacgg	catgggggtga	2040

SEQ ID NO: 21 moltype = AA length = 679

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FEATURE
source Location/Qualifiers
1..679
mol_type = protein
organism = Zea mays

SEQUENCE: 21

MASANNWLGF	SLSGQDNFQP	NQDSSPAAGI	DISGASDFYG	LPTQQGSDGH	LGVPGLRDDH	60
ASYGIMEAYN	RVPQETQDWN	MRLDYNNGG	SELSMLVGSS	GGGGNGKRA	VEDSEPKLED	120
FLGGNSFVSD	QDQSGGYLFS	GVPIASSANS	NSGNTMELS	MIKTWLRNNO	VAQPQPPAPH	180
QPQPEEMSTD	ASGSSFGCSD	SMGRNSMVA	GGSSQSLALS	MSTGSHLPMV	VPSGAASGAA	240
SESTSSSENK	ASGAMDSPGS	AVEAVPRKSI	DTFGQRTSIY	RGVTRHRWTG	RYEAHLWDNS	300
CRREGQSRKG	RQVYLGGYDK	EDKAARAYDL	AALKYWGTTT	TTFNPISNIE	KELEEMKHMT	360
RQEIYIAYLR	NSSGFSRGAS	KYRGVTRHHQ	HGRWQARIGR	VAGNKDLYLG	TFSTEEAAE	420
AYDIAAIKFR	GLNAVTFNFM	SRYDVKSILE	SSTLPVGGAA	RRLKDAVDHV	EAGATIWRAD	480
MDGAVISQLA	EAGMGGYASY	GHHGWPTIAF	QQPSPLSVHY	PYGQPSRGWC	KPEQDAAAAA	540
AHSLQDLQLQ	HLGSAAHNFF	QASSSSTVYN	GGAGASGGYQ	GLGGSSFLM	PSSTVVAAD	600
QGHSTANQG	STCSYGDDHQ	EGKLIGYDAA	MVATAAGGDP	YAAARNGYQF	SQSGSSTVSI	660
ARANGYANMW	SSPFNMGMG					679

SEQ ID NO: 22 moltype = DNA length = 2088

FEATURE
source Location/Qualifiers
1..2088
mol_type = other DNA
organism = Oryza sativa

SEQUENCE: 22

atggccacta	tgaacaactg	gctcgccttc	tcgctctcgc	cgcaggacca	actcccaccg	60
tcgcagacca	atagcactct	catctccgct	gctgcaacca	ccacaaccgc	aggcgattcg	120
tcaacggggc	acgtctgctt	caacatccct	caagactggt	ccatgcgcgg	aagcgagctt	180
agcgtctcgc	tcgcggagcc	caagttggag	gatttcttgg	gaggcatctc	cttctcggag	240
caacagcatc	atcacgcggg	aaagggcggt	gttatcccaa	gctctgctgc	cgcagtctat	300
gcaagctccg	gctccagcgt	gggctacctc	tacctccgcg	cttcatctcc	gtcacttcag	360
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ggtggcggtg	tggtctcagc	agcagcggct	gcagcagctt	cgggtaaatg	cgggattggc	480
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agcctgtcga	tgaacatggc	tggtactact	accgctcaag	gtggaggcgc	aatggcactt	600
ctcgagggcg	ctggcgaaag	aggaaggacc	acaccagcat	ccgagagctc	ctctacttcc	660
gcgcacggag	ccaccacggc	tacaatggct	ggcgggagga	aagagatcaa	cgaggaagga	720
tctggttcgc	ctggtgcgct	ggttgcaagt	ggctcagaat	caggtggttc	cggcgctggt	780
gttgaagctg	gtgcgcgtgc	ggcagcggct	cgggaagagc	ttgatacttt	cggccaaaga	840
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tgggacaaca	gctgtcgccg	caggggccaa	actaggaagg	gaagacaggt	ctatctagga	960
ggatatgaca	aagaggagaa	ggctgcccga	gcgtacgacc	tggccgcggt	gaagtactgg	1020
ggtccaacaa	cgacgaccaa	cttcccgggt	aacaactacg	agaaggagct	ggaagagatg	1080
aagcacatga	cgcggcagga	gttcgtcgct	tctctcaggc	gcaagtcatc	tggtttctcc	1140
agaggtcgct	cgatctatag	aggagttacc	cgccaccacc	agcacggaag	gtggcaggca	1200
agaatcggga	gagtcgcggg	taacaaggac	ctgtacttgg	gaacctcttc	gactcaggag	1260
gaggcagcgg	aagcgtatga	cattgcggcg	atcaagtctc	gcggtctcaa	tgccgtgacc	1320
aaacttcgaca	tgtcacgcta	tgatgtcaag	tcgattctgg	atagcgtgcg	gttgctgtgt	1380
ggaaacggctg	ccaacgcctc	caaggacgcg	gaagcagctg	ccgcgtacga	tgttgccagg	1440
attgcctcac	atctcgtgtg	agatggagct	tacgctgccc	actacgggca	tcataccacc	1500
tctgcagccg	cagcttggtc	tacaatagca	ttccaagcgg	cagcggcttc	tcctccacac	1560
gctgctggtc	tttaccatcc	gtacgcgcaa	cctctccgct	gttggtgtaa	gcaggaaaca	1620
gatcatgcgg	tgattgcggc	tgcacacagc	ttgcaagatc	tgcatcacct	caatctggga	1680
gccgcagcag	tgccccatga	cttcttctca	caagccatgc	agcagcagca	tggcctgggc	1740
agcatagaca	atgcgtctct	ggagcactcc	accggatcga	actcggtggt	gtacaatgga	1800
gacaacggcg	gaggaggtgg	aggttacatc	atggcaccta	tgtcagcggt	ctctgctacc	1860
gctacggcgg	tggcctcatc	ccacgaccac	ggtggagacg	gcggcaagca	ggtccaaatg	1920
ggctacgact	cctaccttgt	gggagctgac	gcttacggcg	gaggaggagc	tggctcgcag	1980
cctagctggg	ccatgacgac	tgcttctgct	cctgcggcta	cgagctcgct	ggatatgaca	2040
ggagtgtgtc	atggcgccca	actgttctcg	gtgtggaatg	atacatag		2088

SEQ ID NO: 23 moltype = AA length = 695

FEATURE
source Location/Qualifiers
1..695
mol_type = protein
organism = Oryza sativa

SEQUENCE: 23

MATMNNWLAF	SLSPQDQLPP	SQTNSTLISA	AATTTTAGDS	STGDVCFNIP	QDWSMRGSEL	60
SALVAEPKLE	DFLGGISFSE	QQHHHGKGG	VIPSSAAACY	ASSGSSVGYL	YPPSSSSSLQ	120
FADSVMVATS	SPVVAHDGVS	GGGMVSAAAA	AAASGNGGIG	LSMIKNWLRS	QPAPQPAQAL	180
SLSMNMAGTT	TAQGGGAMAL	LAGAGERGRT	TPASELSLTS	AHGATTATMA	GGRKEINEEG	240
SGSAGAVVAV	GSESGGSGAV	VEAGAAAAAA	RKSVDTFGQR	TSIYRGVTRH	RWTGRYEHL	300
WDNSCRREGQ	TRKGRQVYLG	GVDKEEKAAR	AYDLAALKYW	GPTTTTNPFP	NNYEKELEEM	360
KHMTREQEFA	SLRRKSSGFS	RGASIYRGVT	RHHQHGRWQA	RIGRVAGNKD	LYLGTFTSTQE	420
EAAEAYDIAA	IKFRGLNAV	NFDMSTRYDV	SILDSALPVP	GTAARKLKDA	EAAAAYDVGR	480
IASHLGGDGA	YAAHYGHHHH	SAAAAWPTIA	FQAAAAPPHP	AAGLYHPYAO	PLRGWCKQEQ	540
DHAVIAAAHS	LQDLHLHLNL	AAAAAHDFFS	QAMQQQHGLG	SIDNASLEHS	TGNSNVVYNG	600

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DNCGGGGGYI MAPMSAVSAT ATAVASSHDH GGDGGKQVQM GYDSYLVGAD AYGGGGAGRM 660
 PSWAMTPASA PAATSSSDMT GVCHGAQLFS VWNNT 695

SEQ ID NO: 24 moltype = DNA length = 2112
 FEATURE Location/Qualifiers
 source 1..2112
 mol_type = other DNA
 organism = Oryza sativa

SEQUENCE: 24
 atggcttctg caaacaactg gctgggcttc tcgctctccg gccaaagagaa tccgcagcct 60
 caccaggata gctcgctcc gccagccatc gacgtctccg gcgcggcgga cttctatggc 120
 ctgcccagct cgcagccgac gccggccgac gcgcacctcg gcgtggcggg gcatcatcac 180
 aacgcctcgt atggcatcat ggaggccttc aataggggag ctcaagaggc acaagattgg 240
 aacatgaggg ggtggacta caacggcgcc gcctcggagc tgtcgatgct cgtcggctcc 300
 agcggcgcca agagggcggc gccggtggag gagaccgagc cgaagctgga ggacttcttc 360
 ggcggaact cgttctctc cgagcaagat catcacgcgg cgggggggctt cctcttctcc 420
 ggcttccga tggccagcag caccacacgc aacagcggga gcaacactat ggagctctcc 480
 atgatcaaga cctggctccg gaacaacggc caggtgcccc cgggccacca gccgcagcag 540
 cagcagccgg cgcccgccgg cgcccgccgg cagcagcagg cgcacgaggg gccggagatg 600
 agcaccgagc cgagcggcag cagcttcggg tgcctctccg acgcgatggg gaggagtaac 660
 aacggcgccg cggtctcggc gccggcgccg gggacgagct cgcagagcct gccgctctcg 720
 atgagcacgg cctgcactc gcacctgcct atcgtcgtcg ccggcgccgg gaacgccagc 780
 ggcgagcgg ccgagagcac atcgtcggag aacaagcggg ccagcggcgc catggattcg 840
 ccggcggtg gcgcgataga gccgtgccc aggaagtcca tcgacacgtt cgggcaagg 900
 acctcgatat atcgaggtgt aacaaggcat agatggacag gccgatatga ggctcatctc 960
 tgggataata gctgtagaag agaagggcag agtcgcaagg gtaggcaagt ttatcttgg 1020
 ggctatgaca aggaggataa agcagcgaga gcttatgatt tggcagctct gaagtattgg 1080
 ggcacaacaa caacaacaaa ttccaata agtaactatg aaaaagagct agatgaaatg 1140
 aaacatatga caggcgagga gtatttgca tacctaagaa ggaatagcag tggattttct 1200
 cgtggtgcat cgaatatatc tgggttaacc aggcaccatc agcatgggag atggcaagca 1260
 aggataggga ggggtgcaag aaacaaggac ctctacttag gcaccttcag caccaggag 1320
 gagggcgccg aggcgtacga catcgccggc atcaagttcc gggggctcaa cgcgcgcacc 1380
 aactttgaca tgagccgcta cgcagtcagg agcatcctg agagcagcac gctgccggtg 1440
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 atctggcgcc ccgcgcacat ggaaggcgcc gccgctcatc ccggcctggc cgcagctcgg 1560
 atggggcgct acgcccctc gtaccaccac caccaccacc acggctggcc gaccatcgcc 1620
 ttccagcagc gcccgccgct cgcctgacg tacccgtacg gccaggcgcc gccggcgccg 1680
 tcgcgcgggt ggtgcaagcc cgagcaggac gccgcgctc ctcgcgcgcc gcacagcctc 1740
 caggacctcc agcagctgca cctcgccagc gccgcgcgcc acaactctt ccaggcgctc 1800
 tcgagctcga cgggtctaaa cggcgccggc ggccgggtatc agggcctcgg tggcaacgcc 1860
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 catggaaaca cctgcagcta cggcaacgag gagcagggga agctcatcgg gtacgacgcc 1980
 atggcgatgg cgagcggcgc gcccgccggc ggggtatcag tgtcgagggg ctcggcgctc 2040
 acggtagaca tcgcgagggc gaacggctac tcggccaact ggagctcgcc ttccaatggc 2100
 gccatgggat ga 2112

SEQ ID NO: 25 moltype = AA length = 703
 FEATURE Location/Qualifiers
 source 1..703
 mol_type = protein
 organism = Oryza sativa

SEQUENCE: 25
 MASANNWLGF SLGQENPQP HQDSSPPAAI DVSGAGDFYG LPTSQPTAAD AHLGVAGHHH 60
 NASYGIMEAF NRGAQEAQDW NMRGLDYNGG ASELMLVGS SGGKRAAAVE ETEPKLEDFL 120
 GGNSFVSEQD HHAAGGFLFS GVPMASTNS NSGNTMELS MIKTWLRNNG QVPAGHPQQ 180
 QQPAAAAAAA QQQAHEAAEM STDASASSFG CSSDAMGRSN NGGAVSAAA GTSSQSLALS 240
 MSTGSHSHLP IVVAGGNGAS GGAESTSSE NKRASGAMDS PGGGAIEAVP RKSIDTFGQR 300
 TSIYRGVTRH RWTGRYEHL WDNCRREGQ SRKGRQVYLG GYDKEDKAAR AYDLAALKYW 360
 GTTTTTFNFI SNYEKELDEM KHMTRQEYIA YLRNSSGFS RGASKYRGVT RHHQHGRWQA 420
 RIGRVAGND LYLGTFTSTEE EAAEAYDIAA IKFRGLNAV NFDMSRYDVK SILESTLPV 480
 GGAARRLKEA ADHAEAAAGAT IWRADMDGA GVISGLADVG MGAYAASYHH HHHHGWP TIA 540
 FQQPPPLAVH YPYGQAPAAP SRGWCKPEQD AAVAAAAHSL QDLQQLHLGS AAHNFFQAS 600
 SSSTVYNGG GGYQLGGNA FLMPASTVVA DQGHSTATN HGNTCSYGN EQGKLIGYDA 660
 MAMASGAAGG GYQLSQGSAS TVSIARANGY SANWSSPFNG AMG 703

SEQ ID NO: 26 moltype = DNA length = 2097
 FEATURE Location/Qualifiers
 source 1..2097
 mol_type = other DNA
 organism = Setaria viridis

SEQUENCE: 26
 atggctactg tgaacaactg gctggcttcc tccctctccc cgcaggatct gccgccttcc 60
 cagacggatt ccacctctat ctccgccacc gccaccgagc aggtctccgg cgactctctc 120
 ttcaacatcc cccaagattg gagcatgagg ggatcggaac tctcagcgct agtcgcccag 180
 ccgaagctgg aggaacttct cggcgccatc aacttctcgg agcagacca caaggccaac 240
 ctcaacgtga tccccagcag tagtaacgct tgctacgcga gctcaggcgc tagcaccggc 300

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taccaccagc	tgtaccacca	ccaaagctcc	gcgctccatt	tcgctgactc	cgtcatggtg	360
gcctcctccg	ccggcggtcca	tgacggcggt	gccagcatgc	tcagtgcagc	ggccacogtc	420
aatggcgcg	ctggcgagc	cagtgccaac	ggcggcagca	tcgggctgtc	catgatcaag	480
aactggctgc	ggagtgcaac	agctccgccc	ctgcccgcgc	cgagagtagt	ggcgggccgca	540
gagggcgcg	aggcgagcga	ggggctttct	ttgtccatga	acatggcagg	ggcgcaaggc	600
gctggcatgc	cgcttctcgc	cggagagcga	ggcggggcgc	ccgagagcgt	gtccacatca	660
gcacaggggtg	gagcagtcgc	cgctcggaag	gaggatagcg	gtggcagcgg	cggtgccggc	720
gcctagtag	ccgtgagcac	agacacgggt	ggcagcgcca	cggttgctga	gacggcgggc	780
aggaagacgg	tagacacgtt	cggccagcgc	acgtcgattt	accgcggcgt	gaccaggcat	840
agatggacag	ggaggtacga	ggctcatctg	tgggacaaca	gttcgagaag	ggaaggacaa	900
actcgcaagg	gtcgtcaagt	ctatctagg	ggttatgaca	aagaggagaa	agctgctagg	960
gcttatgatt	tggctgctct	taagtattgg	ggtcccacga	cgaccacaaa	ttttccagt	1020
agtaactacg	aaaaggagct	ggaggaatg	aagcacatga	caaggcagga	gtttgtagcg	1080
tctctgagaa	aggagagcag	tggcttctcc	agaggtgcat	ccattttacag	aggagtaact	1140
aggcatcacc	aacacgggag	atggcaagca	cggattggac	gagttgcagg	gaacaaggat	1200
ctctacttgg	gcaccttcag	cacgcaggag	gaggcagcag	aagcgtacga	catcgcgggc	1260
atcaagttcc	gcgggctcaa	cgcctgcacc	aacttcgaca	tgagccgcta	cgactggaag	1320
agcatcctgg	acagcagcgc	gctcccccac	ggcagcgccc	ccaagcgcc	caaggaggcc	1380
gagggcgcg	cgtccgcgca	gcacgcccgc	gtggtgagct	acgacgtagg	ccgcatcgcc	1440
tcgcacctgc	gcgacggcgg	cgccttgcca	gcctacggca	cgcactacca	cgacggcgca	1500
gcgccttggc	cgacccatgc	gttccagcgc	agcgcggccc	ccgcggcgcc	gtaccaccgc	1560
tacggcgagc	cgatgcgtgg	gtggtgcaag	caggagcagg	atcacgcggt	gatcgccgcc	1620
gcgcacagcc	gtcacagcgt	caaccacctg	aacctggggc	ccgcggcgcc	tgcgacgcac	1680
ttcttctcgg	cgggggcagg	cgcgatgcac	ggcctcggca	gcctcgacaa	cgctcactc	1740
gagcatagca	cgggtcccaa	ctccgtcgtg	tacaatgggg	ttggagacag	caacgggtggc	1800
gcgctcggcg	ccggtctacat	gatgccgatg	agcgttcggc	cggcgaccac	gacggcaaatg	1860
gttagtcacg	agcaggtgca	tgcgcgggtc	catcaagggg	agcacgacga	agccaccaag	1920
cagggcgcg	agatggggta	cgagagctac	ctggtgaacg	cggaggcgcc	ctacgggtggc	1980
gggggagatg	cttctgggac	gcgggcctcg	gccccggcag	cggcaagcag	caacgacaac	2040
atggccggcg	ttggccatgg	cggcgcgag	ctcttcagtg	tctggaatga	cacataa	2097

SEQ ID NO: 27 moltype = AA length = 698
 FEATURE Location/Qualifiers
 source 1..698
 mol_type = protein
 organism = Setaria viridis

SEQUENCE: 27

MATVNNWLAF	SLSPQDLPPS	QTDSTLISAT	ATDEVSGDVC	FNIPQDWSMR	GSELSALVAE	60
PKLEDPLGGI	NFSQHHKAN	LNVIPSSSNA	CYASSGASTG	YHQLYHHQSS	ALHFADSVMV	120
ASSAGVHDGG	ASMLSAAATV	NGGAGAASAN	GGSIGLSMIK	NWLRSQPAPP	LPQPRVVAIA	180
EGAQAAQGLS	LSMNMAGAQQ	AGMPLLAGER	GRAPESVSTS	AQGGAVAARK	EDSGSGGGAG	240
ALVAVSTD TG	GSQVAETAA	RKTVDTFGQR	TSIYRGVTRH	RWTGRYEHL	WDNSCRREGQ	300
TRKGRQVYLG	GDKKEEKAAR	AYDLAALKYV	GPTTTTNFPV	SNYEKELEEM	KHMTROEFVA	360
SLRRKSSGFS	RGASIYRGVT	RHHQHGRWQA	RIGRVAGNKD	LYLGTFTQJE	EAAEAYDIAA	420
IKFRGLNAV	NFDMSTRYDV	SILDSSALPI	GSAAKRLKEA	EAAASQAQAG	VVSVDVGRIA	480
SHLGDGGLA	AYGTHYHAAA	AAWPTIAFQP	SAAAAGLYHP	YAQPMRGWCK	QEODHAVIAA	540
AHSLHELNL	NLGAAGAHAD	FFSAGQAAMH	GLGSIDNASL	EHSTGSNSV	YNGVGDSSNG	600
AVGGGYMMP	SAAAATTAM	VSHEQVHARS	HQGEHDEATK	QAAQMGYESY	LVNAEAAAYG	660
GRMPSWTPAS	APAAASNDN	MAGVGHGGAQ	LFSVWNDT			698

SEQ ID NO: 28 moltype = DNA length = 2034
 FEATURE Location/Qualifiers
 source 1..2034
 mol_type = other DNA
 organism = Setaria viridis

SEQUENCE: 28

atggcttccg	ccaacaactg	gttgggcttc	tcgctctcgg	gccaggacaa	cccgagggtc	60
aatcaggata	gctcgctcgc	tgccgcccgc	atcgacatct	ccggcgccac	cgacttctac	120
ggcctgcccga	cgcagcaggg	ctccgacggg	catctcggcg	tgccgggccc	gcgggacgat	180
cacgcttctc	atggcatcat	ggaggccttc	aacaggagcc	agcaagaaac	acaagattgg	240
aacatgagag	gtctggacta	caacggcgcc	gcctcggagc	tctcgatgct	cgtcgggctg	300
agcggcgcca	agagggccgt	ggaggacggc	gagcccaagc	tcgaggactt	cctcgggcgc	360
aactcgttcg	tctccgagca	agaccaatcc	ggcggctacc	tcttctcggg	ggtcccgatg	420
gccagcagca	ccaacagcaa	cagcggggag	aacaccatgg	agctctccat	gatcaagacc	480
tggttccgga	acaaccaggt	gccccagcgc	cagccggcgg	ctccgaccca	ggcgggcgcc	540
cagcccgagg	agatgagcac	cgacggccagc	gcgagcagct	tcggctgctc	ggattcgctg	600
gggaggaacg	gcacgggtgg	ggcgggggag	tcgcagagcc	tgccgctgtc	gatgagcagc	660
gggccccagc	accttcccat	ggtcgtggcc	ggcggcgga	gcggcgccag	tgagcgggcg	720
gcctcggaga	gcacgtcgtc	ggagaacaag	cgggagagtg	gcgcatgga	ttcgcccagc	780
agcggcgcca	tagaagccgt	cccagggga	tccatcgaca	cggttgggca	aaggacctct	840
atataccgag	gtgtaaccag	acatagatgg	acagggcggt	atgaggctca	tctctgggat	900
aatagttgta	gaagagaagg	gcagagtcgc	aagggtaggg	aagtttatct	tggtggctat	960
gacaaggagg	acaaggcagc	aagggttat	gatttggcag	ctctcaagta	ttggggcact	1020
actacaacaa	ctaatttccc	gataagtaac	tacgagaaag	agctagagga	aatgaaacat	1080
atgactagac	aggaatacat	tcgctacctc	agaaggaata	gcagcggatt	ttctcgtggg	1140
gcatcaaagt	atcaggaggt	aactagacac	catcagcatg	ggagatggca	agcaaggata	1200

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gggagagttg caggaaacaa ggatctctac ttgggcacat tcagcaccga ggaggaggcg 1260
gcggaggcct acgacatcgc cgccatcaag ttccgcgggc tcaacgcgt caccacattc 1320
gacatgagcc gctacgacgt gaagagcatt ctcgagagca gcacgctgcc tgtcggcggc 1380
gcggccaggc gctcaagga cgccgtggac caegtggagg ccggcgccac catctggcgc 1440
gcgacatgg acgcggcgt gatctcgag ctggcgcag ccgggatggg cgctacgcg 1500
tcgtacggcc accacgctg gccgaccatc gcgtttcagc agccttcgcc gctcaccgtg 1560
cactaccgt acggccagcc gtcccgcggg tgggtgcaag ccgagcagga cgccgcgctg 1620
gccgcgcgcg cgcacagcct gcaggacctg cagcagctgc acctcggcag cgccgccac 1680
aacttcttcc aggcgtcgtc gagctcgacg gtctacaacg ccggcgcgcc cagcgcctgcc 1740
gggtatcagg gcctcggtag cggcgcgccg ggctccttc tgatgccgtc gagcaccgtc 1800
gtggcagacc agggccacag cagcacggcc aaccagggga gcacctgcag ctacggggac 1860
gaccaggatg ggaagctcat cgggtacgac gccatggcgg ccgcgcggcg ggaccctgac 1920
gccgcgcgcg ggagcgggta tcagttctcg cagggtctcg gatccacggt gagcatcgcg 1980
agggcgaaag ggtactccaa caactggagc tctcctttca acggcatggg gtga 2034

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SEQ ID NO: 29      moltype = AA length = 677
FEATURE
source            Location/Qualifiers
                  1..677
                  mol_type = protein
                  organism = Setaria viridis

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SEQUENCE: 29
MASANNWLGF SLSQDNPOA NQDSSPAAAG IDISGATDFY GLPTQQGSDG HLGVPGLRDD 60
HASYGIMEAF NRSQQTQDW NMRGLDYNGG ASELMLVGS SGGKRAVEDG EPKLEDFLGG 120
NSFVSEQDQS GGYLFSGVPM ASSTNSNSGS NTMELSMIKT WLRNNQVPQP QPPAPHQAAP 180
QPEEMSTDAS ASSPGCSDSL GRNGTVAAGS SSQLALSMST GPQHLPMVVA GGGGGASGAA 240
ASESTSSENK RASGAMDSPS SGAIEAVPRK SIDTFGQRTS IYRGVTRHRW TGRYEHLWD 300
NSCRREGQSR KGRQVYLGYY DKEDKAARAY DLAALKYWT TTTTNFPISN YEKELEEMKH 360
MTRQEVYIAYL RRNSGFSRGR ASKYRGVTRH HQHGRWQARI GRVAGNKDLY LGTFSTEEEA 420
AEAYDIAAIK FRGLNAVTFN DMSRYDVKSI LESSTLPVGG AARRLKDAVD HVEAGATIWR 480
ADMDAGVISQ LADAGMGAYA SYGHHGWPTI AFQQPSPLTV HYPYQPSRG WCKPEQDAV 540
AAAAHSLQDL QQLHLGSAH NFPQASSST VYNGGGASAA GYQGLGSGGG GSFLMPSSTV 600
VADQGHSSTA NQGSTCSYGD DQDGKLIGYD AMAAAGDPY AAARSGYQFS QGSGSTVSIA 660
RANGYSNNWS SPFNGMG 677

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SEQ ID NO: 30      moltype = DNA length = 16
FEATURE
source            Location/Qualifiers
                  1..16
                  mol_type = genomic DNA
                  organism = Zea mays

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SEQUENCE: 30
acgtaagcgc ttacgc 16

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SEQ ID NO: 31      moltype = DNA length = 1371
FEATURE
source            Location/Qualifiers
                  1..1371
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 31
acacaggacc aagaacttga agatgcattt gaaggccttt atcttgttga ctcccaaggg 60
ccctagactt tgtaattctt cattttgtgct ctgctgactc ggtctgatac tgatgtaact 120
gatcaatgaa ctaattgtat tagaactgga ttgtactctt ttttctctt atatggtttt 180
ctcataaggc gagttttttc ctagaagggt ttttaataag acagccattg cacaacacgc 240
tataatattt tttttaaggt ctatgagact gactccgtgt gtgctactgc ctactggcta 300
ctactatctg tgaaattgtg acctgtgaac ttgaaatgt gaaatttgtg acttgagaac 360
tatgatttta tgacatatga agttgtgaac tgtgtatttg atacctgtgt gaatttatga 420
cctatttagg ccttgttcgt ttacaccaat ccagctctgg attgacatgg attggaatta 480
aatacatgtc acaatctatg tcccaaaata atccaagcct actcattttt ttatttgggt 540
aaacccatca tagattataa cccaaggatt taggaaattt ttaactatg gaagacatga 600
attctattca tagcttatta ggtatggaat aaatccatga atatattgca caagtttata 660
ttagaattca tgaatcaaaa gaataactag ttttgagaga tacatggatt aaatggtaga 720
tttaactcca ctatgggatt gatgtgtata tatggattta ttcaatccaa atccggatta 780
aatccttggt ggatctatat atattgggtgt gctcttagct cggttgtgta ggtgggcca 840
gtttgacgtg ccgagctggc acgatcggac cttttaccgc tgccgtgtct gtgcaagggg 900
tgttgcccggt caggaggcac cgtgagttaa tcggactcaa ttggaccggt ctctcggat 960
cgcccgctgc cgccgtttgg atttctatac ctgcacctgt ggccgtgtgg gagtggggac 1020
tgccaatgac attcttgcac cctcctcac caatcaaggc ggcaacatac cgccctttg 1080
gccttccatg aacatgaacg cggcggaacg ccacgcggcg gtgcactact cacctgcacg 1140
aattcgccgc ccactcacag cgccaaccca acttgattgc acgcactacc atcaattcgc 1200
cgcccgggac gtaagcgctt acgcccaccc cttctgccag ctgctattta tacgcctcgc 1260
cccgtccag tctcagcaga accaccagtc ctccactcca tcttctactc cgaccacaac 1320
cacagcgacc acgaccgtgc acgtacgtac atcagcacac caggcaacgg c 1371

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SEQ ID NO: 32      moltype = DNA length = 1397
FEATURE
source            Location/Qualifiers
                  1..1397
                  mol_type = other DNA

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organism = synthetic construct
SEQUENCE: 32
acacaggacc aagaacttga agatgcattt gaaggccttt atcttgttga ctcccaaggg 60
ccctagactt tgtaactctt cattttgtgct ctgctgatct ggtctgatac tgatgtaact 120
gatcaatgaa ctaattgtat tagaactgga ttgtactctt tttttccttt atatgggttt 180
ctcataaggc gagtttttac ctagaagggt ttttaataag acagccattg cacaaacagc 240
tataaatatt tatttaaaagt ctatgagact gactccgtgt gtgctactgc ctactggcta 300
ctactatctg tgaattgtg acctgtgaac ttgaaatgt gaaatttgtg acttgagaac 360
tatgatttta tgacatatga agttgtgaac tgtgtatttg atacctgtgt gaatttatga 420
cctatttagg ccttggtcgt ttacaccaat ccagctctgg attgacatgg attggaatta 480
aatacatgtc acaatctatg tcccaaaata atccaagcct actcattttt ttatttgggt 540
aaacccatca tagattataa cccaaggatt taggaaattt ttaactatg gaagacatga 600
attctattca tagcttatta ggtatggaat aaatccatga atattattgca caagtttata 660
ttagaattca tgaatcaaaa gaataactag ttttgagaga tacatggatt aaatggtaga 720
tttaactctc ctatgggatt gagtgtgata tatggattta ttcaatccaa atccggatta 780
aatccttggt ggaatctat atattgggtgt gctcttagct cgggttgtga ggtgggcat 840
gtttgacgtg ccgagctggc acgatcggac cttttaccg tgcgctgtc gtgcaagggg 900
tggtgcccgt caggaggcac cgtgagttaa tcggactcaa ttggaccggt ctctcggat 960
cgcccgctgc cgccgtttgg atttctatac ctgcacctgt ggctgtggg gagtggggac 1020
tgcaaatgac attcttgcat cctctctcac caatcaaggc ggcaacatac cggccctttg 1080
gccttccatg aacatgaacg cggcggaacg ccacgccggc gtgcactact cacctgcattg 1140
aattcgccgc ccactcacag gcgcaaccca acttgattgc acgcactacc atcaattcgc 1200
cgcccgggac gtaagcgtt acgcaccgct ccttacgtaa gcgcttacgc ccatcccttc 1260
tgccagctgc tatttatagc cctcgccccc ctccagcttc agcagaacca ccagtcctcc 1320
actccatctt ctactccgac cacaaccaca gcgaccacga ccgtgcacgt acgtacatca 1380
gcacaccagg caacggc 1397

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SEQ ID NO: 33      moltype = DNA length = 1423
FEATURE           Location/Qualifiers
source            1..1423
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 33
acacaggacc aagaacttga agatgcattt gaaggccttt atcttgttga ctcccaaggg 60
ccctagactt tgtaactctt cattttgtgct ctgctgatct ggtctgatac tgatgtaact 120
gatcaatgaa ctaattgtat tagaactgga ttgtactctt tttttccttt atatgggttt 180
ctcataaggc gagtttttac ctagaagggt ttttaataag acagccattg cacaaacagc 240
tataaatatt tatttaaaagt ctatgagact gactccgtgt gtgctactgc ctactggcta 300
ctactatctg tgaattgtg acctgtgaac ttgaaatgt gaaatttgtg acttgagaac 360
tatgatttta tgacatatga agttgtgaac tgtgtatttg atacctgtgt gaatttatga 420
cctatttagg ccttggtcgt ttacaccaat ccagctctgg attgacatgg attggaatta 480
aatacatgtc acaatctatg tcccaaaata atccaagcct actcattttt ttatttgggt 540
aaacccatca tagattataa cccaaggatt taggaaattt ttaactatg gaagacatga 600
attctattca tagcttatta ggtatggaat aaatccatga atattattgca caagtttata 660
ttagaattca tgaatcaaaa gaataactag ttttgagaga tacatggatt aaatggtaga 720
tttaactctc ctatgggatt gagtgtgata tatggattta ttcaatccaa atccggatta 780
aatccttggt ggaatctat atattgggtgt gctcttagct cgggttgtga ggtgggcat 840
gtttgacgtg ccgagctggc acgatcggac cttttaccg tgcgctgtc gtgcaagggg 900
tggtgcccgt caggaggcac cgtgagttaa tcggactcaa ttggaccggt ctctcggat 960
cgcccgctgc cgccgtttgg atttctatac ctgcacctgt ggctgtggg gagtggggac 1020
tgcaaatgac attcttgcat cctctctcac caatcaaggc ggcaacatac cggccctttg 1080
gccttccatg aacatgaacg cggcggaacg ccacgccggc gtgcactact cacctgcattg 1140
aattcgccgc ccactcacag gcgcaaccca acttgattgc acgcactacc atcaattcgc 1200
cgcccgggac gtaagcgtt acgcaccgct ccttacgtaa gcgcttacgc accgctcctt 1260
acgtaagcgc ttaacgcccac ccttctgccc agctgctatt tatacgcctc gccccgctcc 1320
agtctcagca gaaccaccag tcttccactc catcttctac tccgaccaca accacagcga 1380
ccacgaccgt gcacgtacgt acatcagcac accaggcaac ggc 1423

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SEQ ID NO: 34      moltype = DNA length = 1398
FEATURE           Location/Qualifiers
source            1..1398
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 34
acacaggacc aagaacttga agatgcattt gaaggccttt atcttgttga ctcccaaggg 60
ccctagactt tgtaactctt cattttgtgct ctgctgatct ggtctgatac tgatgtaact 120
gatcaatgaa ctaattgtat tagaactgga ttgtactctt tttttccttt atatgggttt 180
ctcataaggc gagtttttac ctagaagggt ttttaataag acagccattg cacaaacagc 240
tataaatatt tatttaaaagt ctatgagact gactccgtgt gtgctactgc ctactggcta 300
ctactatctg tgaattgtg acctgtgaac ttgaaatgt gaaatttgtg acttgagaac 360
tatgatttta tgacatatga agttgtgaac tgtgtatttg atacctgtgt gaatttatga 420
cctatttagg ccttggtcgt ttacaccaat ccagctctgg attgacatgg attggaatta 480
aatacatgtc acaatctatg tcccaaaata atccaagcct actcattttt ttatttgggt 540
aaacccatca tagattataa cccaaggatt taggaaattt ttaactatg gaagacatga 600
attctattca tagcttatta ggtatggaat aaatccatga atattattgca caagtttata 660
ttagaattca tgaatcaaaa gaataactag ttttgagaga tacatggatt aaatggtaga 720

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ttaaattctca ctatgggatt gagtgtgata tatggattta ttcaatccaa atccggatta 780
aatccttgggt ggatctatat atattgggtgt gctcttagct cggttgtgta ggtgggcat 840
gtttgacgtg cagagctggc acgatcggac cttttaccgc tgccgtgctc gtgcaagggg 900
tgttgcccggt caggaggcac cgtgagttaa tcggactcaa ttggaccgggt ctctcggat 960
cgcccgctgc cgccgtttgg atttctatac ctgcaacctgt ggctgtggg gagtggggac 1020
tgcaaatgac attcttgcac cctctctcac caatcaaggc ggcaacatac cggccctttg 1080
gccttccatg aacatgaacg cggcggaacg ccacgccggc gtgcactact cacctgcattg 1140
aattcgccgc ccactcacag cgccaaccca acttgattgc acgcactacc atcaattcgc 1200
cgcccgggac gtaagcgctt acgcaccgct ccttacgtaa gcgcttacgc ccatcccttc 1260
tgccagctgc tataataaac gcctcgcccc gctccagtct cagcagaacc accagtcttc 1320
cactccatct tctactccga ccacaaccac agcgaccacg accgtgcacg tacgtacatc 1380
agcacaccag gcaacggc

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SEQ ID NO: 35      moltype = DNA length = 1371
FEATURE           Location/Qualifiers
source            1..1371
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 35
acacaggacc aagaacttga agatgcattt gaaggccttt atcttgttga ctccaagggt 60
ccctagacct tgtaattctg cattttgtgt ctgctgatct ggtctgatac tgatgtaact 120
gatcaatgaa ctaattgtat tagaactgga ttgtactctt ttttctctt atattggtttt 180
ctcataaggc gagtttttac ctgaaaagggt ttttaataag acagccattg cacaaacagc 240
tataatattt tatttaaagt ctatgagact gactccgtgt gtgctactgc ctactggcta 300
ctactatctg tgaattgtg acctgtgaac ttgaaatgt gaaatttgtg acttgagaac 360
tatgatttta tgacatatga agttgtgaac tgtgtatttg atacctgtgt gaatttatga 420
cctatttagg ccttgttctg ttacaccaat ccagctctgg attgacatgg attggaatta 480
aatacatgtc acaatctatg tcccaaaata atccaagcct actcattttt ttatttggtt 540
aaacccatca tagattataa cccaaggatt taggaaattt ttaacctatg gaagacatga 600
attctattca tagcttatta ggtatggaat aaatccatga atattattgca caagtttata 660
ttagaattca tgaatcaaaa gaataactag ttttgagaga tacatggatt aaatggtaga 720
ttaaattctca ctatgggatt gagtgtgata tatggattta ttcaatccaa atccggatta 780
aatccttgggt ggatctatat atattgggtgt gctcttagct cggttgtgta ggtgggcat 840
gtttgacgtg cagagctggc acgatcggac cttttaccgc tgccgtgctc gtgcaagggg 900
tgttgcccggt caggaggcac cgtgagttaa tcggactcaa ttggaccgggt ctctcggat 960
cgcccgctgc cgccgtttgg atttctatac ctgcaacctgt ggctgtggg gagtggggac 1020
tgcaaatgac attcttgcac cctctctcac caatcaaggc ggcaacatac cggccctttg 1080
gccttccatg aacatgaacg cggcggaacg ccacgccggc gtgcactact cacctgcattg 1140
aattcgccgc ccactcacag cgccaaccca acttgattgc acgcactacc atcaattcgc 1200
cgcccgggac gtaagcgctt accccatcc cttctgccag ctgctatagc gccgcctcgc 1260
ccgctctcag tctcagcaga acaccagtc ctccactcca tcttctactc cgaccacaac 1320
cacagcgacc acgaccgtgc acgtacgtac atcagcacac caggcaacgg c 1371

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```

SEQ ID NO: 36      moltype = DNA length = 1397
FEATURE           Location/Qualifiers
source            1..1397
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 36
acacaggacc aagaacttga agatgcattt gaaggccttt atcttgttga ctccaagggt 60
ccctagacct tgtaattctg cattttgtgt ctgctgatct ggtctgatac tgatgtaact 120
gatcaatgaa ctaattgtat tagaactgga ttgtactctt ttttctctt atattggtttt 180
ctcataaggc gagtttttac ctgaaaagggt ttttaataag acagccattg cacaaacagc 240
tataatattt tatttaaagt ctatgagact gactccgtgt gtgctactgc ctactggcta 300
ctactatctg tgaattgtg acctgtgaac ttgaaatgt gaaatttgtg acttgagaac 360
tatgatttta tgacatatga agttgtgaac tgtgtatttg atacctgtgt gaatttatga 420
cctatttagg ccttgttctg ttacaccaat ccagctctgg attgacatgg attggaatta 480
aatacatgtc acaatctatg tcccaaaata atccaagcct actcattttt ttatttggtt 540
aaacccatca tagattataa cccaaggatt taggaaattt ttaacctatg gaagacatga 600
attctattca tagcttatta ggtatggaat aaatccatga atattattgca caagtttata 660
ttagaattca tgaatcaaaa gaataactag ttttgagaga tacatggatt aaatggtaga 720
ttaaattctca ctatgggatt gagtgtgata tatggattta ttcaatccaa atccggatta 780
aatccttgggt ggatctatat atattgggtgt gctcttagct cggttgtgta ggtgggcat 840
gtttgacgtg cagagctggc acgatcggac cttttaccgc tgccgtgctc gtgcaagggg 900
tgttgcccggt caggaggcac cgtgagttaa tcggactcaa ttggaccgggt ctctcggat 960
cgcccgctgc cgccgtttgg atttctatac ctgcaacctgt ggctgtggg gagtggggac 1020
tgcaaatgac attcttgcac cctctctcac caatcaaggc ggcaacatac cggccctttg 1080
gccttccatg aacatgaacg cggcggaacg ccacgccggc gtgcactaac gtaagcgctt 1140
acgcaccgct ccttacgtaa gcgcttacgc ctcacctgca tgaattcgcg gccactcac 1200
agcgcacacc caacttgatt gcacgacta ccatcaattc gccgcgcggg ccatcccttc 1260
tgccagctgc tatttatacg cctcgccccg ctccagtctc agcagaacca ccagtctctc 1320
actccatctt ctactccgac cacaaccaca gcgaccacga ccgtgcacgt acgtacatca 1380
gcacaccagg caacggc

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SEQ ID NO: 37      moltype = DNA length = 2000
FEATURE           Location/Qualifiers

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source          1..2000
                mol_type = genomic DNA
                organism = Triticum aestivum

SEQUENCE: 37
ttagaagctt tggcgtgtcg agaagctcaa gcgcttgac ttgatctcat ggaaaataga 60
atcggtgttg cgagtgattc gaagacgggtg gtttcagata ttgaaaaagg tatagaaggg 120
tcgcattctt ccattgtaaa ggagattgca atgcggggcga tggagtttgg tacatgtgac 180
tttattttcg agggctcgagc actcaattgg gaagctcaca gtttagctaa atttgctagt 240
tcttttagatg ttggacgccca tgtttggcta ggggctcctc atgaccccta attatccctg 300
taaaccctgt tactattcaa taaagcttgg tttactgcta aaaagactat tcttttcttt 360
tcttctcagg gaaatttttt ccacgtggc ttcagccgac aaagaccaa tgcgagact 420
gaatgcattg tgtgccactc gaggaggaga gttagacgac ggccctccat tctgcaccgc 480
gtttgaagcg tcttagtctg atgagtatga gttagatatg gacgggtgacg acaccaatg 540
cctattttct caccaccgta agtttgccga taccatcttt cgaatgtcgg tggctagact 600
gcttgagtcg aatggcccat ttgacctgta gcaaccacga gtcgggtggcg acatagcata 660
ctcagcattat aaccocactg gggtagatca gaggtgaaaa atatactttc tctctacact 720
aaatagggtg gtccttaata gaaaaaaacg ataccagagg ggcgagcacc cctcagcata 780
tttttataca agggagcaaga aacgtgggtg aattattcgc tccccatggg gatcaagttg 840
gtcttgaaaa aaattgtatg ggctacttaa caatgcactc cgtggagtag tgtttgtgac 900
ctctcgagtc tctaccgatc ttctgtgcac gttttgtcag agctttgtac atgcgaactt 960
tgaacatcca gactccggtt tccaagtcct agacctgaca ttggtggtcg cattttctcg 1020
gttttatttt agggcattcg acgatgtgcg ttcagtggga agaagacgtt ctgcgcgact 1080
acgaggacgt ttatggcgcg ttcgtcaatc tcaaaatatt gtgctgtgtg catgtccttc 1140
gcatgggtga gtgtatgtgc gtgtagataa gcatccgcat ttgtattgtg tcaaaaaaat 1200
tcaggttctt ttaatccacc cactagctgt tcaattgac aatgctaaag gacaggatcg 1260
ttcccaattt gcaatgtgtc tgaatggagt taaaaggctt ttttttgcg cgcggtgaga 1320
ttaaaggctt tttggtaaac ggaagtgtag aactactcat atttcgagag caggataatg 1380
ttagccatgt ttagaactt caggagaacg aaggaccata ttgtggtctg gatccattct 1440
ggccttgtta acattctag ttgtatagg gatgcattcc tttgctcttg agtaatacat 1500
ttcccttacc cgccagaaaa aaaaactttt acaacttact cgcaaaaaat aatgtggcac 1560
ggttaacagt gagtttatgg cgccctttta cattcgttcc gatcacgatg ttcattatga 1620
cgtacctgtg tacgtgcacg tggatcgtga tccatgcaac atgatcaaac gctgcaggcg 1680
caaatcatca ctgagccgat cggttaccgc cgttcccgcg atcaattatc aagttagcta 1740
gcggggcata aatccgata atcaagctaa ggcacctgca tgcagcatg catgcatggc 1800
gcagcaacgc ccgcgcgccg gtgcgtctcc ccgcgatccc taacagtga acgtacgcgc 1860
gccttctcca cttgcgcttc caggcacgtc gatccactgt ataaacacgc cgctggtgcg 1920
tacgcagagc acgcacacgg ccttgcatg ctactacct tgcgaccaca aatcgccggg 1980
cttctctcgc cggtcgatcc                                     2000

SEQ ID NO: 38      moltype = DNA length = 1783
FEATURE            Location/Qualifiers
source             1..1783
                  mol_type = genomic DNA
                  organism = Triticum aestivum

SEQUENCE: 38
acactttaca aagtttatca ccctagctaa cttagcaaac ttatgggcaa caacgttggc 60
cgaacgactc atccaagaaa ctttaaaatc atacctagaa tgcaaaagct gctttatctc 120
cttgatccat aggcgcgaca cagagtagtc caccctcttg tgattgatct tgttgacaac 180
caccctgcta tcgagcttta catgaagcgt ctcaacgttc atctgaatag ccagctacac 240
cgcgcgtcgg caggccagaa cctccgacga ctcgggatcc gtgatgtcga ggaagaaatg 300
gcaggccctg gcaaggagcg ctccattgtg atctcttgca acgacgcctc cactcctttt 360
atctctagct ctactagtga caccattggc gttgaacttg acccagccct cgactggggc 420
cccccatctc ggtacttgtt ttgccatcac cgtctgctcc ttcgtttgat gcacattctt 480
ccattcctgc ataaaaactgc acatagtttt tgaatttca tgcggcgagg caatcctctt 540
tccatccctc gcatcatttc tagtcatcca taaggatat acaacatgga tcatagcttc 600
cttctcttcc tcaactcgca cagctaacca ggtaagcatc catcgagcct gtgctccatg 660
atcatccatc ggcgtggggg gagcagtaac atcaatgtcc ttctcgttcc gaagaaattt 720
ccaaaaaagc tgagcatgtg gacaagacca aaacttgtgc ataatagtct cctctctccc 780
acacgccaca tagaaaaacc caggtttgat accgcgcgca tggacttcag acccaatagc 840
aagaccatct aactaacatc cacatataaa ttttctgctt tacctggagc attaggcctt 900
gtcaaatggg aggtgcttag ggaagtgcct agaaaaataa accagacttt tcttaagcac 960
cgggtgcttat ttgtacagga tagacgttta actaagcgtt tctcctgtag aaataggcac 1020
cgggtgcttta gaaaaatccg gtttattttt ctaagcatct ctctaagcac ctttcattgt 1080
acaaggcctt agtgtcccaa agagccggtc aatagaactt gaggggtcgg gccgggtcaa 1140
cgcacatttc ttcatcgaca tcttgagatg atatgcggat tgaacattga aggttccatt 1200
ctggccttgt taacattccc agtcgtgcag cgatgcattc ctttgctctt gagtaataca 1260
tttcttttac cctgtaaaaa aaactttagt atttgtaaaa ggcaacgttg cacgggttaac 1320
atactcaaaa aataaaacgt ggcaagggtta actgtcagtt tatggcgccc ttttacattc 1380
gttccgcacg caatctcgat gttcatatat gtagtactac gtgtgtacgt gcacgtggat 1440
cgtgatccat gcaacatgat cagacgctgc aggcgctaat catcagtgag ccgatcggct 1500
accgcggttc ccgcgatcaa tcgagttcgc tagcggggca tctatacgca caatcaagct 1560
aggcaccatg catcatgca tgcattggcg agcaacgccc gcccggtcgt cteccccgcg 1620
gcattcctga aacgtacgcg cgcctcctcc acttcgcctt ccaggcacgt cgatccacta 1680
tataaacacg ccgctagtgc gtctgcagag cagcacacgc accttgcatc actagagcta 1740
ccttccagcc acaaatcgcc gggcatctct cgccggtcga tgcg                                     1783

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-continued

SEQ ID NO: 39 moltype = DNA length = 1986
 FEATURE Location/Qualifiers
 source 1..1986
 mol_type = genomic DNA
 organism = Triticum aestivum

SEQUENCE: 39

gtgtcgagaa	gctcaagcgc	ttgcacttga	tctcatggaa	aatagaatcg	ttgttgcgag	60
tgattcgaa	acggtggttt	cagataattga	aaaaggtata	gaagggtcgc	attcttccat	120
tgtaaaaggag	attgcaatgc	gggcatgga	gtttggtaca	tgtgacttta	ttttcgaggg	180
tcgagcactc	aattgggaag	ctcacagttt	agctaaattt	gctagttctt	tagatgttgg	240
acgccatggt	tggctagggg	ctcctcatga	cccctaatta	tccctgtaaa	ccttgttact	300
attcaataaa	gcttggttta	ctgctaaaaa	gactattctt	ttcttttctt	ctcagggaaa	360
ttttttccat	cgtggcttca	gccgacaaag	accaaagtgc	gagactgaat	gcatttgtgtg	420
ccactcgagg	aggagatgta	gacgacggcc	tccattctcg	caccgcgttt	gaagcgttct	480
agtcctgatga	gtatgagttg	agatatgacg	gtgacgacac	ccaatgccta	tttctccacc	540
accgtaagtt	tggcgatacc	atctttcgaa	tgtcggtggc	tagactgctt	gagtccaatg	600
gcccatttga	cctgtagcaa	ccacgagtcg	gtgggcacat	agcatactca	gcataataacc	660
cactaggggt	acatcagagg	tgaataaatat	actttcttcc	tacactaaat	aggttgggtcc	720
ttaatagaaa	aaaacgatac	cagaggggcg	agcaccctcc	agcatatttt	tatacaagga	780
gcaagaaaacg	tggttgaaat	atctgctccc	catggggatc	aagttggtct	tgaaaaaaat	840
tgatatgggct	acttaacaat	gcactccgtg	gagtagtgtt	ttgtacctct	cgagtctcta	900
ccgatcttcg	tgcacggttt	tgtcagagct	ttgatcatgc	gaactttgaa	catccagact	960
ccggttttca	agtcctagac	ctgacatttg	tggtcgcatt	ttctggtttt	tatttttaggc	1020
cattcgacga	tgtgcgttca	gtgggaagaa	gacgttctcg	ccgactacga	ggacgtttat	1080
ggcggcttcg	tcaatctcaa	aatattgtgt	cgtgtgcatg	tccttcgcat	ggttgagtgt	1140
atgtgcgtgt	agataagcat	ccgcatttgt	atttgtgtcca	aaaaattcag	gttcttttaa	1200
tcaccgcact	agctgttcaa	ttgatcaatg	ctaaaggaca	ggatcgttcc	caatttgcaa	1260
tggtgctgat	ggagattaaa	aggctttttt	tttgcgcgcg	gtgagattaa	aaggcttttg	1320
gttaacggaa	gtgtagaact	actcatattt	cgagagcagg	ataatgttag	ccatgtttta	1380
gaacttcagg	agaacgaagg	accatattgt	ggtctggatc	cattctggcc	ttgttaacat	1440
tcctagtttg	tataggggat	cattcctttg	ctcttgagta	atacatttcc	cttaccggcc	1500
agaaaaaaa	actttttcaa	cttactcgca	aaataaaaatg	tggcacgggt	aacagtgtgt	1560
tatggcgcc	ctttttacat	cgttccgatc	acgatgttca	tatgtacgta	cctgtgtacg	1620
tgcacgtgga	tcgtgtacca	tcaaacgctg	caggcgcaaa	tcatactga		1680
gccgatcggg	taccgctggt	cccgcgatca	attatcaagt	tagctagcgg	ggcatcaata	1740
cgcataaatca	agctaaaggca	cctgcgatga	tgcatgcgat	catggcgacg	caacgcccgc	1800
gcgcccgctg	gtctccccc	catccctaac	agtgaaacgt	acgcgcgcct	tctccacttc	1860
gccttccagg	cacgtgcgat	cactgtataa	acacgccgct	ggtgcgtacg	cagagcacgc	1920
acacggcctt	gcattgtctg	ctaccttgcg	accacaaatc	gccgggcttc	tctcgccggt	1980
cgatcc						1986

SEQ ID NO: 40 moltype = DNA length = 1132
 FEATURE Location/Qualifiers
 source 1..1132
 mol_type = genomic DNA
 organism = Triticum aestivum

SEQUENCE: 40

tgtagctgcc	caatgacttg	gaggctcgat	ttgagccttc	aagatttgtc	cgtgtgtctc	60
atgtagcatg	tggttaactca	aaaatcttgc	gcactgtcaa	aactagtact	agtatctctg	120
aagttgtgca	aatctgtact	acgttgttaga	ggactccatg	catactccac	ttggctcggg	180
ggggatatac	ctcagttggg	agagctccgc	tcttgcaatt	gggtcgttgc	gattacgggt	240
tggtctgtcta	attgtccaag	cggtgttagg	atctgaggaa	gcttagtttg	gatgtggaa	300
cattcgctgt	tctgattggt	gagataccgc	aaatccactt	atccaacaat	ttcagttcat	360
tttctacttt	tttctgcagc	ccgttgcaag	gaaggaaaaga	gcaagaacta	gggctaagtg	420
ctgagttacg	gttggttagc	cggaattgta	agccgaagaa	gcattctgtg	gacctggcgc	480
tatgaatggt	ttcttttcca	ataccagagc	taaataatta	ggcatcaaaa	ttcaaagaag	540
gtgatacttt	tcacgtactc	ctttggtttg	gaggaatttc	ataggaattc	tagatgatag	600
gattcttata	ggattttttt	cctatagagc	catttggttc	ataggaatgg	attcctattt	660
ctacatagga	ttggtttcta	tccttcatat	ttaataggaa	aataaaaaata	aacctagact	720
caatggaaaa	attccttttg	tgtcaaccaa	atgacatctc	gtttcctatt	cctactcata	780
ggatttgaga	tacatatcat	ctcatcttcc	ataaaattca	tattcctacg	ataattttat	840
cctatgaacc	aaaagaggcc	tcaactactc	gtaccatgaa	agaaagggga	ttttgggggg	900
agatgagaaa	cgaggggatc	acgccccgat	taattttctg	tggatagagt	aacgtaggaa	960
ggagaggaag	gggcaacgca	ttcgcgttta	attaagataa	ttttgattcg	tttgacttc	1020
cacgttctcg	ttaagccttt	tcgcttacct	gacgtttcgg	ttacttgcta	tgataagtag	1080
tagtacgtca	gcctcctcac	tcggtagcac	cagctagagt	cgacctcaga	tc	1132

SEQ ID NO: 41 moltype = DNA length = 1863
 FEATURE Location/Qualifiers
 source 1..1863
 mol_type = genomic DNA
 organism = Triticum aestivum

SEQUENCE: 41

atggcggagc	ctaagctgga	ggacttcttc	ggcggcatat	cctactccga	ccaccaccac	60
caccacaagc	aggccggcgg	caacaacatg	gtcgtccctg	caggtagcgg	cagcggcgcc	120
gccgcatgct	acacgacctc	ggcgggcagt	agcgtgggct	acctgtacca	ccccagctcg	180

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gcttccctcc agtttgccga ctccgtcatg gtggttctct cggcggtgg cgtccaccac 240
gacggcgctg gcatcatggc gaacaccaca gccaacggtg accttaataa tggcagcggc 300
gggggtggca tcgggctgtc catgatcaag agctggctgc ggagccagcc atccccggca 360
caaccaccgg agcagcagag ggcggatgcg gcggcacagg gtctctcgct gtccatgaac 420
atggcgcgct gcatggcgcc gctcggtggc ggggagcggt gtgtgcccga gttggcgatt 480
gtccggaaag atgacacggc cgttgccagc agcgctggtt ctggcgccgt ggtgtcggcc 540
ggtggtgctg atagcactgg cgaagcagt ggcgtggtgg tggagacgcc ggcgtccagg 600
aagacggctg acacgtttgg gcagcgacc tcgatttacc gtggtgtgac caggcataga 660
tggactggga gatatgagg tcacttatgg gacaacagt gcagaagaga agggcaaca 720
cgcaagggtc gtcaagtcta tctaggtggc tatgacaaag aggagaaagc tgcaagagct 780
tatgatttgg ctgctctcaa atattgggt gcgacgacga cgacaaattt cccagtaaac 840
aactacgaaa aagagctaga ggagatgaag cacatgacaa ggcaggagt tgtggcgctc 900
ttgcgaagga agagcagtgg tttctccaga ggcgcacca tttatcgtgg agtaactagg 960
caccaccaac atgggagatg gcaagcgagg ataggagag ttgcagggaa caaggatctc 1020
tacttaggca ccttcagcac gcaggaggag gctgcggagg cgtacgacat cgcggcgatc 1080
aagttccggg ggctcaatgc tgtcacaac ttgcacatga cgcgtacga tgtcaagagc 1140
atcctcgaca gcacgcgct ccccatcggt agcgctgcca agcgctcaa ggaacgcagg 1200
ggcgctacgg cgtcgtccct cgcgtgcgag cagcagcagc acgcaggcgt ggtgagcgcc 1260
tatgacatgg gcgcctggc gcctacgggt gccgcatacc accaccatca cccctctgtc 1320
ggcgcgctg ttgctgtggc gaccatcgcg ttccaggcgc gcctcatca ggtttccagc 1380
ggcggtggtg gacacatgta ccccccctac gcccatgccc agcgctgccc cgggtggtgt 1440
aagcaggagc cggaccacgc ggtgatcgcg gctgcgcaca gcctgcagga gctccaccac 1500
ctcaacctcg gcgcggtgg cggcgacacat gacttcttct cgcagcatgc ccaggcgatg 1560
cagcagcagc acggtggcct cggcagcatc cacaacgcgg ccggcgccctc gctcgagcac 1620
agcaccggct ccaactccgt cgtgtacaac aacggcgggc tcgagagcta catgcttccc 1680
atgagcacca caccacacgc gacggcgagc cagccccgcg agcaggtggc agcgcagtc 1740
cacgcacgcg cgtggcgggc acaggcatac agcggcgcca acatgagcag ggcacacgac 1800
gacgggaaga tcgcctacga gaactacctc gtgagcaccg cagaagccta cggcgggggg 1860
tag

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SEQ ID NO: 42      moltype = AA length = 620
FEATURE           Location/Qualifiers
source            1..620
                  mol_type = protein
                  organism = Triticum aestivum

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SEQUENCE: 42
MAEPKLEDFL GGISYSDHHH HHKQAGGNM VVPAGSGSGG AACYTTSGGG SVGYLYHPSS 60
ASLQFADSVM VASSGGGVHH DGAGIMANTT ANGDLNNGSG GGGIGLSMIK SWLRSQPSPA 120
QPPEQQRADA AAQGLSLSMN MAACMPPLVG GERGVPELAI VRKDDTAGGS SAGSGAVVSA 180
GGADSTGGSS GVVVETPASR KTADTFGQRT SIYRGVTRHR WTRGYEAHLW DNSCRREGQT 240
RKGRQVYLGg YDKEEKAARA YDLAALKYWG ATTTTNFPVN NYEKELEEMK HMTRQEFVAS 300
LRRKSSGFSR GASIYRGVTR HHQHGRWQAR IGRVAGNKDL YLGTFFSTQEE AAEAYDIAAI 360
KFRGLNAVTN FDMTRYDVKS ILDTALPIG SAAKRLKDAE AATASSLALQ QQQHAGVVSg 420
YDMGALAAAY AAYHHHHPV SV AGAVAWPTIA FQAPPHQVSS GGGGHMYHPY AHAQPLRGWC 480
KQEPDHA VIA AAHSLQELHH LNLGAGGGAH DFFSQHAQAM QQQHGGGLSI HNAAGASLEH 540
STGSNSVVYN NGVESYMLP MSTTTTATAS HAREQVAHA HARAVAAQAY SGNMMSRDHD 600
DGKIAIENYL VSTAEAYGGG

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SEQ ID NO: 43      moltype = DNA length = 1473
FEATURE           Location/Qualifiers
source            1..1473
                  mol_type = genomic DNA
                  organism = Triticum aestivum

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SEQUENCE: 43
atggcggagc ctaagctgga ggacttctct ggcggcatat cctactccga ccaccaccac 60
caccacaagc aggcggcgcc cggcagcgcc gggggtggca tcgggctgtc catgatcaag 120
agctggctgc ggagccagcc atcctcgccc ggtggtgctg atagcactgg cgaagcagt 180
ggcgtggtgg tggagacgccc ggcgtccagg aagacggctg acacgtttgg gcagcggacc 240
tcgatttacc gtggtgtgac caggcataga tggactggga gatatgagg tcacttatgg 300
gacaacagtt gcagaagaga agggcaaca cgcaagggtc gtcaagtcta tctaggtggc 360
tatgacaaag aggagaaagc tgcaagagct tatgatttgg ctgctctcaa atattgggt 420
gcgacgacga cgacaaattt cccagtaaac aactacgaaa aagagctaga ggagatgaag 480
cacatgacaa ggcaggagt tgtggcgctc ttgcgaagga agagcagtgg tttctccaga 540
ggcgcatcca tttatcgtgg agtaactagg caccaccaac atgggagatg gcaagcgagg 600
atagggagag ttgcagggaa caaggatctc tacttaggca ccttcagcac gcaggaggag 660
gctgcggagg cgtacgacat cgcggcgatc aagttccggg ggctcaatgc tgtcacaac 720
ttcgacatga cgcgtacga tgtcaagagc atcctcgaca gcaccgcgt ccccatcggt 780
agcgctgcca agcgctcaa ggaacgcagg cggcgtacgg cgtcgtccct cgcgtgcgag 840
cagcagcagc acgcaggcgt ggtgagcgcc tatgacatgg gcgcctggc cgcctacgg 900
gccgcatacc accaccatca cccctctgtc gccggcgctg ttgctgtggc gaccatcgcg 960
ttccaggcgc gcctcatca ggtttccagc ggcggtggtg gacacatgta ccccccctac 1020
gcccacgccc agcgctgccc cgggtggtgt aagcaggagc cggaccacgc ggtgatcgcg 1080
gctgcgcaca gctgcagga gctccaccac ctcaacctcg gcgcgggtgg cggcgacat 1140
gacttcttct cgcagcatgc ccaggcgatg cagcagcagc acggtggcct cggcagcatc 1200
cacaacgcgg ccggcgctc gctcgagcac agcaccggct ccaactccgt cgtgtacaac 1260
aacggcgggc tcgagagcta catgcttccc atgagcacca ccaccaccgc gacggcgagc 1320

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cacgccccgcg agcaggtggc agcgcgatgcc cacgcacgcg cggtggcgcc acaggcatac 1380
agcggcgcgca acatgagcag ggatcacgac gacgggaaga tcgcctacga gaactacctc 1440
gtgagcaccg cagaagccta cggcgggggg tag 1473

```

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SEQ ID NO: 44      moltype = AA  length = 490
FEATURE           Location/Qualifiers
source            1..490
                  mol_type = protein
                  organism = Triticum aestivum

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SEQUENCE: 44
MAEPKLEDFL GGISYSDHHH HHKQAGGSG GGGIGLSMIK SWLRSQPSSA GGADSTGGSS 60
GVVETPASR KTADTFGQRT SIYRGVTRHR WTGRYEHLW DNSCRREGQT RKGRQVYLGG 120
YDKEEKAARA YDLAALKYWG ATTTTNFPVN NYEKELEEMK HMTRQEFVAS LRRKSSGFSS 180
GASIRGVTR HHQHGRWQAR IGRVAGNKDL YLGTFTSTQEE AAEAYDIAAI KFRGLNAVNT 240
FDMTRYDVKS ILDTALPIG SAAKRLKDAE AATASSLALQ QQQHAGVVSQ YDMGALAAYG 300
AAYHHHPVS AGAVAWPTIA FQAPPHQVSS GGGGHMYHPY AHQPLRGWC QEPDHAIVIA 360
AAHSLQELHH LNLGAGGGAH DFFSQHAQAM QQQHGLGSI HNAAGASLEH STGNSVVVYN 420
NGGVESYMLP MSTTTTATAS HAREQVAHA HARAVAQAAY SGGNMSRDHD DGKIAYENYL 480
VSTAEAYGGG 490

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SEQ ID NO: 45      moltype = DNA  length = 1656
FEATURE           Location/Qualifiers
source            1..1656
                  mol_type = genomic DNA
                  organism = Triticum aestivum

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```

SEQUENCE: 45
atggccgaga ccgagcccaa gcttgaggac ttcctcgttg gcaactcgtt cgtctccgag 60
caagatcagg ccggtggcct ctgttctccc ggtgtcccga tggcgaccag caccaacagc 120
aacagcagga gcaacacccat ggaagctctc atgatacaaga gctggctccc tgacaaccag 180
gtaccccaag cgccgcaccc gcagtcgcag accggggcgc ccgcgacga gccgcaaccc 240
cacgaggaga tgggcaccga cgcgagcagc ttcgatccgc tggggagaaa cggcgcggtg 300
gtgggtggcg ggaagctcgca gagcctggcg ctctcgatga gcacgggctc aggatcgcac 360
tgcccgatgg ctgtgctcgg cggcagtcct agtgggggag tctcagagag cactcctcgc 420
gagaacaagc gggccagcgg cgccatggat tcgcccggcg gtgcggtaga agctgtcgcg 480
aggaagtcca tcgacacatt cgggcaaaag acgtcgatat atcgaggtgt aacaagacat 540
agatggacag ggcgttatga ggctcatctc tgggacaata gctgcagaag agaagggcag 600
agtcgcaagg ctaggcaagt ttaatctggg ggctatgaca aggaggacaa ggcagcaagg 660
gcttatgatt tggcagcact caagtatggg ggtacaacaa caacaacaaa ttcccaatt 720
aatacctacg aaaaagaggt ggatgaaatg aaacatatga ctaggcagga atacatcgca 780
tacctaagaa ggaatagcag cggattttcc cgtggtgcgt caaaatatcg tgggtgaact 840
aggcaccatc agcatgggag agggcaagca agaattggga ggggtgcagg aaacaaggat 900
ctctacttag gcaccttcag caccgaggag gaggcgccgg aggcctacga catcgccggc 960
atcaagttcc gtgggtcgaa cgccgtcacc aacttcgaca tgagtcgcta cgacgtcaag 1020
accatcctcg agagcagcac cctgcgggtg ggtggcgccg cgaggcgccc caaggaggtc 1080
gctgatcatc cagaggctgg cgccacgatc tggcgggccc gcatggacgg cggcgctcat 1140
tcccagctga ccgagctcgg gatgggcacg tacgcgtcgt accaccacgg ctggccaacc 1200
atcgcattec agcagccctc gcgcgtgctg gtgcactacc catacgtcgc ccagcgtct 1260
cgccgggtgt gcaagcccca gcaggacgcc agcgtcgcca cgcaaacctt gcaggacct 1320
cagcagctgc acctcgggac cgccgctcac aactatttcc aggcctcgtc gagctccacg 1380
gtctacaacg gcggcgggta ccagcaaggc accagtggca acgccttctt gatgcgggt 1440
agcaccgtgg tggatgaaca ggggcacagc agcactgcca ccaaccaggg aagcacatgc 1500
agctacgggg acgaagaggg gaagctcaat ctcggttagc atgcctatgg gatggcgagc 1560
accggcgccg acccatagcg ggcggcgcca actgtgagca ttgcgagggc gaacgggtac 1620
tccaacaact ggagctcgcc gttcaatggc atgtga 1656

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SEQ ID NO: 46      moltype = AA  length = 551
FEATURE           Location/Qualifiers
source            1..551
                  mol_type = protein
                  organism = Triticum aestivum

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SEQUENCE: 46
MAETEPKLED FLVGNFVSE QDQAGGFLFS GVPMATSTNS NSRNTMELS MIKSWLRDNG 60
VPQAPHPQSQ TGAPAQPPQ HEEMGTDASS FDPLGRNGAL VVAGSSQSLA LSMSTGSGSH 120
LPMVLGGSP SGGVSESTSI ENKRASGAMD SPGGAWEAVA RKSIDTFGR TSIYRGVTRH 180
RWTGRYEHL WDNCRREGQ SRKGRQVYLG GYDKEDKAAR AYDLAALKYW GTTTTNFPI 240
NTYEKEVDEM KHMTRQEFIA YLRNNSGFS RGASKYRGVT RHHQHGRWQA RIGRVAGNKD 300
LYLGTFTSTEE EAAEAYDIAA IKFRGLNAVNT NFDMRSYDVK TILESSTLPV GGAARRPKEV 360
ADHPEAGATI WRAGMDGGVI SQLTDVGMGT YASYHHGWPT IAFQQPSPLS VHYPYVAQPS 420
RGWCKPEQDA SVATQSLQDL QQLHLGTAH NYFQASSSST VYNGGGYQQG TSGNAFLMPA 480
STVVDEQGH SSTATNQGSTC SYGDEEGKLN LGYDAMAMAS TGADPYAAP TVSIARANGY 540
SNWSSPFNG M 551

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SEQ ID NO: 47      moltype = DNA  length = 1587
FEATURE           Location/Qualifiers
source            1..1587
                  mol_type = genomic DNA

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organism = *Triticum aestivum*

SEQUENCE: 47

atggccgaga	cgcagcccaa	gcttgaggac	ttcctcgttg	gcaactcgtt	cgtctccgag	60
caagatcagg	cgggtggctt	cttgttctcc	ggtgtcccga	tggcgaccag	caccaacagc	120
aacagcagga	gcaacacccat	ggagctctct	atgatcaaga	gctggctccg	tgacaaccag	180
gtaccccaag	cgcgcacccc	gcagtcgcag	accggggcgc	cgcgcagca	gccgcaaccc	240
cacgaggaga	tgggcaccga	cgcgagcagc	ttcgatccgc	tggggagaaa	cggcgcgttg	300
gtgggtggcg	ggagctcgca	gagcctggcg	ctctcgatga	gcacgggctc	aggatcgcac	360
ttgccgatgg	ctgtgctcgg	cgcgagtcgc	agtgggggag	tctcagagag	caactcctcg	420
gagaacaagc	gggcccagcg	cgcctatgat	tcgccggggc	gtgcggtaga	agctgtcgcg	480
aggaagtcca	tcgacacatt	cgggcaaaag	acgtcgatat	atcgagggtg	aacaagacat	540
agatggacag	ggcgttatga	ggctcatctc	tgggacaata	gctgcagaag	agaagggcag	600
agtgcgaagg	gtaggcaagt	ttatctgggt	ggctatgaca	aggaggacaa	ggcagcaagg	660
gcttatgatt	tggaagcact	caagtattgg	ggtacaacaa	caacaacaaa	tttcccaatt	720
aatacctacg	aaaaagaggt	ggatgaaatg	aaacatatga	ctaggcagga	atacatcgca	780
tacctaagaa	ggaatagcag	cggattttcc	cgtggtgcgt	caaaatatcg	tggtgtaact	840
aggcaccatc	agcatgggag	atggcaagca	agaattggga	gggttgacag	aaacaaggat	900
ctctacttag	gcaccttcag	caccgaggag	gaggcgggcg	aggcctacga	catcgcgggc	960
atcaagttcc	gtgggttgaa	cgcgctcacc	aacttcgaca	tgagtcgcta	cgacgtcaag	1020
accatcctcg	agagcagcac	cctgcgggtg	ggtggcgcg	cgaggcgccc	caaggaggtc	1080
gctgatcatc	cagagggtgg	cgcacagcat	tggcgggccg	gcatggacgg	cggcgctcat	1140
tcccagctga	cgcagctcgg	gatgggcacg	tacgcgtcgt	accaccacgg	ctggccaacc	1200
atcgcatctc	agcagccctc	gccgctgtcg	gtgcactacc	catacgtcgc	ccagccgtct	1260
cgcgggtggt	gcaagcccca	gcaggacgcc	agcgtcgcca	cgcaaacgct	gcaggacctc	1320
cagcagctgc	acctcgggac	cgcgctccac	aactatttcc	aggcatcgtc	gagctccacg	1380
gtctacaacg	cggcggggta	ccagcaaggc	accagtggca	acgccttctt	gatgccgggt	1440
agcaccgtgg	tggatgaaca	ggggcacagc	agcactgcca	ccaaccaggg	aagcacatgc	1500
agctacgggg	acgaagaggg	gaagtcfaat	ctcgggtacg	atgccatggc	gatggcgagc	1560
accgcgccgc	accatacgcg	ggcgtga				1587

SEQ ID NO: 48 moltype = AA length = 528
FEATURE Location/Qualifiers
source 1..528
 mol_type = protein
 organism = *Triticum aestivum*

SEQUENCE: 48

MAETEPKLED	FLVGNFSVSE	QDQAGGFLFS	GVPMTSTNS	NSRSNTMELS	MIKSWLRDNQ	60
VPQAPHPQSQ	TGAPAQPPQP	HEEMGTDASS	FDPLGRNGAL	VVAGSSQSLA	LSMSTGSGSH	120
LPMAVLGGSP	SGGVSESTSS	ENKRASGAMD	SPGGAVEAVA	RKSIDTFGQR	TSIYRGVTRH	180
RWTGRYEHL	WDNSCRREGQ	SRKGRQVYLG	GVDKEDKAAR	AYDLAALKYW	GTTTTTNFPI	240
NTYEKEVD	KHMTROEYIA	YLRNSSGFS	RGASKYRGVT	RHHQHGRWQA	RIGRVAGNKD	300
LYLGTFTST	EAAEAYDIAA	IKFRGLNAVT	NFDMSRYDVK	TILESSTLPV	GGAARRPKEV	360
ADHPEAGATI	WRAGMDGGVI	SQLTDVGMGT	YASYHHGWPT	IAFQQPSPLS	VHYPYVAQPS	420
RGWCKPEQDA	SVATQSLQDL	QQLHLGTAAH	NYFQASSSST	VYNGGGYQQG	TSGNAFLMPA	480
STVVDEQGHS	STATNQGSTC	SYGDEEGKLN	LGYDAMAMAS	TGADPYAA		528

SEQ ID NO: 49 moltype = DNA length = 20
FEATURE Location/Qualifiers
source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 49

gtaatcctaa	aataatccgt	20
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SEQ ID NO: 50 moltype = DNA length = 20
FEATURE Location/Qualifiers
source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 50

gagtaagtgg	actctctgct	20
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SEQ ID NO: 51 moltype = DNA length = 20
FEATURE Location/Qualifiers
source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 51

gcacgacatg	tccaaagcgt	20
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SEQ ID NO: 52 moltype = DNA length = 20
FEATURE Location/Qualifiers
source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 52

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gtagtgctga caccagtgtgta 20

SEQ ID NO: 53 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 53
 gggctagtagtac atccaaatgt 20

SEQ ID NO: 54 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 54
 ggctagtaca tccaaatgtgta 20

SEQ ID NO: 55 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 55
 gttaaaccctt gtcaggggatt 20

SEQ ID NO: 56 moltype = DNA length = 20
 FEATURE Location/Qualifiers
 source 1..20
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 56
 gtccgacata ggagtgttaa 20

SEQ ID NO: 57 moltype = DNA length = 3937
 FEATURE Location/Qualifiers
 source 1..3937
 mol_type = genomic DNA
 organism = Triticum aestivum

SEQUENCE: 57

atggcgggga	gtgggaagag	gaggcgcgcg	acgggtgctcg	acgacgacga	gcggcggggg	60
cgggcgaggc	tggaggaggc	ggccctgctc	ctccacaaga	tcaaagggtgc	gaatcccctt	120
ccctctctct	gatcgactgt	aaaccctcgt	aatccccttc	gaattgggggt	tgcaatcgga	180
gttaacgcgg	gtttgcgcg	ccgcgcgcgt	gggctgtcgg	cgcagggctg	gtgggctggg	240
tggtcgcgga	ggtcagcgcc	ggcgcgaccc	cttccctggc	tctccaccgc	taccagaact	300
actgcgcctg	cgctcccgcc	cgccgcgcgg	ccgcgtccct	gtccacatgg	tttggctccc	360
cctccctctc	tgcgcttcaa	atctctgcac	accaaaccctg	tcagatccct	accgggtgtg	420
gcgcgtccgc	gggggtgcggg	cgcgcggtgca	gcgcctgcag	ctacgacgcc	cccgtcggca	480
cggacgtctc	ctccctcctc	cgcaagggaat	tccacgcctc	ccgactcagt	acggactcgc	540
gtagaccaaa	ctggctcgat	tctcatcaat	ctctcgttgt	tactggctcat	gctaatacgt	600
ttgggatgcg	tcgttgatgt	gcagatgtgc	ttctcaggg	cctgctcgtg	gtgcagcaac	660
tcctgcagga	gaataagcac	tgctccaaga	gggacatcta	ctacatgtac	ccctccatgt	720
tcgtaggctc	gtgctggtct	gcgggttgcca	tgtctgatcc	atctatgcgt	agtgtgagtt	780
tcagggtgtc	tctttctttg	ctgtgtgagt	gacaattaca	agtctcactg	cagaacaagc	840
gattgttgat	cgtgccatca	atgatatctg	catactcttc	aagtcgagcc	ggcataatct	900
caacgtggta	tgtgctaccg	cttgtctgcc	atctttgcag	aagcttcagt	tatgtgttta	960
gaagtcattt	gtgcattgga	gaataagcct	ctggctgtta	ttagattata	tatggagtac	1020
aatgtgcgcg	actgcgactt	gttactccct	ggttgttcc	ccattatgg	gtaccactgt	1080
cctatctgat	gtaggattat	ttctttaatc	tgtcttttgt	gatgcattat	ttcttgaggt	1140
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atttttgttc	cttcatcctc	cctaccactt	gtcttgcttt	acttaaaaca	tgatctggct	1440
aagcagtggt	atgctgcgct	tttagcagac	atttagagca	tcataatctc	ctgtgtccat	1500
acgatggctc	tagttatttc	tgaacatttt	cccgtatgt	actcgtcaat	atctttattg	1560
ttccagtttg	gtgatgggct	ggataagatt	tgtggagggt	gaaaagaaag	tgtattgtat	1620
aacaaacgct	aatgctgtaa	gcttctatga	ttctatcgaa	atctatctaa	ttactcaata	1680
tgatctgttt	gtactatgca	ttaatttggc	tagccttttt	ccttcttcaa	atgcaggctt	1740
tccccattcc	agttagactt	gaggcaatca	aagggtgttt	aacaatttga	aaactcataa	1800
taactcttgc	cattcataga	cctttattat	ttcatatctg	ttccttagg	aaaatcagtt	1860
ttaacatctc	ttagcaagtc	aatgttcagt	ttaaatttgt	aatagatagt	gattgatgta	1920
tcagaaagtg	ggatttcgac	aaagcaccac	attttggaca	tcattgtgaa	gagtaatgag	1980
aattcatctt	tattttttgt	ctcaaacata	tatggagagg	agtatccttg	ttattccaca	2040
aactatatgg	cataggtttg	ctgtcttttt	atacatacac	tatatctctt	tcatatctct	2100
aattactatg	cagtaaacct	aagctgtcta	agttttttct	tcagatggtt	gttagtggtg	2160

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ctcactacat	ccttgttgtt	gagaaggagg	caggtgtgag	gattaaagac	ccctatctg	2220
catatgattt	ttctattcct	tgagtccttc	ctgcagtttg	attctgtatg	ccatttgcag	2280
tgttccagcg	tttggccaat	gacaagttct	gcgaaaagaa	tcgctgcatt	gttattacag	2340
taactgctaa	cgctgcacat	agtgttatac	tgccattatt	tttcacatag	catgtttgat	2400
ggttttcttc	gttttactca	gggaaggagg	taccagata	ttccaacaag	aaggcatgtg	2460
ctctttcgta	tatttgtttc	tcctgtatat	taaaactgaa	ttgtctattg	aaacgtacat	2520
tccttgtaat	gttgacagat	cttgcgctac	agctgcgaac	agctgcactt	gcctgcttat	2580
tgcttagtgg	attcagatcc	gtagtggttt	gatattctgg	ctacotataa	atttggttcc	2640
atggtagggc	caaaatatct	atctaatact	tagttattta	ttgaagtcct	aatagcatca	2700
aaggcagtg	aaatttatca	gtgcaatttc	aatttttttg	ttaagccaag	tgaaatgtga	2760
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aaattagaat	catttatgta	gttaatagcc	ggttcttttt	cctgaagatt	tttctctctg	3780
aaatattcac	gactatataa	tccttggcag	gttgaagttg	gaagccatgc	tggaaatggg	3840
tgtaagttt	gagattgaag	catgtgcagc	aagttccatt	tcgtttctgt	cgcaagagta	3900
catccccc	cagatcaggc	tagggaggta	tatatag			3937

SEQ ID NO: 58 moltype = DNA length = 3928
 FEATURE Location/Qualifiers
 source 1..3928
 mol_type = genomic DNA
 organism = Triticum aestivum

SEQUENCE: 58

atggcgggga	gtgggaagag	gagggcgcg	acgggtgctg	acgacgacga	gcggcggggc	60
cgggcgaggc	tggaggaggc	ggccctgctc	ctccacaaga	tcaaagggtg	agatccccct	120
ttccctctct	cgatcgactg	taaacctctg	tactccctct	cgaattgggg	ttgcaatcgg	180
agttaacgcg	ggttcgcgcc	gcgcgcgcgc	tgggtgtctg	gagcagggct	gggtgggctg	240
gtggtcgctg	agatcagcgc	cggccgatcc	ccttccctgg	cgctccaccc	ctaccagaac	300
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gcacggagct	cctctccctc	ctccgcaagg	aattccacgc	ctccgcctcc	agtagcgatt	540
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agtttgggat	gcgtcgttga	tgtgcagatg	tgtctctcag	ggctcgtctc	gtgggtgcagc	660
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tgttcgtagg	ttctgtctgc	ttctgcgggt	cgatgcctga	tccatctatg	cgtagtgtga	780
gtttcagggt	ttctgtttct	ttgctgtgtg	agtgacaatt	acaagttcca	ctgcagaaca	840
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ttagaagcca	tttgtgcatt	ggagaataag	cctctggctg	ttatttccatt	atgggtgtagt	1020
gggtgtacaca	atgtgcacaca	ctgccacttg	ttactccctg	gttgttctct	gtcctatctg	1080
atgtaggagc	attttcttaa	tctgtctttt	gtgatgcatt	attcatttag	gtcctcattt	1140
cttcatgaat	cgtcaggagc	tgttttcatt	agttctctca	gcagttctgc	tgcatttatt	1200
tcatacgctt	ttcactctgc	ttttagttaac	catctacaat	ttctgctcac	ttggctgctt	1260
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aacatgttta	atccttcgat	gctgttactt	aactaatcag	catttttggt	ccttcatcct	1380
ccctaccgct	tgtcctgctt	tacttaaaaa	atgatctggg	tgtaaaacag	gggatgctgc	1440
ttcttttagca	gagatttaga	gcatacacaat	ctcctatgtc	catacaatgg	ttcttagttat	1500
ttccgaacat	tttcccgctg	tgtactcgtc	atgtctttat	tggtccagtt	tggtgatggg	1560
ctggataaga	tttgtggagg	gtgaaaagaa	agtgactgcg	ataacaaacg	tcaatactgt	1620
aagcttctat	gattccatct	aatcacccag	tatgatatgt	ttctagtatg	cattaatttt	1680
gctagccttt	ttccttcttc	aaatgcaggc	tttcccccatt	ccagtttagca	ttgaagcaat	1740
caaaaggtgt	tttaacaattt	gaaaactcat	aacaactctt	gtcttcatag	ttttttatta	1800
tttcatatct	gtttcccttag	gaaaaacagt	tttataaatc	gtgatgtaca	caattagctt	1860
atatctctta	gcaagtcaat	gttcagttta	aatttgtaat	agatagtgtg	tgatgtatta	1920
gaaagtggga	tttcgacaaa	caaccaaat	ttggacagca	tgtgaaagag	taatgaggat	1980
tcactcttat	tttttgtctc	aaacaaatat	ggagaagagt	atccttggtta	ttccacaaac	2040
tgtagtggcat	aggtttgtctg	tcttgtttata	tgtacactat	atttctttca	tattctcaat	2100
tactatgcag	taactttaag	ctgtctaaagt	ttttcttgcg	agatgttggt	agtgttgctc	2160
actacatcct	tggtgttgag	aaggaggcag	gtgtgaggat	gaaagggccc	ctgtctgcac	2220
atgatttttc	tattcttctg	agtcttccca	cagttttctt	ctgtatgcca	tttgcagtat	2280
ttccagcttt	ggccaatgac	aagttctgcg	aaaagaatcg	ctgcattggt	attacagtaa	2340

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ctgctaacgc	tgcattctagt	gttatactgc	cattatTTTT	cacatagcat	gtttgatggt	2400
tttcttctgt	ttactcaggg	aagaggctac	ccagatattc	caacaagaag	gcatgtgctc	2460
tttcatatat	ttgtttctct	tgtatattag	aactgaattg	ctatttgaaa	tgtacattcc	2520
ttgtaatggt	gcagattctt	gcgctacett	gtcgaacagc	tgcacttgcc	tgcttattgc	2580
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Location/Qualifiers

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 organism = Triticum aestivum

SEQUENCE: 63

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SEQ ID NO: 64      moltype = DNA length = 4203
FEATURE           Location/Qualifiers
source            1..4203
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                  organism = Triticum aestivum

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SEQ ID NO: 65 moltype = DNA length = 1559
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 mol_type = genomic DNA
 organism = Triticum aestivum

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Location/Qualifiers

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 cttaa 3845

SEQ ID NO: 88 moltype = DNA length = 4099
 FEATURE Location/Qualifiers

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source          1..4099
                 mol_type = genomic DNA
                 organism = Zea mays

SEQUENCE: 88
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cgcagctggt ggagaccgtc acgctgcggc gaggaacggg taccagttct cgcagggctc 4020
gggatccacg gtgagcatcg cgaggcgcaa cgggtacgct aacaactgga gctctccttt 4080
caacaacggc atgggggtga                                     4099

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SEQ ID NO: 89 moltype = DNA length = 2002
 FEATURE Location/Qualifiers

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source          1..2002
                mol_type = genomic DNA
                organism = Zea mays

SEQUENCE: 89
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gggaggggga ggcccagcta gcggtgcagt taccgagata aaaacacaca aggcgcggagg 180
cccctatctt aacctccact tctcctccc ccgtaaaagag agattaaaaa aggattagtt 240
cttctgtact agcctggcgt ccgagagaga gagagagaga gagcaccogg ccttggttaac 300
aattcgctct ggaggatcct atgcgaggtt tttagggtgt tttttgctca cttctccatg 360
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tgcttgactt cttctcttcc ctaacctttg cactgtccaa aatggcttcc tgatccctc 1920
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ctgcacttcc agataatcaa cc 2002

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SEQ ID NO: 90      moltype = DNA length = 2001
FEATURE            Location/Qualifiers
source             1..2001
                  mol_type = genomic DNA
                  organism = Zea mays

SEQUENCE: 90
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gttctatccc tggaaagcta tacctcctta cccctatcag cttctcctca catctcctct 1920
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caacgccctt cccttagcta g                                     2001

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SEQ ID NO: 91      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 91
tggtcatact tacgcggtga                                     20

```

```

SEQ ID NO: 92      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 92
gttacctggt cactattacg                                     20

```

```

SEQ ID NO: 93      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 93
ccagttgttc acagtggcca                                     20

```

```

SEQ ID NO: 94      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 94
agcgagccag ttgttcacag                                     20

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```

SEQ ID NO: 95      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 95
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SEQ ID NO: 96      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 96
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```

SEQ ID NO: 97      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 97
tgcccttgctt gtgatcagag                                    20

```

```

SEQ ID NO: 98      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 98
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```

```

SEQ ID NO: 99      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 99
ggagcccttc cttcccgcca                                     20

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-continued

SEQ ID NO: 100	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 100		
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SEQ ID NO: 101	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 101		
atctaagacc agtgaagact		20
SEQ ID NO: 102	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 102		
tgggatgatg aatggagctc		20
SEQ ID NO: 103	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 103		
cgtacgtaca ggagcacacc		20
SEQ ID NO: 104	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 104		
agcacaccag gcaacggcaa		20
SEQ ID NO: 105	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 105		
ggtaccagct caccacctca		20
SEQ ID NO: 106	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 106		
agctggctgc cgtagccgtg		20
SEQ ID NO: 107	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = Triticum aestivum	
SEQUENCE: 107		
LSMIKNWLRS		10
SEQ ID NO: 108	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	organism = Triticum aestivum	
SEQUENCE: 108		
KLEDPLGGIS FSE		13
SEQ ID NO: 109	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	

-continued

organism = *Triticum aestivum*

SEQUENCE: 109
LSMIKTWLFN 10

SEQ ID NO: 110
FEATURE
source
moltype = AA length = 13
Location/Qualifiers
1..13
mol_type = protein
organism = *Triticum aestivum*

SEQUENCE: 110
KLEDFLVGNS FVS 13

SEQ ID NO: 111
FEATURE
source
moltype = AA length = 202
Location/Qualifiers
1..202
mol_type = protein
organism = *Triticum aestivum*

SEQUENCE: 111
RKSIDTFGQR TSIYRGVTRH RWTGRYEHL WDNSCRREGQ XRKGRQVYLG GYDKEXKAAR 60
AYDLAALKYW GTTTTTNFPX SNYEKELEEM KHMTRQEXXA XLRRXSSEGFS RGASXYRGVT 120
RHHQHGRWQA RIGRVAGNKD LYLGTFTSTZE EAAEAYDIAA IKFRGLNAVT NFDMSRYDVK 180
SILXSXLFPV GGAAXRLKXA EX 202

SEQ ID NO: 112
FEATURE
source
moltype = AA length = 690
Location/Qualifiers
1..690
mol_type = protein
organism = synthetic construct

SEQUENCE: 112
MAXXNNWLGF SLSXQXNPXP XQDSSPXAAI DXXXXXXXXX XXXDXXXXXX XXEAXBXCX 60
NXPQDWNMRG XXXXXSSELS XLXXXXVAEP KLEDFLGXS FXSZQDXXXG GXLFSGVPXA 120
XXXXXXXXXX XXXXXXXXS XNSNSGSXTX XLSMIKWLR SXPXPQXPXX XXAXAQXLX 180
SXBXXMXGCX XXXXXXLXA GXSSQSLALS MSTGXXHLP XVVXGGGXXX XSGGXSESXS 240
XXNXRASGAM DSXXXPPXAX EAVARKSIDT FGQRTSIYRG VTRHRWTGRY EHLWDNSCR 300
REGQXRKGRQ VYLGGYDKEX KAARAYDLAA LKYWGTTTTT NFPXSNYEKE LEEMKHMTRQ 360
EXXAXLRRXS SGFSRGASXY RGVTRHHQHG RWQARIGRVA GNKDLYLGT FSTZEEAAEAY 420
DIAAIKFRGL NAVTNFDMRS YDVKSILXSS XLPVGGGAAXR LKXAEXXXEA GATIWRAXXY 480
DXGXVXSQXL GMGAYAAXYX XHXHHXWPTI AFQQPXPLXV XXPYAQXXPS RGWCKXEQDX 540
AVIAAAHSIQ DLQXLXLGAA XAHBFFXASX XXXXXXXXXX XXSXSXVYNG GGGXXXGXGXG 600
GGYFLMPMST VVAXXXQGHS SXXNQGXTCS YGDXXXXXXX IGYDAYLVXA XAXXGGGRMX 660
SWGASTVSI ARAXXXPSX WXXPFNGXMG 690

SEQ ID NO: 113
FEATURE
source
moltype = AA length = 782
Location/Qualifiers
1..782
mol_type = protein
organism = synthetic construct

SEQUENCE: 113
MAXXNBWLGF SLSQEXEPXX XXXXXXXXXX DXXGAXXXYX JPYXXXXXXX XXXXXXXXXX 60
XXXXXXXXXX XXXXDWNMR GXXXXXXXX XSXLXXXXX XXXXXXXXMA EXEPKLEDFL 120
XGXSXXXXXX XXXXXXXXXX XXXXXXXXXX XXXXXXXXXX XXXXXXXXXX 180
XXXXXGGGXX XGXXXXXXX XXXXXNXSX XXXXLSMIK SWLRXXXPX XPXXQXXXXA 240
XAQXXXXXXX MXXXXXXXP LXXXXXXVX GXXXXXXXX XXXXXXXXXX AXXXXXXXG 300
GXXXXXXXX XXXXXGXD SX GGGXXXXXXX XXXRXXDTF GQRTSIYRGV TRHRWTGRYE 360
AHLWDNSCR EGQXRKGRQV YLGGYDKEXK AARAYDLAAL KYWGXTTTTN FPXNXYEKEX 420
XEMKHMTRQE XXAXLRRXS GFSRGASXYR GVTRHHQHGR WQARIGRVAG NKDLYLGTFS 480
TZEEAAEAYD IAAIKFRGLN AVTNFDMXRY DVKXILXSXX LPXGXAAXRX KXXXXXXX 540
XXJXXXXXXX GVXXXXXXX XXXYXAXYXX XXXXXXXXXX WPTIAFQXPX XXXXXXXXXX 600
XXHXXXXXQP XRGWCKXEXD XXVXXXXXSL QXLXLXLGX XXXAHBFXPX XXXXXXXXXX 660
GXXXXXXX XXXXXXXXXX XXXXXXXXXX XXXXPXTXX XXXXXXXXXX XXXXXXXXXX 720
XXXXXXGXXX SXXXXXGXJX XXYXXXXXXX XXAXYXXGG YSNWSSPFN GMQLFSVWND 780
SN 782

SEQ ID NO: 114
FEATURE
source
moltype = DNA length = 495
Location/Qualifiers
1..495
mol_type = genomic DNA
organism = *Zea mays*

SEQUENCE: 114
atggaagcac cagcccgccg tctcccatc gtgcgcgcgc tgcggcgccg gctgctcctc 60
gtctcgcgcc acgcccacgg cgctgcctcc tcccgcgctg gcggcccgac gcccgctgcc 120
gtgcggccct ggggtgcggc gctgcaggca ctccacgacc tgcctgtgct cggcgagcag 180
cagcaggcgg gcgggttcgg gaagtgcctg gccgcgctga tggggctgag ctctgtctac 240
ggcgagatcc tgctctctct cgtcaacggc gagtcgtaca tcggggccca ctgctgcgtc 300
gccatccgcg gcgccacggc ctactgctgg ccgcccatgc tcgcctccgt cggtctcacc 360
gccgaggagg ccgacgtgct gcgcggcttc tgcgacggcg aggaggcgga ggccaccgcg 420

-continued

```
gcgcgcgcgc ctgctgtgcc cgtctctccc gcccggggac tgggccagct ggetgccgta 480
gccgtgaggg ggtga 495
```

```
SEQ ID NO: 115      moltype = DNA length = 483
FEATURE            Location/Qualifiers
source             1..483
                   mol_type = genomic DNA
                   organism = Zea mays
```

```
SEQUENCE: 115
atggcattcc ctgggtgtgc cctcgccgtg ctcagctgcc tggcgctgct cgccgcgacg 60
gccgcgcacc gggcgcccat caccggggccg tcgccctcgg ctctcgcgcg cctgacgaca 120
ctggcgagagc ggctggaggg cgccggagac cagcagtgct gggaggcgct ggttgagatc 180
aagtcgtgca cgggcgagat catcctctc ttcctcaggg gcgaggcggt cctggggccc 240
ggctgctgcc gcgccatccg cgtcatcgag caaagctgct gggccgccga ctcgatgatg 300
tctatcatcg gtttcacccc gcaggaaggg gacatgctca aggggtactg cgacgcgggc 360
gacgacaacg ccaccggcgg gcagagcggg tcgccgccgc ctcgcggtgc tgatgacgcg 420
gtcggtgctg tccgcggaag cttcgccgcc gtggcgggaa ggaagggtc catgcaccgt 480
tag 483
```

```
SEQ ID NO: 116      moltype = AA length = 10
FEATURE            Location/Qualifiers
source             1..10
                   mol_type = protein
                   organism = synthetic construct
```

```
SEQUENCE: 116
LSMIKXWLRS 10
```

```
SEQ ID NO: 117      moltype = AA length = 10
FEATURE            Location/Qualifiers
source             1..10
                   mol_type = protein
                   organism = synthetic construct
```

```
SEQUENCE: 117
LSMIKTWLRN 10
```

```
SEQ ID NO: 118      moltype = AA length = 10
FEATURE            Location/Qualifiers
source             1..10
                   mol_type = protein
                   organism = synthetic construct
```

```
SEQUENCE: 118
LSMIKNWLRS 10
```

```
SEQ ID NO: 119      moltype = AA length = 13
FEATURE            Location/Qualifiers
source             1..13
                   mol_type = protein
                   organism = synthetic construct
```

```
SEQUENCE: 119
KLEDPLGGXS FXS 13
```

```
SEQ ID NO: 120      moltype = AA length = 13
FEATURE            Location/Qualifiers
source             1..13
                   mol_type = protein
                   organism = synthetic construct
```

```
SEQUENCE: 120
KLEDPLVGNS FVS 13
```

```
SEQ ID NO: 121      moltype = AA length = 13
FEATURE            Location/Qualifiers
source             1..13
                   mol_type = protein
                   organism = synthetic construct
```

-continued

SEQUENCE: 121
KLEDPLGGIS FSE

13

SEQ ID NO: 122 moltype = AA length = 202
FEATURE Location/Qualifiers
source 1..202
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 122
RKSIDTFGQR TSIYRGVTRH RWTGRYEAHL WDNSCRREGQ XRKGRQVYLG GYDKEXKAAR 60
AYDLAALKYW GTTTTTNFPX SNYEKELEEM KHMTRQEXXA XLRRXSSGFS RGASXYRGVT 120
RHHQHGRWQA RIGRVAGNKD LYLGTFTSTZE EAAEAYDIAA IKFRGLNAVT NFDMSRYDVK 180
SILXSSXLPV GGAAXRLKXA EX 202

1. A method of expressing a gene product in an egg cell of a plant, comprising:

introducing into the plant a polynucleotide comprising a polynucleotide sequence selected from the group consisting of:

- (a) a polynucleotide sequence of SEQ ID NO: 37, 38, or 39;
- (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 37, 38, or 39;
- (c) a fragment of the polynucleotide of (a) or (b), wherein the fragment retains its ability to drive expression of a heterologous polynucleotide sequence in an egg cell;

wherein said polynucleotide sequence drives expression of an operably linked heterologous polynucleotide sequence.

2. The method of claim 1, wherein the operably linked heterologous polynucleotide sequence is a Babyboom (BBM) polynucleotide that encodes a polypeptide that, when expressed in a female gametophyte, induces haploid induction.

3. The method of claim 2, wherein the operably linked heterologous polynucleotide sequence encodes a BBM polypeptide or functional fragment thereof.

4. The method of claim 1, wherein the plant is a monocot or dicot plant.

5. The method of claim 1, wherein the plant is a wheat, cotton, sunflower, safflower, tobacco, *Arabidopsis*, soy, barley, oats, rice, maize, triticale, sorghum, canola, *cannabis*, sugarcane, rye, millet, turf grass, or a flax plant.

6. A plant comprising a promoter operably linked to a heterologous polynucleotide sequence, wherein the promoter is selected from the group consisting of:

- (a) a polynucleotide sequence of SEQ ID NO: 37, 38, or 39;
- (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 37, 38, or 39;
- (c) a fragment of the polynucleotide of (a) or (b), wherein the fragment retains its ability to drive expression of the heterologous polynucleotide sequence in an egg cell of a plant.

7. A method of inducing parthenogenesis, the method comprising:

expressing, in a female gametophyte, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in a plant egg cell, wherein the Babyboom (BBM) polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof, wherein the female gametophyte is

rendered parthenogenic and forms a haploid embryo without pollination, and wherein the Babyboom (BBM) polynucleotide is not naturally expressed in a female gametophyte.

8. The method of claim 7, wherein the nucleotide sequence encoding the Babyboom (BBM) polypeptide or fragment thereof is selected from the group consisting of: BBM1, BBM2, BMN2, BMN3, ODP2, and BBML.

9. The method of claim 7, wherein the Babyboom (BBM) polynucleotide is selected from the group consisting of:

- (a) a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (b) a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (c) a nucleotide sequence that has at least 85% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (d) a fragment of any of the nucleotide sequences of (a), (b), or (c), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity;
- (e) a fragment of any of the nucleotide sequences of (a), (b), (c), or (d), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity, wherein the amino acid fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

10. The method of claim 7, wherein the Babyboom (BBM) polypeptide is selected from the group consisting of:

- (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and
- (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

11. The method of claim 7, wherein the promoter that expresses in the plant egg cell is an egg cell-specific promoter or an egg cell-preferred promoter.

12. The method of claim 7, wherein the promoter that expresses in the egg cell comprises a polynucleotide comprising:

- (a) a polynucleotide sequence of SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86;
- (b) a polynucleotide sequence having at least 90% sequence identity to SEQ ID NO: 15, 16, 17, 18, 19, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, or 86;
- (c) a fragment of the polynucleotide of (a) or (b), wherein the fragment retains its ability to drive expression of the operably linked polynucleotide sequence in an egg cell;

wherein said promoter drives expression of the operably linked Babyboom (BBM) polynucleotide encoding the Babyboom (BBM) polypeptide or the fragment thereof.

13. The method of claim 7, wherein the female gametophyte has not been fertilized prior to or during expression of the Babyboom (BBM) polynucleotide in an egg cell.

14. The method of claim 7, further comprising: modifying a regulatory region of an endogenous Babyboom (BBM) polynucleotide so that the Babyboom (BBM) polynucleotide expresses in a female gametophyte.

15. The method of claim 7, the method comprising:

expressing the Babyboom (BBM) polynucleotide from a modified endogenous genomic BBM locus, wherein the modified endogenous genomic BBM locus comprises a modified regulatory region of an endogenous polynucleotide encoding a BBM polypeptide, wherein one or more nucleotides in the regulatory region have been modified so that the BBM polypeptide expresses in a female gametophyte.

16. The method of claim 7, the method comprising:

expressing the Babyboom (BBM) polynucleotide from a modified endogenous genomic egg cell locus, wherein the egg cell's coding or genomic sequence in the endogenous genomic egg cell locus has been modified so that it encodes a BBM polypeptide, wherein the BBM polypeptide expresses in a female gametophyte.

17. The method of claim 7, wherein the Babyboom (BBM) polynucleotide has been modified from its native form using a gene editing technique.

18. (canceled)

19. The method of claim 7, wherein the female gametophyte is a monocot or dicot female gametophyte.

20. The method of claim 7, wherein the female gametophyte is a wheat, cotton, sunflower, safflower, tobacco, *Arabidopsis*, *cannabis*, canola, sugarcane, soy, barley, oats, rice, maize, triticale, sorghum, turf grass, rye, millet, or a flax female gametophyte.

21. The method of claim 7, further comprising:

- (a) contacting the haploid embryo with a chromosome doubling agent for a period sufficient to generate a doubled haploid embryo;
- (b) isolating the doubled haploid embryo; and
- (c) regenerating a doubled haploid plant from the doubled haploid embryo of step (b).

22. The method of claim 7, further comprising:

- (a) regenerating a parthenogenic plant from a haploid embryo comprising the Babyboom (BBM) polynucleotide operably linked to the promoter that expresses in an egg cell;

- (b) pollinating the parthenogenic plant of (a) with pollen from a non-haploid inducer; and

- (c) rescuing a haploid embryo from the parthenogenic plant of (b).

23. A method for obtaining a wheat plant producing clonal, non-reduced, non-recombined gametes, the method comprising:

- (a) suppressing in a wheat plant cell the activity of:

- (1) all endogenous Spo11 or Prd1, Prd2, or Prd3 polynucleotides or polypeptides;
- (2) all endogenous Rec8 polynucleotides or polypeptides;
- (3) all endogenous Osd1 polynucleotides or polypeptides;

- (b) expressing, in a female gametophyte derived from the wheat plant cell, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell, wherein the Babyboom (BBM) polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof, wherein the female gametophyte is rendered parthenogenic and forms an embryo; and

- (c) obtaining a plant from the embryo, wherein the embryo comprises the egg cell expressed Babyboom (BBM) polynucleotide and suppressed Spo11, Rec8, Osd1, Prd1, Prd2, or Prd3 polynucleotides or polypeptides, thereby producing a wheat plant producing clonal, non-reduced, non-recombined gametes.

24. The method of claim 23, wherein the plant is a first filial generation hybrid plant.

25. The method of claim 23, wherein the activity of the endogenous Spo11 polynucleotides or polypeptides, Rec8 polynucleotides or polypeptides, Osd1 polynucleotides or polypeptides, Prd1, Prd2, or Prd3 polynucleotides or polypeptides, or combinations thereof is suppressed by introducing a nucleotide modification into its polynucleotide sequence or an amino acid modification into its polypeptide sequence.

26.-29. (canceled)

30. The method of claim 23, wherein the nucleotide sequence encoding the Babyboom (BBM) polypeptide or fragment thereof is selected from the group consisting of BBM1, BBM2, BMN2, BMN3, ODP2, and BBML.

31. The method of claim 23, wherein the Babyboom (BBM) polynucleotide is selected from the group consisting of:

- (a) a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (b) a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (c) a nucleotide sequence that has at least 85% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (d) a fragment of any of the nucleotide sequences of (a), (b), or (c), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity;
- (e) a fragment of any of the nucleotide sequences of (a), (b), (c), or (d), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity.

tion activity, wherein the amino acid fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

32. The method of claim **23**, wherein the Babyboom (BBM) polypeptide is selected from the group consisting of:

- (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and
- (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

33. A method of obtaining a clonal apomictic plant from one or more gametophytic cells in a plant in the absence of egg cell fertilization comprising:

- (a) expressing, in one or more gametophytic cell, a Babyboom (BBM) polynucleotide operably linked to a promoter that expresses in an egg cell, wherein the Babyboom (BBM) polynucleotide comprises a nucleotide sequence encoding a Babyboom (BBM) polypeptide or a fragment thereof that retains haploid induction activity, wherein the Babyboom (BBM) polynucleotide is not naturally expressed in a female gametophyte;
- (b) developing an embryo from the gametophytic cell in the absence of egg cell fertilization; and
- (c) obtaining a progeny plant from one or more gametophytic cells wherein the progeny plant contains the chromosomes from the gametophytic cell of (a), thereby achieving propagation of a flowering plant in the absence of egg cell fertilization.

34. The method of claim **33**, wherein the embryo is formed from an unreduced plant cell, wherein the unreduced plant cell is an egg cell or formed from a somatic cell.

35.-37. (canceled)

38. The method of claim **33**, wherein the Babyboom (BBM) polynucleotide is selected from the group consisting of:

- (a) a nucleotide sequence as set forth in SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (b) a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (c) a nucleotide sequence that has at least 85% sequence identity to any one of SEQ ID NO: 1, 3, 5, 7, 9, 11, 13, 20, 22, 24, 26, 28, 41, 43, 45, 47, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 87, or 88;
- (d) a fragment of any of the nucleotide sequences of (a), (b), or (c), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity;
- (e) a fragment of any of the nucleotide sequences of (a), (b), (c), or (d), wherein the nucleotide sequence encodes an amino acid fragment that has haploid induction activity, wherein the amino acid fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

39. The method of claim **33**, wherein the Babyboom (BBM) polypeptide is selected from the group consisting of:

- (a) an amino acid sequence as set forth in SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (b) an amino acid sequence that has at least 95% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (c) an amino acid sequence that has at least 85% amino acid identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 21, 23, 25, 27, 29, 42, 44, 46, or 48;
- (d) a fragment of any of the amino acid sequences of (a), (b), or (c), wherein the fragment has haploid induction activity; and
- (e) a fragment of any of the amino acid sequences of (a), (b), or (c) or (d), wherein the fragment comprises an A motif, a B motif, or an AP2 DNA binding domain, or combinations thereof.

* * * * *